

SUMS

**SMART UNIVERSITY
MANAGEMENT SYSTEM**

Software Requirements Specification

Large-Scale Project Programming

Table of Contents

Table of Contents	2
List of Figures	4
List of Tables	5
1. Introduction	6
1.1 Purpose	6
1.2 Document Scope	6
1.3 Product Scope.....	6
1.4 Intended Audience	6
1.5 Source Authority	6
1.6 Relationship to Approved Baselines.....	6
1.7 Document Conventions.....	6
1.8 Identifier Rules	6
1.9 Priority Definitions	6
1.10 Definitions and Abbreviations	6
2. Overall Description	7
2.1 Product Perspective.....	7
2.2 Product Vision.....	7
2.3 Product Functions Summary	7
2.4 User Classes.....	7
2.5 Stakeholder Summary.....	7
2.6 Operating Context	8
2.7 Product Constraints.....	8
2.8 Assumptions	8
2.9 Dependencies.....	8
2.10 Release 1.0 Boundary	8
2.11 Out-of-Scope Items.....	10
2.12 Future Enhancements	10
2.13 Team Information	10
3. System Actors and User Classes	11
3.1 Actor-Function Responsibility Matrix	12
4. Functional Requirements.....	13
4.0 Requirement Record Model.....	13
Functional Requirement Profiles.....	14
4.1 Identity and Access	15
4.2 Academic Organization	16
4.3 Curriculum and Study Plans	16
4.4 Terms, Sections, and Scheduling	17
4.5 Student and Instructor Records.....	18
4.6 Registration and Advising.....	18
4.7 Attendance.....	19
4.8 Assessment, Grading, GPA, and Standing	20
4.9 Academic Records, Transcripts, and Graduation.....	21
4.10 Reports, Notifications, Audit, and Search	22
4.11 Supporting Operations and Experience	23
5. Business Rules	24
6. Non-Functional Requirements	27
7. Security Requirements	29
8. Data Requirements	31
9. Interface Requirements	33
10. Use Case Catalogue	34
11. Detailed Use Case Specifications	36
UC-01 - Sign In	36
UC-02 - Manage Accounts and Roles.....	36
UC-03 - Manage Academic Organization	37
UC-04 - Manage Course Catalogue.....	38

UC-05 - Manage Study Plan	39
UC-06 - Manage Academic Term.....	39
UC-07 - Manage Section and Schedule	40
UC-08 - Cancel or Change Section	41
UC-09 - Manage Student Record and Status	41
UC-10 - Manage Instructor Record and Status	42
UC-11 - Submit Registration Request	43
UC-12 - Review Registration Request	43
UC-13 - Confirm or Intervene in Registration.....	44
UC-14 - Add Drop or Withdraw.....	46
UC-15 - Manage Course Repeat	46
UC-18 - Record Attendance	47
UC-19 - Manage Laboratory Absence and Makeup	48
UC-20 - Define Assessments and Enter Scores	48
UC-21 - Calculate Submit Approve and Publish Grades.....	49
UC-22 - Correct Published Grade	51
UC-24 - Close Academic Term.....	51
UC-25 - View Academic Record and GPA.....	52
UC-26 - View Unofficial Transcript	53
UC-27 - Issue Official Transcript.....	53
UC-30 - View Academic Report	54
UC-32 - Review Audit History	54
UC-36 - Search Authorized Records	55
UC-37 - Manage Own Profile	56
12. Acceptance Criteria.....	57
12.1 Rule	57
12.2 Complete Acceptance-Criteria Catalogue.....	57
13. Requirements Prioritization	79
14. Requirements Traceability.....	80
14.A OCR --> CAP -> Requirement	80
14.B CAP -> FR / UC Coverage.....	81
14.C Stakeholder -> Requirement	83
14.D Use Case -> FR / BR / SEC / DR / CAP	84
14.E Requirement -> Acceptance Criteria	85
14.F Must Requirement -> Planned Test Coverage Placeholder	85
14.G Decision / Assumption -> Requirement	86
Decision Coverage	86
Assumption Coverage	86
15. Assumptions, Dependencies, and Open Issues.....	88
15.1 Preserved Assumptions	88
15.2 Dependencies	88
15.3 Open Issues Requiring Master Project Control	88
15.4 Approved SRS-Phase Decisions.....	89
16. Constraints and Exclusions.....	90
17. Validation and Quality Review.....	90
17.1 Quality Conclusion	91
17.2 Must-Scope Feasibility Review	91
17.2.1 Must-Scope Feasibility Control.....	92
17.3 SRS Approval and Next-Phase Gate	93
Appendix A - Glossary and Acronyms	94
Appendix B - Requirement Writing Rules	95
Appendix C - Requirement Catalogue Index.....	96
Appendix D - Use Case Catalogue Summary	96
Appendix E - CAP-01-CAP-70 Coverage Matrix.....	98
Appendix F - Open Issues and Pending Validation Targets	98
Appendix G - Baseline Preservation and Change-Control Rules	99

List of Figures

Figure 1 - SUMS System Context Overview7

Figure 2 - SUMS Scope Boundary.....9

Figure 3 - High-Level Capability Map..... 13

Figure 4 - High-Level Course Registration Workflow45

Figure 5 - High-Level Grade Workflow50

List of Tables

Table 1 - Confirmed Team Information	10
Table 2 - System Actor Catalogue	11
Table 3 - Actor-Function Responsibility Matrix	12
Table 4 - Functional Requirement Profiles - Profile	14
Table 5 - Functional Requirements - Identity and Access	15
Table 6 - Functional Requirements - Academic Organization	16
Table 7 - Functional Requirements - Curriculum and Study Plans	16
Table 8 - Functional Requirements - Terms, Sections, and Scheduling	17
Table 9 - Functional Requirements - Student and Instructor Records	18
Table 10 - Functional Requirements - Registration and Advising	18
Table 11 - Functional Requirements - Attendance	19
Table 12 - Functional Requirements - Assessment, Grading, GPA, and Standing	20
Table 13 - Functional Requirements - Academic Records, Transcripts, and Graduation	21
Table 14 - Functional Requirements - Reports, Notifications, Audit, and Search	22
Table 15 - Functional Requirements - Supporting Operations and Experience	23
Table 16 - Business Rules Catalogue	24
Table 17 - Non-Functional Requirements Catalogue	27
Table 18 - Security Requirements Catalogue	29
Table 19 - Conceptual Data Requirements Catalogue	31
Table 20 - Interface Requirements Catalogue	33
Table 21 - Use Case Catalogue	34
Table 22 - Acceptance Criteria Catalogue	57
Table 23 - Requirements Prioritization - Catalogue	79
Table 24 - Traceability Matrix - 14.A OCR -> CAP -> Requirement	80
Table 25 - Traceability Matrix - 14.B CAP -> FR / UC Coverage	81
Table 26 - Traceability Matrix - 14.C Stakeholder -> Requirement	83
Table 27 - Traceability Matrix - 14.D Use Case -> FR / BR / SEC / DR / CAP	84
Table 28 - Decision Coverage - Decision Range	86
Table 29 - Assumption Coverage - Assumption	86
Table 30 - Preserved Assumptions Log	88
Table 31 - SRS Quality Audit	90
Table 32 - Must-Scope Feasibility Review	91
Table 33 - Must-Scope Work-Package Planning	92

1. Introduction

1.1 Purpose

Establish the complete requirements baseline for SUMS.

1.2 Document Scope

This document covers system actors, FR, BR, NFR, SEC, DR, IR, UC, acceptance criteria, priority, traceability, assumptions, constraints, and the next-phase gate.

1.3 Product Scope

The product supports one generic university and the seven official areas plus approved necessary supporting capabilities. University of Palestine / جامعة فلسطين is a configurable demonstration name only.

1.4 Intended Audience

Master Project Control, SUMS Project Team, Course Instructor/Evaluator, later designers, developers, database designers, UI/UX designers, testers, documenters, and presenters.

1.5 Source Authority

Authority order is: approved Scope Baseline DOCX; approved Scope Baseline PDF; Approved Course Knowledge Baseline; official project instructions; official project list; course slides; illustrative example.

1.6 Relationship to Approved Baselines

OCR-01-OCR-29, STK-01-STK-11, CAP-01-CAP-70, ASM-01-ASM-12, RSK-01-RSK-15, DEC-15-DEC-235, approved objectives, success criteria, policies, and priorities are inherited without renumbering or reinterpretation.

1.7 Document Conventions

- **Shall:** mandatory Must obligation.
- **Should:** conditional capability after Must stability.
- **May:** optional Could capability.
- Requirements describe behavior and constraints, not implementation.

1.8 Identifier Rules

FR-01-FR-140; BR-01-BR-66; NFR-01-NFR-25; SEC-01-SEC-24; DR-01-DR-21; IR-01-IR-16; UC-01-UC-43; AC-001-AC-140. No identifier is reusable or renumberable after approval.

1.9 Priority Definitions

Must = committed Release 1.0; Should = conditional after Must completion/stability; Could = optional; Proposed = pending validation and not an approved threshold.

1.10 Definitions and Abbreviations

SUMS: Smart University Management System. GPA: Grade Point Average. RTL/LTR: right-to-left/left-to-right. W: Withdrawn. I: Incomplete. P/F: Pass/Fail. Registrar: University Registrar / Academic Affairs Representative. Section: term-specific offering of a course. Closed term: finalized and locked academic term.

2. Overall Description

2.1 Product Perspective

A self-contained academic management application with no required public portal, native mobile app, offline mode, or external operational integrations.

2.2 Product Vision

Provide a coherent, auditable, secure, user-friendly academic management application aligned with the approved generic university model.

2.3 Product Functions Summary

Academic organization; curriculum and plans; courses; terms; sections and schedules; student and instructor records; registration/advising; attendance; assessments/grades; GPA/standing; records/transcripts; reports; access control; audit; conditional notifications/import/localization; optional backup and advanced support.

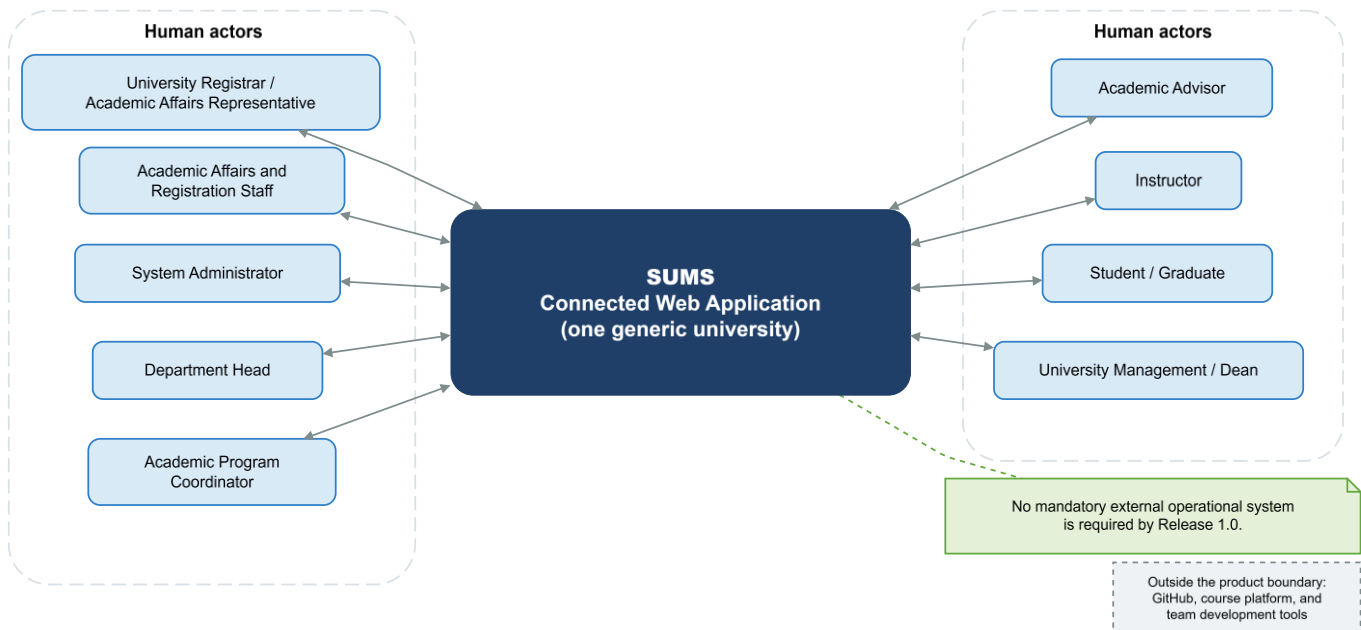


Figure 1 - SUMS System Context Overview

Figure 1 places SUMS at the center of the connected web application and shows the nine approved human actor groups. It also makes the product boundary explicit: no mandatory external operational system is required, while GitHub, the course platform, and development tools remain outside the product.

2.4 User Classes

Registrar, Registration Staff, System Administrator, Department Head, Program Coordinator, Academic Advisor, Instructor, Student/Graduate, University Management, and supporting automated system processes.

2.5 Stakeholder Summary

STK-01-STK-09 are operational/product stakeholders; STK-10 is the delivery team; STK-11 is the course evaluator. A stakeholder is not automatically a system actor.

2.6 Operating Context

Connected browser access; one university; fictional demonstration data; Gregorian calendar; Palestine time for academic operations; Sunday-Thursdays academic working days.

2.7 Product Constraints

Closed system, role-based access, OOP and relational database in implementation, original work, no hard deletion of approved academic records, no mandatory external integration, and no technology selection in this phase.

2.8 Assumptions

ASM-01-ASM-12 are preserved by identifier and are updated only in status and clarified wording according to the Master Project Control resolution recorded in Section 15. ASM-01 and ASM-12 are resolved. ASM-11 is partially resolved because the four team-member names and compliant team size are confirmed while role allocation remains pending.

2.9 Dependencies

Approved organization/curriculum must precede offerings; offerings precede registration; registration precedes attendance/grades; approved grades precede GPA/standing/transcripts. Technology-stack selection, architecture design, hosting/deployment assumptions, database design, test-data volume definition, performance-test planning, and technical feasibility validation precede any final approval of NFR-13-NFR-15. The detailed accessibility checklist depends on the UI/UX Design phase. Team-role allocation is required before repository setup and implementation, but it is not a prerequisite for SRS approval.

2.10 Release 1.0 Boundary

All 49 Must capabilities and their Must requirements. Should and Could are not acceptance blockers.

A. Inside Release 1.0 Must Boundary

Identity and access
 Academic organization
 Programs and study plans
 Students and instructors
 Courses and prerequisites
 Academic terms
 Sections and schedules
 Registration and advising
 Attendance
 Assessment and grading
 GPA and academic records
 Transcripts
 Academic reports
 Audit and accountability
 [M] General Accessibility Basics

B. Conditional Should / Could Scope

[S] Notifications
 [S] Waitlists
 [C] Automatic waitlist promotion
 [S] Program transfer
 [S] Degree progress and graduation-readiness views
 [S] Bilingual and RTL support
 [S/C] Bulk import
 [S] Attachments
 [C] Backup and restore
 Other approved Should or Could capabilities

C. Outside Scope

Admissions
 Finance and payments
 HR and payroll
 Library
 Full LMS
 Native mobile applications
 Offline operation
 External SMS/email gateways
 Biometric attendance
 Government integration
 Multi-university tenancy
 Automatic timetable generation

Must: Release 1.0 acceptance scope	Should: conditional after Must stability
Could: optional; never an acceptance blocker	Out of Scope: excluded unless approved change control

Figure 2 - SUMS Scope Boundary

Figure 2 distinguishes the committed Release 1.0 Must boundary from conditional Should and optional Could scope and from explicitly excluded product areas. Conditional or optional items do not become Release 1.0 acceptance blockers.

2.11 Out-of-Scope Items

Admissions, HR/payroll, finance/tuition/payments, library/hostel/transport/clinic/research/alumni/facilities, full LMS, native apps, offline operation, external SMS/email/biometric/SSO/government/LMS integration, automatic timetable generation, multi-university tenancy, real-data migration, and production certification.

2.12 Future Enhancements

Admissions, finance, LMS, native apps, multi-tenant operation, biometric/RFID attendance, external messaging, SSO, and government integration.

2.13 Team Information

The confirmed SUMS project team contains four members and complies with the official 3–4 student requirement. No role, responsibility, ownership, testing duty, or documentation duty is inferred in this SRS.

Table 1 - Confirmed Team Information

Team Member	Assigned Role	Main Responsibilities
Abd Al-Malik		
Amina		
Sanaa		
Hossam		

Resolved items are team size, team-member names, and compliance with the official team-size rule. Pending items are final role allocation, main responsibilities, module ownership, testing responsibilities, and documentation responsibilities. These pending allocations do not block SRS approval. Master Project Control shall complete them before repository setup and implementation begin.

3. System Actors and User Classes

Table 2 - System Actor Catalogue

Actor ID	Actor / Stakeholder	Description and Goal	Permitted Scope / Responsibilities	Prohibited Authority	Capabilities	Authentication	Data Boundary
ACT-01	University Registrar / Academic Affairs Representative / STK-01	Final university-wide academic authority	Approve general exceptions, grade reopening/corrections, curriculum activation, graduation; review institution-wide records	Cannot perform technical administration unless separately assigned; cannot bypass preserved policy rules	CAP-01,03,07,10,17-21,25,35,37	Authentication required	Institution-wide academic scope
ACT-02	Academic Affairs and Registration Staff / STK-02	Operational academic administration	Manage student records, registration operations, holds, withdrawals, official transcripts, terms, sections as authorized	Cannot directly change grades or make final Registrar decisions	CAP-03,04,06,07,12,13,18,23,31,37,40,47-49,53,54,61	Authentication required	Authorized student/term/section records
ACT-03	System Administrator / STK-03	Technical operation and support	Manage accounts, roles, login logs, settings, backups/restores, technical support	No academic decision authority; cannot modify grades/GPA/academic outcomes	CAP-15,20,27,28,49-52,60,62,70	Authentication required	Technical/security data and authorized account data
ACT-04	Department Head / STK-04	Department academic authority	Approve sections, instructors, department exceptions, attendance bars/late edits, grades, appeal recommendations, makeup labs, department reports	No university-wide final authority outside approved scope	CAP-01-10,14-16,19,20,25,26,30,33,37,54-56,59,63,64	Authentication required	Own department and delegated scope
ACT-05	Academic Program Coordinator / STK-05	Program governance	Manage plans, course classification, prerequisite/co-requisite proposals, degree progress, transfer re-evaluation, program reports	Plan changes require Department Head approval; no direct grade modification	CAP-01,02,14,32,33,35,40,45,46,58,63	Authentication required	Own program(s)
ACT-06	Academic Advisor / STK-06	Assigned-student advising and registration review	Review and approve registration, recommend exceptions, view plans/standing, maintain advising notes	No access to unassigned students except approved temporary access; no final general exception authority	CAP-02,05-08,13,21-23,32,34-36,45-47,53,57,65	Authentication required	Assigned students or approved temporary scope
ACT-07	Instructor / STK-07	Academic delivery user	Manage assigned sections, attendance, assessments, draft grades, correction requests, course history	No access to unassigned active sections; no final grade/appeal approval	CAP-04,09,10,15,16,19,25,26,30,33,55,58,66	Authentication required	Assigned sections and permitted read-only history
ACT-08	Student / STK-08	Primary end user	Request registration, view own attendance/grades/GPA/records, submit appeals/requests, manage permitted profile data	No access to another student's data; cannot approve own requests or alter official records	CAP-05-13,16-18,21-26,29,31,32,34-36,41-47,53,57,67,69	Authentication required	Own data and requests only
ACT-09	University Management / Dean / STK-09	Executive oversight	View aggregated dashboards, trends, and institutional indicators	Read-only; no direct academic record modification	CAP-01,19,38,68	Authentication required	Approved aggregate institutional scope
ACT-10	System / Scheduled Process / No stakeholder equivalent	Supporting non-human actor	Evaluate deadlines, recalculate derived values, create audit/login events, deliver system notifications, execute optional automatic promotion/backup	Cannot create policy or bypass approval; actions remain traceable	CAP-20,24,37,50	Not a human login	Only approved automated rules

3.1 Actor-Function Responsibility Matrix

The matrix summarizes the approved authority boundaries in the actor catalogue.

Manage means create or update only within the actor's approved scope; **Approve** means a formally approved decision authority; **Submit** means initiate a request or result set; **Review** means evaluate and recommend or return; **View / Read Only** never permits modification; **Manage Own** is limited to the actor's own permitted profile data; **No Authority** means the actor cannot perform that function. Where a summary cell could be read broadly, the detailed ACT-01-ACT-10 catalogue and linked requirements control.

Table 3 - Actor-Function Responsibility Matrix

Actor	Identity and Accounts	Academic Organization	Curriculum and Plans	Terms and Sections	Student Records	Registration	Attendance	Grades and GPA	Transcripts	Reports	Audit	Technical Operations
Registrar	Review	Approve	Approve	Approve	Manage	Approve	Approve	Approve	Approve	View	Review	No Authority
Registration Staff	No Authority	View	View	Manage	Manage	Manage	View	Read Only	Manage	View	View	No Authority
System Administrator	Manage	No Authority	No Authority	No Authority	No Authority	No Authority	No Authority	No Authority	No Authority	No Authority	Review	Manage
Department Head	No Authority	Review	Approve	Approve	View	Approve	Approve	Approve	Read Only	View	Review	No Authority
Program Coordinator	No Authority	View	Manage	Review	View	Review	No Authority	Read Only	Read Only	View	View	No Authority
Academic Advisor	No Authority	View	View	View	View	Approve	View	View	View	View	Read Only	No Authority
Instructor	Manage Own	No Authority	View	View	View	View	Manage	Submit	No Authority	View	Read Only	No Authority
Student / Graduate	Manage Own	View	View	View	View	Submit	View	View	View	View	No Authority	No Authority
University Management	No Authority	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	No Authority	View	No Authority	No Authority

4. Functional Requirements

CAP-01-CAP-70 Grouped Capability Domains	
1. Identity, Accounts, Roles, and Access CAP-27, CAP-28, CAP-43, CAP-44, CAP-49, CAP-51, CAP-52, CAP-62, CAP-65, CAP-66, CAP-67, CAP-69, CAP-70 11 Must / 1 Should / 1 Could	7. Attendance and Laboratory Rules CAP-09, CAP-16, CAP-30 3 Must / 0 Should / 0 Could
2. Academic Organization and Governance CAP-01, CAP-59, CAP-63, CAP-64 3 Must / 0 Should / 1 Could	8. Assessments, Grades, GPA, and Standing CAP-10, CAP-21, CAP-25, CAP-26 3 Must / 1 Should / 0 Could
3. Curriculum, Courses, and Study Plans CAP-02, CAP-14, CAP-32, CAP-33, CAP-34, CAP-36, CAP-45, CAP-46 5 Must / 3 Should / 0 Could	9. Academic Records, Transcripts, and Graduation CAP-17, CAP-18, CAP-35 2 Must / 1 Should / 0 Could
4. Terms, Sections, Scheduling, Rooms, and Laboratories CAP-03, CAP-04, CAP-05, CAP-31, CAP-37, CAP-54, CAP-55, CAP-56 8 Must / 0 Should / 0 Could	10. Reports, Notifications, Search, and Audit CAP-11, CAP-19, CAP-20, CAP-29, CAP-38, CAP-39, CAP-58, CAP-68 3 Must / 4 Should / 1 Could
5. Students and Instructors CAP-12, CAP-13, CAP-15, CAP-47, CAP-48, CAP-61 6 Must / 0 Should / 0 Could	11. Supporting Operations and User Experience CAP-40, CAP-41, CAP-42, CAP-50, CAP-60 0 Must / 3 Should / 2 Could
6. Registration, Advising, Holds, and Waitlists CAP-06, CAP-07, CAP-08, CAP-22, CAP-23, CAP-24, CAP-53, CAP-57 5 Must / 2 Should / 1 Could	
Concise domain view only. The complete CAP catalogue and exact CAP-to-FR/UC trace remain in Section 14.B and Appendix E.	

Figure 3 - High-Level Capability Map

Figure 3 groups CAP-01-CAP-70 into eleven readable domains and shows the priority distribution in each domain. The complete capability wording, priority, Release treatment, and exact FR/UC trace remain controlling in Section 14.B and Appendix E.

4.0 Requirement Record Model

Each FR row is completed by its **Profile**. The profile supplies the required preconditions, inputs, validation, required behavior context, outputs, failure/exception behavior, BR/SEC/UC relations, and dependencies. The row supplies ID, title, atomic statement, priority, status, actors, source trace, and linked acceptance criterion. Together they constitute the complete FR record; no field is intentionally blank.

Functional Requirement Profiles

Table 4 - Functional Requirement Profiles - Profile

Profile	Module / Rationale	Preconditions and Inputs	Validation / Required Behavior	Outputs and Failure Behavior	Related BR / SEC / UC	Dependencies
P1	Identity and Access	Authenticated or account-administration context; inputs are account identifier, credential, state, role, reason, or profile values.	Validate identity format, password/account rules, role authority, state, and scope.	Output is authorized session/account/profile state and audit where sensitive; on failure deny, preserve data, show safe message.	BR-15,BR-56,BR-62-BR-64; SEC-01-SEC-12,SEC-14-SEC-17; UC-01,UC-02,UC-37,UC-43	D1: approved identity/account model; user/person record as applicable.
P2	Academic Organization	Authorized academic-governance context; inputs are institution, college, department, program, status, and authority data.	Validate single-university hierarchy, active references, authority, and historical use.	Output is updated organization record/history; on failure reject invalid hierarchy or unauthorized change.	BR-01-BR-03,BR-56; SEC-09-SEC-11,SEC-14,SEC-17; UC-03	D2: P1 authorization.
P3	Curriculum and Study Plans	Authorized program/department context; inputs are course/plan/rule/version data.	Validate course code/credits/type, plan integrity, prerequisites/co-requisites, approval, and historical version constraints.	Output is approved catalogue/plan/rule state; on failure reject and preserve existing version.	BR-02,BR-17,BR-18,BR-51,BR-56; SEC-09-SEC-13,SEC-17; UC-04,UC-05,UC-28,UC-38	D3: P2 organization and P1 authorization.
P4	Terms, Sections, and Scheduling	Authorized term/section context; inputs are dates, windows, course, instructors, meetings, room/lab, capacity, status, reason.	Validate code/date/state, active course/instructor/place, conflicts, capacity, authority, and history.	Output is term/section/schedule state; on failure reject, preserve data, and audit sensitive changes.	BR-04-BR-12,BR-56,BR-61; SEC-09-SEC-14,SEC-18,SEC-19; UC-06-UC-08	D4: P2/P3 records and P1 authorization.
P5	Student and Instructor Records	Authorized official-record context; inputs are identity, program/department, status, dates, reason, and contact data.	Validate identifier uniqueness/format, authority, status transition, reference integrity, and archive policy.	Output is official profile/status/history; on failure reject and preserve prior state.	BR-13-BR-15,BR-56,BR-64; SEC-08-SEC-14,SEC-17; UC-09,UC-10,UC-43	D5: P1 identities and P2 organization.
P6	Registration and Advising	Open-term, Active-student, offered-section context unless exception path; inputs are sections, request, advisor decision, hold/exception/reason.	Validate prerequisites, co-requisites, conflicts, capacity, load, term/student/section status, holds, repeat, advisor and approval rules.	Output is enrollment/request/waitlist state and audit; on failure reject atomically, preserve seats/data, show rule result.	BR-16-BR-27,BR-61; SEC-09-SEC-14,SEC-18,SEC-19; UC-11-UC-17	D6: P3 curriculum, P4 term/sections, P5 student/advisor data.
P7	Attendance	Assigned-instructor and enrolled-student/session context; inputs are session/status/evidence/reason/approval.	Validate assignment, enrollment, session state, statuses, edit window, threshold, laboratory and makeup rules.	Output is attendance record/percentage/warning/bar/history; on failure reject unauthorized/late/invalid change and preserve data.	BR-28-BR-35; SEC-09,SEC-10,SEC-13,SEC-14,SEC-17; UC-18,UC-19	D7: P4 section schedule, P5 actors, P6 enrollment.
P8	Assessment, Grading, GPA, and Standing	Assigned-section and approved workflow context; inputs are assessments, weights, scores, grade actions, correction/appeal data.	Validate weights/precision/completeness, grading population/formula/fallback/floor, workflow authority, repeat/GPA/standing rules.	Output is draft/submitted/published/locked result, recalculated GPA/standing, or formal decision; on failure reject/rollback and preserve history.	BR-36-BR-50,BR-56,BR-61; SEC-09-SEC-14,SEC-17-SEC-19,SEC-21; UC-20-UC-25	D8: P3 course rules, P4 sections/term, P6 registrations.
P9	Academic Records, Transcripts, and Graduation	Authorized record/closed-term context; inputs are transcript request, degree evaluation, approval/cancellation reason.	Validate fixed academic history, scope, graduation conditions/status sequence, transcript control, and no-hard-delete.	Output is record view, transcript, progress/eligibility, or fixed graduation state; on failure deny and preserve official history.	BR-51-BR-56; SEC-09-SEC-11,SEC-14,SEC-17; UC-25-UC-29	D9: P8 approved results and P3 applicable plan.
P10	Reports, Notifications, Audit, and Search	Authorized scoped-query or sender context; inputs are criteria, scope, audience, format, content, filters.	Validate role/data scope, report/notification type, sensitive-data minimization, audit immutability, and export authority.	Output is scoped view/export/notification/audit result; on failure deny without revealing unauthorized data.	BR-54-BR-57; SEC-09,SEC-10,SEC-14-SEC-16,SEC-20,SEC-21; UC-30-UC-36	D10: source records from P2-P9 and P1 authorization.
P11	Supporting Operations and Experience	Authorized supporting context; inputs are profile/import/file/backup/maintenance/language/delegation values.	Validate file/row/identifier rules, authority, retention, concurrency, atomicity, browser/language policy, and delegation limits.	Output is updated profile/import/result/backup state/interface behavior/delegation; on failure reject, preserve data, and audit where required.	BR-58-BR-66; SEC-03,SEC-05,SEC-09,SEC-10,SEC-13,SEC-14,SEC-20-SEC-24; UC-37-UC-42	D11: P1 authorization and relevant source module.

4.1 Identity and Access

Table 5 - Functional Requirements - Identity and Access

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-01	Authenticate User	The system shall authenticate each approved user using the account identifier assigned to that user and a password.	Must	Approved — Committed Release 1.0	All users	P1	CAP-27,CAP-51,CAP-52; DEC-77,107,108	AC-001
FR-02	Enforce Unique Accounts	The system shall maintain one unique account for each person and prevent duplicate active accounts for the same university or employee identity.	Must	Approved — Committed Release 1.0	System Administrator	P1	CAP-27,CAP-52; DEC-72,108,172	AC-002
FR-03	Issue Temporary Initial Password	The system shall issue a temporary password for a newly activated account and require its replacement at first successful sign-in.	Must	Approved — Committed Release 1.0	System Administrator,All users	P1	CAP-27; DEC-80	AC-003
FR-04	Change Password	The system shall allow an authenticated user to change the user's own password after validating the current credential and the approved password rules.	Must	Approved — Committed Release 1.0	All users	P1	CAP-27,CAP-43; DEC-99,157	AC-004
FR-05	Reset Password	The system shall allow an authorized administrator to reset a password after identity verification and shall require the user to replace the temporary reset password at the next sign-in.	Must	Approved — Committed Release 1.0	System Administrator	P1	CAP-27; DEC-80,161	AC-005
FR-06	Apply Failed-Login Lockout	The system shall lock an account for the approved period after the approved number of consecutive failed sign-in attempts and shall permit authorized manual unlock.	Must	Approved — Committed Release 1.0	System Administrator,All users	P1	CAP-27; DEC-158-159	AC-006
FR-07	End Inactive Sessions	The system shall end an authenticated session after the approved inactivity period and require re-authentication.	Must	Approved — Committed Release 1.0	All users	P1	CAP-27; DEC-160	AC-007
FR-08	Manage Account State	The system shall allow authorized administrators to set an account to Active, Locked, or Disabled and shall enforce the access effect of the selected state.	Must	Approved — Committed Release 1.0	System Administrator	P1	CAP-27,CAP-49,CAP-70; DEC-72,105,126,163	AC-008
FR-09	Assign Multiple Roles	The system shall allow an authorized account to hold one or more approved roles without creating duplicate accounts.	Must	Approved — Committed Release 1.0	System Administrator	P1	CAP-28; DEC-74	AC-009
FR-10	Control Role Changes	The system shall allow only authorized administrators to add or remove roles, require a reason, and record the change in the audit history.	Must	Approved — Committed Release 1.0	System Administrator	P1	CAP-27,CAP-28,CAP-20; DEC-162	AC-010
FR-11	Enforce Role and Data Scope	The system shall authorize each requested operation according to the user's active roles and the approved data scope for those roles.	Must	Approved — Committed Release 1.0	All users	P1	CAP-27,CAP-51,CAP-62,CAP-65,CAP-66,CAP-67; DEC-117-124	AC-011
FR-12	Separate Technical and Academic Authority	The system shall prevent technical administration roles from approving or modifying grades, GPA, registration exceptions, standing, graduation, or other academic decisions.	Must	Approved — Committed Release 1.0	System Administrator	P1	CAP-62; DEC-118	AC-012
FR-13	Provide Graduate Record Access	The system should provide a graduate with restricted access to the graduate's own records, transcript, notifications, and permitted profile data.	Should	Approved — Conditional after Must stability	Graduate	P1	CAP-69; DEC-125	AC-013
FR-14	Apply Archived-Account Access Policy	The system shall apply read-only or blocked access to a withdrawn, inactive, or otherwise archived account according to the recorded administrative reason.	Must	Approved — Committed Release 1.0	System Administrator,Registration Staff	P1	CAP-49,CAP-70; DEC-105,126	AC-014

4.2 Academic Organization

Table 6 - Functional Requirements - Academic Organization

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-15	Manage Institution Display Information	The system shall allow authorized staff to maintain the configurable institution display name and general institutional information for the single-university installation.	Must	Approved — Committed Release 1.0	Registrar, System Administrator	P2	CAP-01; DEC-17,18,227,233	AC-015
FR-16	Manage Colleges	The system shall allow authorized academic staff to create, update, activate, and deactivate college records within the single university.	Must	Approved — Committed Release 1.0	Registrar	P2	CAP-01; DEC-45	AC-016
FR-17	Manage Departments	The system shall allow authorized academic staff to create, update, activate, and deactivate departments and associate each department with one college.	Must	Approved — Committed Release 1.0	Registrar, Department Head	P2	CAP-01; DEC-45	AC-017
FR-18	Manage Academic Programs	The system shall allow authorized academic staff to create, update, activate, and deactivate academic programs and associate each program with one department.	Must	Approved — Committed Release 1.0	Registrar, Program Coordinator	P2	CAP-01, CAP-63; DEC-45, 119	AC-018
FR-19	Preserve Organization History	The system shall preserve historical references to academic organization records that are used by approved academic records even after those organization records are deactivated.	Must	Approved — Committed Release 1.0	Registrar	P2	CAP-01, CAP-49; DEC-86, 119	AC-019
FR-20	Enforce Organization Authority	The system shall restrict academic organization and program-governance changes to the approved Registrar, Department Head, and Program Coordinator authority boundaries.	Must	Approved — Committed Release 1.0	Registrar, Department Head, Program Coordinator	P2	CAP-63, CAP-64; DEC-119, 120, 127-129	AC-020

4.3 Curriculum and Study Plans

Table 7 - Functional Requirements - Curriculum and Study Plans

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-21	Manage Course Catalogue	The system shall allow authorized academic staff to create, update, activate, and deactivate course definitions containing the approved conceptual course information.	Must	Approved — Committed Release 1.0	Department Head, Program Coordinator	P3	CAP-14; DEC-57	AC-021
FR-22	Classify Course Delivery Type	The system shall classify each course as Theoretical, Practical/Laboratory, or Mixed.	Must	Approved — Committed Release 1.0	Department Head, Program Coordinator	P3	CAP-33; DEC-89	AC-022
FR-23	Validate Course Credits	The system shall accept only positive whole-number credit hours for a course.	Must	Approved — Committed Release 1.0	Department Head, Program Coordinator	P3	CAP-14; ASM-06	AC-023
FR-24	Manage Prerequisites	The system shall allow authorized program staff to define and revise prerequisite relationships and the required minimum grade for each relationship.	Must	Approved — Committed Release 1.0	Program Coordinator, Department Head	P3	CAP-02, CAP-14; DEC-46, 57, 142	AC-024
FR-25	Manage Co-requisites	The system shall allow authorized program staff to define and revise co-requisite relationships and the approved effect of withdrawing from a co-requisite.	Must	Approved — Committed Release 1.0	Program Coordinator, Department Head	P3	CAP-02, CAP-32; DEC-88, 200	AC-025
FR-26	Manage Study Plans	The system shall allow authorized program staff to define a study plan containing required and elective courses, levels, required credits, and applicable academic rules.	Must	Approved — Committed Release 1.0	Program Coordinator, Department Head	P3	CAP-02, CAP-63; DEC-46, 119	AC-026
FR-27	Version Study Plans	The system shall preserve approved historical study-plan versions and associate each student with the applicable plan version.	Must	Approved — Committed Release 1.0	Program Coordinator, Department Head	P3	CAP-02, CAP-63; DEC-46, 102, 119, 129	AC-027
FR-28	Classify Plan Courses	The system shall allow authorized program staff to classify plan courses as required or elective and assign their program level and credit contribution.	Must	Approved — Committed Release 1.0	Program Coordinator	P3	CAP-02; DEC-46	AC-028
FR-29	Identify Eligible Courses	The system should identify courses that a student is eligible to request based on the approved plan, completion history, prerequisite, co-requisite, status, and term rules.	Should	Approved — Conditional after Must stability	Student, Advisor	P3	CAP-36; DEC-92	AC-029
FR-30	Manage Program Transfer Request	The system should support the authorized processing of a student's transfer from one academic program to another.	Should	Approved — Conditional after Must stability	Student, Advisor, Program Coordinator, Registrar	P3	CAP-45; DEC-101	AC-030
FR-31	Re-evaluate Plan after Transfer	The system should preserve the previous program history and determine which completed courses apply to the new approved study plan.	Should	Approved — Conditional after Must stability	Program Coordinator, Registration Staff	P3	CAP-46; DEC-102	AC-031

4.4 Terms, Sections, and Scheduling

Table 8 - Functional Requirements - Terms, Sections, and Scheduling

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-32	Manage Academic Terms	The system shall allow authorized staff to create and maintain academic terms for First Semester, Second Semester, and Summer Semester using the approved code convention.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P4	CAP-03,CAP-31; DEC-47,138,148	AC-032
FR-33	Manage Term Dates and Windows	The system shall allow authorized staff to configure each term's start and end dates and the registration, add/drop, withdrawal, appeal, and grade deadlines.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P4	CAP-03,CAP-23; DEC-47,139,144	AC-033
FR-34	Manage Term State	The system shall allow authorized staff to control the approved lifecycle state of an academic term and shall permit term-dependent operations only in a compatible state.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P4	CAP-03; DEC-47,93	AC-034
FR-35	Create Course Offerings and Sections	The system shall allow authorized staff to create a course offering and one or more sections for an approved course and academic term.	Must	Approved — Committed Release 1.0	Department Head,Registration Staff	P4	CAP-04; DEC-48	AC-035
FR-36	Generate and Validate Section Identity	The system shall assign or validate a unique section identifier following the approved course-term-section convention.	Must	Approved — Committed Release 1.0	Registration Staff	P4	CAP-04; DEC-173	AC-036
FR-37	Manage Section Capacity and Status	The system shall allow authorized staff to maintain a section's capacity and operational status subject to approved room and registration constraints.	Must	Approved — Committed Release 1.0	Department Head,Registration Staff	P4	CAP-04; DEC-48,140	AC-037
FR-38	Manage Section Instructors	The system shall allow authorized department staff to assign one active Primary Instructor and zero or more active Co-Instructors to a section.	Must	Approved — Committed Release 1.0	Department Head	P4	CAP-04,CAP-15; ASM-08; DEC-48,58	AC-038
FR-39	Manage Section Meetings	The system shall allow authorized staff to maintain one or more weekly meeting schedules for each section.	Must	Approved — Committed Release 1.0	Department Head,Registration Staff	P4	CAP-04; ASM-07; DEC-48	AC-039
FR-40	Manage Rooms and Laboratories	The system shall allow authorized staff to maintain buildings, rooms, and laboratories with their type, capacity, and active status.	Must	Approved — Committed Release 1.0	Registration Staff,Department Head	P4	CAP-56; ASM-04; DEC-112	AC-040
FR-41	Assign Meeting Places	The system shall allow authorized staff to assign an active room or laboratory to each section meeting and validate compatibility with meeting type.	Must	Approved — Committed Release 1.0	Registration Staff,Department Head	P4	CAP-56; DEC-112,140	AC-041
FR-42	Validate Schedule Conflicts	The system shall detect overlapping student, instructor, room, and laboratory meetings during section scheduling and registration.	Must	Approved — Committed Release 1.0	Registration Staff,Department Head,Advisor,Student	P4	CAP-05,CAP-56; DEC-48,50,112	AC-042
FR-43	Validate Room Capacity	The system shall prevent a section capacity from exceeding the assigned room or laboratory capacity unless an approved exception is recorded.	Must	Approved — Committed Release 1.0	Registration Staff,Department Head	P4	CAP-04,CAP-56; DEC-140	AC-043
FR-44	Cancel a Section	The system shall allow authorized staff to cancel a section with a reason and date, stop further registration, preserve history, and initiate the approved handling of affected registrations.	Must	Approved — Committed Release 1.0	Department Head,Registration Staff	P4	CAP-54; DEC-110,201	AC-044
FR-45	Change Section Instructor or Meeting	The system shall allow an approved instructor reassignment or meeting time/place change, preserve old and new values, revalidate conflicts, and notify affected users according to priority.	Must	Approved — Committed Release 1.0	Department Head,Registration Staff	P4	CAP-55,CAP-04,CAP-20; DEC-111,202	AC-045

4.5 Student and Instructor Records

Table 9 - Functional Requirements - Student and Instructor Records

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-46	Create Student Academic Record	The system shall allow authorized registration staff to create a student academic record after admission using the approved university identifier and required official academic information.	Must	Approved — Committed Release 1.0	Registration Staff	P5	CAP-12,CAP-13,CAP-61; DEC-25,56,73,78-79	AC-046
FR-47	Maintain Student Academic Profile	The system shall maintain each student's program, plan version, level, admission date, advisor, status, earned credits, semester GPA, and cumulative GPA.	Must	Approved — Committed Release 1.0	Registration Staff,Advisor,Student	P5	CAP-13,CAP-17; DEC-56,62	AC-047
FR-48	Manage Student Status	The system shall allow authorized staff to change a student among Active, Suspended, On Leave, Graduated, and Withdrawn while preserving the effective date, reason, authority, and history.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P5	CAP-12; DEC-55	AC-048
FR-49	Manage Student Leave	The system shall record an approved leave period and reason, block registration during leave, and support approved reactivation.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P5	CAP-47; DEC-103	AC-049
FR-50	Manage Student Suspension	The system shall record an approved suspension reason and duration, block registration during suspension, and support approved reactivation.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P5	CAP-48; DEC-104	AC-050
FR-51	Archive Academic Accounts and Records	The system shall preserve graduate, withdrawn, inactive-student, and inactive-instructor records while preventing unauthorized new academic operations.	Must	Approved — Committed Release 1.0	System Administrator,Registration Staff	P5	CAP-49; DEC-105	AC-051
FR-52	Create and Maintain Instructor Record	The system shall allow authorized staff to create and maintain an instructor record using the approved employee identifier and official assignment information.	Must	Approved — Committed Release 1.0	System Administrator,Department Head	P5	CAP-15,CAP-52; DEC-58,108,225	AC-052
FR-53	Manage Instructor Status	The system shall allow authorized staff to set an instructor to Active, On Leave, or Inactive and preserve status history.	Must	Approved — Committed Release 1.0	System Administrator,Department Head	P5	CAP-15; DEC-58	AC-053
FR-54	Prevent Assignment of Non-Active Instructors	The system shall prevent a new section assignment to an instructor whose status is not Active.	Must	Approved — Committed Release 1.0	Department Head	P5	CAP-15; DEC-58	AC-054

4.6 Registration and Advising

Table 10 - Functional Requirements - Registration and Advising

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-55	Start Registration Request	The system shall allow an Active student to select offered sections and submit a registration request during an open registration window.	Must	Approved — Committed Release 1.0	Student	P6	CAP-06; DEC-49	AC-055
FR-56	Validate Registration Preconditions	The system shall validate term state, student status, registration holds, section status, and other approved preconditions before accepting a registration request.	Must	Approved — Committed Release 1.0	Student,Advisor,Registration Staff	P6	CAP-06,CAP-07,CAP-53; DEC-50,109	AC-056
FR-57	Validate Prerequisites	The system shall reject registration when the student has not completed each prerequisite with at least the approved minimum grade and shall not permit a prerequisite override.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-07; DEC-34,142	AC-057
FR-58	Validate Co-requisites	The system shall validate each co-requisite relationship during registration and during withdrawal from a related course.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-07,CAP-32; DEC-88,200	AC-058
FR-59	Validate Registration Conflicts	The system shall reject a registration request that creates an unapproved schedule conflict for the student.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-05,CAP-07; DEC-50	AC-059
FR-60	Validate Section Capacity	The system shall reject direct registration in a full section and apply the approved waitlist behavior when the waitlist capability is available.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-07,CAP-08; DEC-50,51	AC-060
FR-61	Validate Credit Load	The system shall calculate the requested term load and enforce the approved regular, high-GPA, warning, probation, summer, and final-term limits.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-07,CAP-22,CAP-31; DEC-81,87,219	AC-061
FR-62	Require Advisor Approval	The system shall place a valid student registration request in Pending Advisor Approval and shall not confirm it until the assigned advisor approves it.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-06,CAP-65; DEC-38,49	AC-062

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-63	Review Registration Request	The system shall allow the assigned advisor to approve, reject, or return a pending registration request with a recorded reason and advice.	Must	Approved — Committed Release 1.0	Advisor	P6	CAP-06,CAP-65; DEC-49,121	AC-063
FR-64	Escalate Delayed Advising	The system shall identify a registration request that remains pending for two working days and make it available for approved Department Head escalation.	Must	Approved — Committed Release 1.0	Advisor,Department Head	P6	CAP-06; DEC-131	AC-064
FR-65	Confirm Registration	The system shall convert an approved and still-valid request into Registered enrollment atomically and revalidate seat, conflict, load, status, and hold conditions at confirmation time.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-06,CAP-07; DEC-49,50,187-190	AC-065
FR-66	Support Authorized Registration Intervention	The system shall allow authorized registration staff to perform or correct registration operations within their authority and shall record the actor, reason, and result.	Must	Approved — Committed Release 1.0	Registration Staff	P6	CAP-06,CAP-61,CAP-20; DEC-40,49,117	AC-066
FR-67	Process Registration Exceptions	The system shall route a permitted registration exception through advisor recommendation and Department Head or Registrar approval according to scope, while prohibiting prerequisite override.	Must	Approved — Committed Release 1.0	Advisor,Department Head,Registrar	P6	CAP-07; DEC-41,128,142	AC-067
FR-68	Manage Registration Holds	The system shall allow authorized staff to apply, review, and release Academic, Disciplinary, Missing Documents, and Administrative holds and shall block registration while an applicable hold is active.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P6	CAP-53; DEC-109	AC-068
FR-69	Add or Drop during Window	The system shall allow an eligible student or authorized staff member to add or drop a course during the configured add/drop window without recording W.	Must	Approved — Committed Release 1.0	Student,Registration Staff	P6	CAP-23; DEC-68,139	AC-069
FR-70	Withdraw after Add/Drop	The system shall record W when an authorized withdrawal is completed after add/drop and before the configured withdrawal deadline, without GPA or earned-credit effect.	Must	Approved — Committed Release 1.0	Student,Advisor,Registration Staff	P6	CAP-23; DEC-31,68	AC-070
FR-71	Process Late Registration Action	The system shall require the approved exception workflow for an add, drop, or withdrawal requested after the applicable deadline.	Must	Approved — Committed Release 1.0	Advisor,Department Head,Registrar,Registration Staff	P6	CAP-23,CAP-07; DEC-68,139	AC-071
FR-72	Enforce Repeat and Parallel-Attempt Rules	The system shall enforce the approved repeat-attempt limit, prevent parallel active attempts, and prevent enrollment in more than one section of the same course in one term.	Must	Approved — Committed Release 1.0	Student,Advisor	P6	CAP-34; DEC-90,197-199	AC-072
FR-73	Join Section Waitlist	The system should allow an otherwise eligible student to join a full section's waitlist in first-come-first-served order.	Should	Approved — Conditional after Must stability	Student	P6	CAP-08; DEC-51,141	AC-073
FR-74	Maintain Waitlist Eligibility and Order	The system should preserve the approved waitlist ordering, permit temporary load excess while waiting, and reassess eligibility when a seat becomes available.	Should	Approved — Conditional after Must stability	Student,Advisor,Registration Staff	P6	CAP-08; DEC-132,141,150	AC-074
FR-75	Automatically Promote Waitlisted Student	The system may atomically register the first eligible waitlisted student when a seat becomes available and stop automatic promotion after add/drop.	Could	Approved — Optional; not an acceptance blocker	Student,Registration Staff	P6	CAP-24; DEC-69,151,188,190	AC-075

4.7 Attendance

Table 11 - Functional Requirements - Attendance

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-76	Create Attendance Session	The system shall allow an assigned instructor to create or confirm each scheduled attendance session for the instructor's section.	Must	Approved — Committed Release 1.0	Instructor	P7	CAP-09,CAP-66; DEC-52,122	AC-076
FR-77	Record Attendance Status	The system shall allow an assigned instructor to record Present, Absent, Late, or Excused for each enrolled student in a non-cancelled session.	Must	Approved — Committed Release 1.0	Instructor	P7	CAP-09; DEC-52	AC-077
FR-78	Calculate Attendance Percentage	The system shall calculate attendance percentage using counted sessions, excluding Cancelled sessions and approved Excused absences from effective absence as defined by policy.	Must	Approved — Committed Release 1.0	Instructor,Student,Department Head	P7	CAP-09,CAP-16; DEC-52,59,153,193	AC-078
FR-79	Convert Late Records	The system shall treat every three Late records as one absence for attendance-rule evaluation.	Must	Approved — Committed Release 1.0	Instructor,Student	P7	CAP-16; DEC-152	AC-079

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-80	Apply Laboratory Attendance Rules	The system shall separately evaluate the approved unexcused-absence and makeup requirements for Practical/Laboratory and Mixed course laboratory meetings.	Must	Approved — Committed Release 1.0	Instructor,Department Head,Student	P7	CAP-16,CAP-30; DEC-60,76,143	AC-080
FR-81	Issue Attendance Warning	The system shall identify an approaching or breached attendance threshold and present the approved warning before an Attendance Barred decision.	Must	Approved — Committed Release 1.0	Instructor,Student,Department Head	P7	CAP-16; DEC-61	AC-081
FR-82	Record Attendance Bar	The system shall allow an authorized decision to set Attendance Barred after a threshold breach and shall preserve the reason and approval history.	Must	Approved — Committed Release 1.0	Department Head,Registrar	P7	CAP-16; DEC-61	AC-082
FR-83	Control Attendance Edits	The system shall allow an assigned instructor to edit attendance within seven days and shall require reason, Department Head approval, and audit for a later change.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P7	CAP-09,CAP-20; DEC-192	AC-083
FR-84	Cancel Attendance Session	The system shall retain a cancelled session as Cancelled, exclude it from attendance calculation, record a reason, and notify affected students when notifications are delivered.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P7	CAP-09,CAP-16; DEC-193	AC-084
FR-85	Process Makeup Laboratory	The system shall support instructor review, Department Head approval, and instructor recording of the approved makeup-laboratory outcome.	Must	Approved — Committed Release 1.0	Instructor,Department Head,Student	P7	CAP-30; DEC-84	AC-085

4.8 Assessment, Grading, GPA, and Standing

Table 12 - Functional Requirements - Assessment, Grading, GPA, and Standing

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-86	Define Assessment Scheme	The system shall allow an assigned instructor to define section assessments and weights subject to the approved default or an approved alternative totaling 100 percent.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P8	CAP-10; DEC-35	AC-086
FR-87	Enter Assessment Scores	The system shall allow an assigned instructor to enter and revise draft assessment scores with up to two decimal places before grade submission.	Must	Approved — Committed Release 1.0	Instructor	P8	CAP-10,CAP-66; DEC-53,154	AC-087
FR-88	Calculate Raw Course Score	The system shall calculate each completed student's raw course score from the approved assessment weights using the exact stored assessment values.	Must	Approved — Committed Release 1.0	Instructor,Student	P8	CAP-10; DEC-35,155	AC-088
FR-89	Select Grading Model	The system shall apply relative grading when the approved population and variation conditions are satisfied and otherwise apply the approved absolute fallback.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P8	CAP-10; DEC-29,207-208	AC-089
FR-90	Calculate Relative Letter Grade	The system shall calculate Z and T scores from the approved section population and map the T score to the approved letter-grade thresholds.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P8	CAP-10; DEC-29,222-223	AC-090
FR-91	Apply Absolute Letter Grade	The system shall map the raw score to the approved absolute letter-grade thresholds when the fallback conditions apply.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P8	CAP-10; DEC-207-209	AC-091
FR-92	Enforce Fail Floor	The system shall assign F to a raw score below 45 regardless of the relative result while retaining that numeric score in the approved section statistics.	Must	Approved — Committed Release 1.0	Instructor,Department Head	P8	CAP-10; DEC-209,223	AC-092
FR-93	Submit Section Grades	The system shall allow the assigned instructor to submit a complete section grade set for department review and shall prevent publication while required results are incomplete.	Must	Approved — Committed Release 1.0	Instructor	P8	CAP-10; DEC-53	AC-093
FR-94	Approve and Publish Section Grades	The system shall allow the Department Head to approve the complete section result set, publish all results together, and lock the published grades.	Must	Approved — Committed Release 1.0	Department Head	P8	CAP-10,CAP-64; DEC-53,130,191	AC-094
FR-95	Process Grade Correction	The system shall process a grade correction through	Must	Approved — Committed	Instructor,Department	P8	CAP-25,CAP-20; DEC-	AC-095

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
		instructor request, Department Head review, Registrar approval, audit history, and automatic recalculation of affected academic results.		Release 1.0	Head,Registrar		70,211	
FR-96	Process Grade Appeal	The system should allow a student to submit one appeal within seven days of publication and route it through Instructor review, Department Head recommendation, and Registrar final decision.	Should	Approved — Conditional after Must stability	Student,Instructor,Department Head,Registrar	P8	CAP-26; DEC-71,82-83,220-221	AC-096
FR-97	Manage W, I, and Pass/Fail Results	The system shall record and apply the approved GPA, credit, completion, and deadline effects of W, I, P, and F results.	Must	Approved — Committed Release 1.0	Instructor,Registrar,Registration Staff	P8	CAP-17,CAP-10; DEC-31-32,145	AC-097
FR-98	Apply Course Repeat to GPA	The system shall use the last completed attempt for GPA, retain every attempt, and award earned credit only once according to the approved repeat policy.	Must	Approved — Committed Release 1.0	Student,Advisor,Registration Staff	P8	CAP-17,CAP-34; DEC-30,90,197	AC-098
FR-99	Calculate Semester and Cumulative GPA	The system shall calculate credit-weighted semester and cumulative GPA on the 4.00 scale and truncate the result to two decimal places without rounding.	Must	Approved — Committed Release 1.0	Student,Registration Staff	P8	CAP-17,CAP-21; DEC-28-29,156	AC-099
FR-100	Determine Academic Standing	The system shall determine Good Standing, Academic Warning, or Academic Probation from the official truncated cumulative GPA after semester closure.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff,Advisor,Student	P8	CAP-21; DEC-66,218	AC-100
FR-101	Recommend Possible Dismissal	The system shall identify two consecutive Probation terms for human review and shall not dismiss a student automatically.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff	P8	CAP-21; DEC-217	AC-101

4.9 Academic Records, Transcripts, and Graduation

Table 13 - Functional Requirements - Academic Records, Transcripts, and Graduation

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-102	Maintain Complete Academic Record	The system shall preserve all attempts, result states, credits, GPA values, standing, eligibility, and approved changes as a fixed academic history.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff,Student	P9	CAP-17,CAP-49; DEC-62,86	AC-102
FR-103	Close Academic Term	The system shall finalize approved grades, GPA, standing, attendance, progress, and required report snapshots and lock closed-term records.	Must	Approved — Committed Release 1.0	Registrar,Registration Staff,Department Head	P9	CAP-37; DEC-93,215	AC-103
FR-104	Provide Unofficial Transcript	The system shall allow a student to view the student's own unofficial transcript containing the approved academic record information.	Must	Approved — Committed Release 1.0	Student	P9	CAP-18,CAP-67; DEC-63,123	AC-104
FR-105	Issue Official Transcript	The system shall allow authorized registration staff to issue an official transcript with a unique number, issue time, issuer, language, and Valid status.	Must	Approved — Committed Release 1.0	Registration Staff	P9	CAP-18,CAP-61; DEC-63,212-214	AC-105
FR-106	Cancel and Reissue Official Transcript	The system shall allow authorized staff to cancel an incorrect official transcript without deletion and issue a corrected replacement while preserving both records.	Must	Approved — Committed Release 1.0	Registration Staff,Registrar	P9	CAP-18,CAP-20; DEC-213-214	AC-106
FR-107	Provide Degree Progress	The system should show required and earned credits, completed and remaining courses, completion percentage, and graduation readiness under the applicable study plan.	Should	Approved — Conditional after Must stability	Student,Advisor,Program Coordinator	P9	CAP-35; DEC-91	AC-107
FR-108	Evaluate Graduation Eligibility	The system should evaluate the approved program-completion, GPA, result-state, and minimum-grade conditions and assign the appropriate graduation-eligibility status.	Should	Approved — Conditional after Must stability	Registration Staff,Program Coordinator,Student	P9	CAP-35; DEC-33,203,205	AC-108
FR-109	Process Graduation	The system should support staff review,	Should	Approved — Conditional	Registration Staff,Program	P9	CAP-35; DEC-204,206	AC-109

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
	Approval	Program Coordinator confirmation, Registrar approval, and creation of a fixed graduation record.		after Must stability	Coordinator,Registrar			

4.10 Reports, Notifications, Audit, and Search

Table 14 - Functional Requirements - Reports, Notifications, Audit, and Search

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-110	Provide Academic Reports	The system shall provide the approved performance, registration, enrollment, attendance-risk, grade-distribution, GPA/standing, graduation, teaching-load, and program/department reports within the user's scope.	Must	Approved — Committed Release 1.0	Authorized users	P10	CAP-19; DEC-64,216	AC-110
FR-111	Apply Report Scope	The system shall automatically restrict each report to the approved self, section, department, program, or institution aggregate scope of the requesting user.	Must	Approved — Committed Release 1.0	Authorized users	P10	CAP-19,CAP-62,CAP-65,CAP-66,CAP-67; DEC-216	AC-111
FR-112	View and Export Reports	The system should allow authorized users to view reports in the system and export permitted outputs as PDF, Excel, or CSV with the approved metadata and data minimization.	Should	Approved — Conditional after Must stability	Authorized users	P10	CAP-29; DEC-75,185-186	AC-112
FR-113	Provide Role Dashboards	The system should provide each major role with a dashboard summarizing permitted tasks, alerts, and academic indicators.	Should	Approved — Conditional after Must stability	All direct users	P10	CAP-38; DEC-94	AC-113
FR-114	Provide Executive Read-Only Analytics	The system may provide University Management with read-only aggregated dashboards and reports without record-modification actions.	Could	Approved — Optional; not an acceptance blocker	University Management	P10	CAP-68; DEC-124	AC-114
FR-115	Deliver In-App Notifications	The system should deliver in-app notifications with recipient scope, severity, time, read/unread state, and a permitted related-action reference.	Should	Approved — Conditional after Must stability	All operational users	P10	CAP-11; DEC-24,54,85	AC-115
FR-116	Publish Targeted Announcements	The system should allow authorized academic staff to publish announcements to an approved university, college, department, program, section, or role audience.	Should	Approved — Conditional after Must stability	Registrar,Registration Staff,Department Head,Program Coordinator,Instructor	P10	CAP-58; DEC-114	AC-116
FR-117	Manage Urgent Notification Accountability	The system should display urgent notifications prominently, record the read time, and allow the authorized sender to view the recipient read count.	Should	Approved — Conditional after Must stability	Authorized sender,Recipient	P10	CAP-11; DEC-194	AC-117
FR-118	Record Audit Events	The system shall create an immutable audit event for each approved sensitive operation containing actor, action, time, affected record, old and new values, reason, and result.	Must	Approved — Committed Release 1.0	System,Authorized reviewer	P10	CAP-20; DEC-65	AC-118
FR-119	Review Audit History	The system shall allow authorized reviewers to search and filter audit history without modifying or deleting it.	Must	Approved — Committed Release 1.0	Registrar,System Administrator,Department Head	P10	CAP-20; DEC-65,166	AC-119
FR-120	Record Login Activity	The system shall record login, logout, result, account, and lock reason and restrict login-log review to the System Administrator.	Must	Approved — Committed Release 1.0	System Administrator	P10	CAP-20,CAP-27; DEC-164	AC-120
FR-121	Search and Filter Authorized Data	The system shall provide search and filtering over relevant records while enforcing the requester's authorization and data scope.	Must	Approved — Committed Release 1.0	All direct users	P10	CAP-39; DEC-95	AC-121

4.11 Supporting Operations and Experience

Table 15 - Functional Requirements - Supporting Operations and Experience

ID	Title	Atomic Requirement Statement	Priority	Status	Primary Actor(s)	Profile	Source Trace	Acceptance
FR-122	Manage User Profile	The system shall allow a user to update permitted contact, address, password, and profile information while reserving official identity and academic data for authorized staff.	Must	Approved — Committed Release 1.0	All users,Registration Staff	P11	CAP-43; DEC-99	AC-122
FR-123	Manage Profile Image	The system may allow an optional profile image and permit authorized moderation or removal.	Could	Approved — Optional; not an acceptance blocker	All users,Registration Staff	P11	CAP-44; DEC-100	AC-123
FR-124	Bulk Import Students and Courses	The system should allow authorized staff to import student and course data from CSV or spreadsheet files.	Should	Approved — Conditional after Must stability	Registration Staff,System Administrator,Program Coordinator	P11	CAP-40; DEC-96,116	AC-124
FR-125	Validate and Partially Apply Import	The system may preview an import, detect duplicates and row errors, apply only valid rows when authorized, and produce a final result report.	Could	Approved — Optional; not an acceptance blocker	Registration Staff,System Administrator,Program Coordinator	P11	CAP-60; DEC-116,171	AC-125
FR-126	Validate Controlled Identifiers	The system shall enforce uniqueness of University ID, Employee ID, Course Code, Semester Code, and section identifier within the approved scope.	Must	Approved — Committed Release 1.0	Authorized staff	P11	CAP-27,CAP-52,CAP-14,CAP-03,CAP-04; DEC-172-173,225	AC-126
FR-127	Manage Supporting Attachments	The system should allow approved PDF, JPG, JPEG, or PNG attachments of at most 5 MB for appeals, excused absences, transfers, and exceptions.	Should	Approved — Conditional after Must stability	Student,Instructor,Advisor, Authorized reviewers	P11	CAP-26,CAP-30,CAP-45,CAP-07; DEC-174-175	AC-127
FR-128	Protect and Retain Attachments	The system should restrict an attachment to its owner and authorized reviewers and retain it with the associated record.	Should	Approved — Conditional after Must stability	Authorized users	P11	CAP-20; DEC-183-184	AC-128
FR-129	Create Backup	The system may allow the System Administrator to create daily automatic and authorized manual backups with configurable retention defaulting to 30 days.	Could	Approved — Optional; not an acceptance blocker	System Administrator	P11	CAP-50; DEC-106,167-168	AC-129
FR-130	Restore Backup	The system may allow only the System Administrator to restore an approved backup after confirmation and shall record the result and verify post-restore integrity.	Could	Approved — Optional; not an acceptance blocker	System Administrator	P11	CAP-50; DEC-169-170	AC-130
FR-131	Reject Stale Concurrent Update	The system shall reject a save based on stale data and require the user to refresh and retry rather than silently overwriting a newer change.	Must	Approved — Committed Release 1.0	All modifying users	P11	CAP-20; DEC-187	AC-131
FR-132	Apply Atomic Sensitive Operations	The system shall roll back the complete sensitive operation when any required step fails, leaving no partial academic change.	Must	Approved — Committed Release 1.0	System	P11	CAP-06,CAP-10,CAP-20; DEC-189	AC-132
FR-133	Control Maintenance State	The system shall announce a maintenance window, prevent new operations during the window, and preserve already saved data.	Must	Approved — Committed Release 1.0	System Administrator,All users	P11	CAP-51; DEC-182	AC-133
FR-134	Provide Clear Error Feedback	The system shall present clear, localized, non-technical error messages to users and keep technical details in authorized internal records.	Must	Approved — Committed Release 1.0	All users	P11	CAP-39,CAP-41; DEC-196	AC-134
FR-135	Support Arabic and English	The system should support Arabic and English user-facing content.	Should	Approved — Conditional after Must stability	All users	P11	CAP-41; DEC-97	AC-135
FR-136	Support RTL and LTR Direction	The system should use right-to-left direction for Arabic and left-to-right direction for English.	Should	Approved — Conditional after Must stability	All users	P11	CAP-42; DEC-98	AC-136
FR-137	Support Responsive Web Access	The system shall provide connected web access on supported desktop, tablet, and mobile browser sizes without requiring a native application.	Must	Approved — Committed Release 1.0	All users	P11	OCR-14; DEC-19,177; ASM-09	AC-137
FR-138	Support Approved Browsers	The system shall support current mainstream versions of Chrome, Edge, Firefox, and Safari, with release verification prioritizing Chrome and Edge.	Must	Approved — Committed Release 1.0	All users	P11	OCR-14; DEC-178	AC-138
FR-139	Maintain Academic Advising Notes	The system should allow an assigned advisor to maintain private advising notes and, when explicitly chosen, publish a separate student-facing summary.	Should	Approved — Conditional after Must stability	Advisor,Student	P11	CAP-57; DEC-113	AC-139
FR-140	Delegate Approval Authority	The system may allow an authorized approver to delegate specified approval operations for a limited period and scope while preserving accountability.	Could	Approved — Optional; not an acceptance blocker	Registrar,Department Head,Registration Staff	P11	CAP-59; DEC-115,133	AC-140

5. Business Rules

Each BR is normative. Unless a rule explicitly states otherwise, its trigger is an attempted operation within its stated scope; its inputs are the referenced academic facts; its outcome is the stated result; exceptions require an approved path and audit. Testability is by boundary, positive, negative, and authorization cases against the linked FR/UC.

Table 16 - Business Rules Catalogue

ID	Title	Rule Statement	Priority	Approved Source	Related FR / UC
BR-01	Single-University Model	SUMS serves one generic university. The institution display name is configurable and does not imply affiliation with a real institution.	Must	DEC-17,18,227,233	FR-15-FR-20; UC-03
BR-02	Academic Hierarchy	The academic hierarchy is University -> College -> Department -> Academic Program.	Must	DEC-45	FR-15-FR-20; UC-03
BR-03	Academic Authority Boundaries	Registrar, Department Head, Program Coordinator, Advisor, Registration Staff, System Administrator, Instructor, Student, and Management shall act only within the approved authority boundaries.	Must	DEC-117-130,134	FR-15-FR-20; UC-03
BR-04	Academic Year and Codes	Terms are First Semester, Second Semester, and Summer; term codes use YYYY-1, YYYY-2, and YYYY-S; dates use the Gregorian calendar and Palestine time.	Must	DEC-137,138,147,148	FR-32-FR-45; UC-06-UC-08
BR-05	Configurable Term Windows	Registration, add/drop, withdrawal, appeal, and grade deadlines are configured per term and govern related actions.	Must	DEC-139,144	FR-32-FR-45; UC-06-UC-08
BR-06	Section Identifier	A section identifier follows COURSECODE-SEMESTER-SECTION and is unique within its term context.	Must	DEC-173	FR-32-FR-45; UC-06-UC-08
BR-07	Active Instructor Rule	Only an instructor with Active status may receive a new section assignment.	Must	DEC-58	FR-32-FR-45; UC-06-UC-08
BR-08	Section Teaching Structure	A section may have one Primary Instructor, one or more Co-Instructors, and multiple weekly meetings.	Must	ASM-07,ASM-08	FR-32-FR-45; UC-06-UC-08
BR-09	Schedule Conflict Rule	Student, instructor, room, and laboratory meeting conflicts are prohibited unless an approved exception exists; automatic timetable generation is excluded.	Must	DEC-48,112	FR-32-FR-45; UC-06-UC-08
BR-10	Capacity Rule	Section capacity shall not exceed the assigned room or laboratory capacity without an approved exception.	Must	DEC-140	FR-32-FR-45; UC-06-UC-08
BR-11	Section Cancellation Rule	A cancelled section stops new registration, preserves reason/date/history, and after term start requires transfer or cancellation without W or GPA effect.	Must	DEC-110,201	FR-32-FR-45; UC-06-UC-08
BR-12	Section Change Rule	Instructor reassignment or time/place change requires Department Head approval, conflict revalidation, notification according to priority, and audit history.	Must	DEC-111,202	FR-32-FR-45; UC-06-UC-08
BR-13	Student Identifier	Student University ID uses GYYYYNNNN, where G is 1 for male or 2 for female, YYYY is admission year, and the five-digit sequence restarts annually.	Must	DEC-73,78,79	FR-46-FR-54; UC-09,UC-10
BR-14	Employee Identifier	Employee ID uses EYYYYNNNN and is unique.	Must	DEC-225	FR-46-FR-54; UC-09,UC-10
BR-15	Student Status Rule	Student statuses are Active, Suspended, On Leave, Graduated, and Withdrawn; only Active students register normally.	Must	DEC-55	FR-46-FR-54; UC-09,UC-10
BR-16	Normal Registration Flow	Normal registration follows Student selection -> system validation -> mandatory advisor approval -> confirmation; authorized staff may intervene.	Must	DEC-38,40,49	FR-55-FR-75,FR-98; UC-11-UC-17
BR-17	Prerequisite Rule	A prerequisite is satisfied only with at least C unless a higher approved minimum applies; prerequisite override is prohibited.	Must	DEC-34,142	FR-55-FR-75,FR-98; UC-11-UC-17
BR-18	Co-requisite Rule	Co-requisites are validated at registration and withdrawal; withdrawal effects follow the relationship-specific rule and advisor review.	Must	DEC-88,200	FR-55-FR-75,FR-98; UC-11-UC-17
BR-19	Enrollment States	Enrollment states are Pending Advisor Approval, Registered, Waitlisted, Dropped, Withdrawn, and Cancelled.	Must	DEC-149	FR-55-FR-75,FR-98; UC-11-UC-17
BR-20	Regular Credit Load	Regular minimum is 18 credits; GPA below 3.00 permits at most 18; Good Standing with	Must	DEC-81	FR-55-FR-75,FR-98; UC-11-

ID	Title	Rule Statement	Priority	Approved Source	Related FR / UC
		GPA at least 3.00 permits 18-21; Warning/Probation permits at most 12; final-term lower load requires approved conditions.			UC-17
BR-21	Summer Credit Load	Summer maximum is 9 credits; Probation maximum is 6.	Must	DEC-87,219	FR-55-FR-75,FR-98; UC-11-UC-17
BR-22	Registration Holds	Supported holds are Academic, Disciplinary, Missing Documents, and Administrative; Financial Hold is excluded.	Must	DEC-109	FR-55-FR-75,FR-98; UC-11-UC-17
BR-23	Advisor Delay Escalation	A registration request pending for two working days may be escalated to the Department Head.	Must	DEC-131	FR-55-FR-75,FR-98; UC-11-UC-17
BR-24	Waitlist Order	Waitlists are first-come-first-served using precise completion time and University ID as tie-breaker; the last seat goes to the first completed valid registration.	Should	DEC-141,188,190	FR-55-FR-75,FR-98; UC-11-UC-17
BR-25	Waitlist Eligibility	A student may join despite potential load excess; promotion occurs only when eligible, needs no new advisor approval if already approved, and automatic promotion ends after add/drop.	Should/Could	DEC-132,150,151	FR-55-FR-75,FR-98; UC-11-UC-17
BR-26	Add Drop Withdrawal	Add/drop during its window produces no W; later authorized withdrawal before deadline records W; late action requires approved exception.	Must	DEC-68,139	FR-55-FR-75,FR-98; UC-11-UC-17
BR-27	Repeat Limit	A course may be attempted two additional times; further attempt requires Department Head approval; parallel active attempts and multiple sections of the same course in one term are prohibited.	Must	DEC-197-199	FR-55-FR-75,FR-98; UC-11-UC-17
BR-28	Attendance Statuses	Attendance statuses are Present, Absent, Late, and Excused; Cancelled sessions are retained and excluded from calculation.	Must	DEC-52,193	FR-76-FR-85; UC-18,UC-19
BR-29	General Attendance Threshold	The minimum general attendance threshold is 75 percent.	Must	DEC-59	FR-76-FR-85; UC-18,UC-19
BR-30	Late Conversion	Every three Late records equal one absence for rule evaluation.	Must	DEC-152	FR-76-FR-85; UC-18,UC-19
BR-31	Excused Absence	An excused absence requires evidence and approval as applicable, remains visible, and is excluded from effective absence.	Must	DEC-153	FR-76-FR-85; UC-18,UC-19
BR-32	Laboratory Attendance	At most one unexcused laboratory absence is permitted; an approved excused absence may require makeup.	Must	DEC-60,76	FR-76-FR-85; UC-18,UC-19
BR-33	Mixed-Course Attendance	Theory percentage and laboratory absence rules are evaluated independently for Mixed courses.	Must	DEC-143	FR-76-FR-85; UC-18,UC-19
BR-34	Attendance Warning and Bar	A warning precedes Attendance Barred; the bar requires threshold breach and authorized approval.	Must	DEC-61	FR-76-FR-85; UC-18,UC-19
BR-35	Attendance Edit Window	Instructor edits are allowed for seven days; later changes require reason, Department Head approval, and audit.	Must	DEC-192	FR-76-FR-85; UC-18,UC-19
BR-36	Assessment Weights	Default weights are coursework 20 percent, midterm 30 percent, and final 50 percent; approved alternatives must total 100 percent.	Must	DEC-35	FR-86-FR-103; UC-20-UC-25
BR-37	Score Precision	Assessment and raw scores support two decimals; calculations use the exact stored raw values.	Must	DEC-154,155	FR-86-FR-103; UC-20-UC-25
BR-38	Relative Grading Formula	For score X , mean μ , and standard deviation σ , $Z = \frac{X-\mu}{\sigma}$ and $T = 50 + 10Z$. Relative grading applies only with at least 10 completed numeric grades and σ not equal to zero.	Must	DEC-29,207,208,222	FR-86-FR-103; UC-20-UC-25
BR-39	Relative Grade Thresholds	$T \geq 65$ A; $60 < 65$ A-; $55 < 60$ B+; $50 < 55$ B; $45 < 50$ B-; $40 < 45$ C+; $35 < 40$ C; $30 < 35$ C-; $25 < 30$ D+; $20 < 25$ D; $T < 20$ F.	Must	DEC-29; Approved Scope Baseline v1.0 — Appendix A	FR-86-FR-103; UC-20-UC-25
BR-40	Absolute Fallback Thresholds	90-100 A; 85-89.99 A-; 80-84.99 B+; 75-79.99 B; 70-74.99 B-; 65-69.99 C+; 60-64.99 C; 55-59.99 C-; 50-54.99 D+; 45-49.99 D; below 45 F.	Must	DEC-207-209; Approved Scope Baseline v1.0 — Appendix A	FR-86-FR-103; UC-20-UC-25
BR-41	Fail Floor and Population	Raw score below 45 is F regardless of T; completed numeric grades including low scores form the population, while W, I, and incomplete cases are excluded.	Must	DEC-209,222,223	FR-86-FR-103; UC-20-UC-25

ID	Title	Rule Statement	Priority	Approved Source	Related FR / UC
BR-42	Grade Publication	The Department Head approves; all section results publish together and then lock.	Must	DEC-53,130,191	FR-86-FR-103; UC-20-UC-25
BR-43	Letter Grade Points	A=4.00, A-=3.70, B+=3.30, B=3.00, B-=2.70, C+=2.30, C=2.00, C-=1.70, D+=1.30, D=1.00, F=0.00.	Must	DEC-29; Approved Scope Baseline v1.0 — Appendix A	FR-86-FR-103; UC-20-UC-25
BR-44	GPA Formula and Precision	GPA is Sum(grade point x credits)/Sum(attempted GPA credits); official GPA is truncated to two decimals without rounding and the truncated value drives rules.	Must	DEC-28,29,156	FR-86-FR-103; UC-20-UC-25
BR-45	Repeat GPA Effect	The last completed attempt counts in GPA; every attempt remains; earned credit is awarded once.	Must	DEC-30,90,197	FR-86-FR-103; UC-20-UC-25
BR-46	W I Pass Fail Effects	W has no GPA or credit effect; I is temporary then becomes final grade or F; P at C-equivalent grants credit without GPA; F is 0.00 and no credit.	Must	DEC-31,32,145	FR-86-FR-103; UC-20-UC-25
BR-47	Grade Correction	Correction follows Instructor request -> Department Head review -> Registrar approval -> audit -> recalculation.	Must	DEC-70,211	FR-86-FR-103; UC-20-UC-25
BR-48	Grade Appeal	Appeal is filed within seven days; route is Student -> Instructor -> Department Head -> Registrar final decision; outcome may increase, decrease, or preserve grade; no duplicate final appeal unless reopened.	Should	DEC-71,82,83,220,221	FR-86-FR-103; UC-20-UC-25
BR-49	Academic Standing	Good Standing is GPA>=2.00; Warning is 1.50-1.99; Probation is below 1.50; standing recalculates after closure.	Must	DEC-66,218	FR-86-FR-103; UC-20-UC-25
BR-50	Dismissal Recommendation	Two consecutive Probation terms may trigger a human dismissal recommendation; dismissal is never automatic.	Must	DEC-217	FR-86-FR-103; UC-20-UC-25
BR-51	Graduation Eligibility	Eligibility requires program and required-course completion, cumulative GPA>=2.00, no required F, no I, and applicable minimum-C rules.	Should	DEC-33,205	FR-102-FR-112,FR-118-FR-120; UC-25-UC-32
BR-52	Graduation Workflow and Record	Statuses are Not Eligible, Potentially Eligible, Eligible, Approved for Graduation, and Graduated; workflow is system check -> staff review -> coordinator confirmation -> Registrar approval; final term/date/GPA/program/credits become fixed.	Should	DEC-203-206	FR-102-FR-112,FR-118-FR-120; UC-25-UC-32
BR-53	Transcript Control	Official transcripts carry unique number, issue time, issuer, language, and Valid/Cancelled status; incorrect copies are cancelled and reissued, not deleted.	Must	DEC-212-214	FR-102-FR-112,FR-118-FR-120; UC-25-UC-32
BR-54	Report Scope and Metadata	Reports are automatically scoped; official PDF outputs include institution, title, issue time, issuer, and Official/Unofficial status.	Must/Should	DEC-185,216	FR-102-FR-112,FR-118-FR-120; UC-25-UC-32
BR-55	Audit Content	Sensitive audit records include actor, action, time, affected record, old/new values, reason, and result and are immutable to ordinary users.	Must	DEC-65	FR-102-FR-112,FR-118-FR-120; UC-25-UC-32
BR-56	No Hard Delete and Retention	Approved academic records and transcripts are permanent; audit records remain for system life; cancellation, disabling, or correction replaces hard deletion.	Must	DEC-86,146	FR-102-FR-112,FR-118-FR-120; UC-25-UC-32
BR-57	Notification Retention	In-app notifications have read state, time, link, severity, targeted delivery, and default retention of two years configurable.	Should	DEC-54,85,165	FR-115-FR-140; UC-34-UC-43
BR-58	Bulk Import Validation	Student/course import provides preview, duplicate checks, row errors, optional valid-row partial import, and final report.	Should/Could	DEC-96,116,171	FR-115-FR-140; UC-34-UC-43
BR-59	Attachment Rules	Approved attachments are PDF/JPG/JPEG/PNG up to 5 MB; access is owner/reviewer/authorized user and retention follows the parent record.	Should	DEC-174,175,183,184	FR-115-FR-140; UC-34-UC-43
BR-60	Backup and Restore	Daily automatic and manual pre-sensitive-operation backups have configurable 30-day default retention; only System Administrator restores with confirmation, audit, result, and integrity check.	Could	DEC-167-170	FR-115-FR-140; UC-34-UC-43
BR-61	Concurrent Edit and Atomicity	Stale saves are rejected; sensitive operations roll back completely on failure.	Must	DEC-187,189	FR-115-FR-140; UC-34-UC-43
BR-62	Password and Session Policy	Password is at least 8 characters with upper, lower, number, and special; lock after 3 failures for 15 minutes; session timeout after 30 minutes inactivity.	Must	DEC-157-160	FR-115-FR-140; UC-34-UC-43
BR-63	Role and Scope Policy	Users cannot self-assign roles; role changes require authorization, reason, and audit; advisors, instructors, students, management, and technical administrators remain within	Must	DEC-118,121-124,162	FR-115-FR-140; UC-34-UC-43

ID	Title	Rule Statement	Priority	Approved Source	Related FR / UC
		approved scopes.			
BR-64	Profile Data Policy	Users may edit contact, address, image, and password; official identity, program, and academic data are staff-controlled.	Must/Could	DEC-99,100	FR-115-FR-140; UC-34-UC-43
BR-65	Language and Direction	Arabic uses RTL and English uses LTR; academic symbols remain unchanged in transcript language variants.	Should	DEC-97,98,212	FR-115-FR-140; UC-34-UC-43
BR-66	Maintenance and Errors	During announced maintenance no new operations begin and saved data is preserved; user errors are localized, clear, non-technical, and technical details are internally logged.	Must	DEC-182,196	FR-115-FR-140; UC-34-UC-43

6. Non-Functional Requirements

Table 17 - Non-Functional Requirements Catalogue

ID	Category	Requirement Statement	Measurement / Acceptance Threshold	Priority / Status	Source Trace
NFR-01	Usability	The system shall enable authorized users to complete each core task through clear terminology, visible status, consistent validation, and actionable feedback.	Task-based review: all Must use cases completed by a trained project reviewer without undocumented workaround.	Must	OCR-14; DEC-94,95,196
NFR-02	Responsive Access	The system shall remain usable on supported desktop, tablet, and mobile browser viewport classes.	UI verification at representative viewport classes; no loss of Must task completion.	Must	DEC-177
NFR-03	Browser Compatibility	The system shall support current mainstream Chrome, Edge, Firefox, and Safari versions, with release verification prioritizing Chrome and Edge.	Execute Must use-case smoke suite on all four; full regression on Chrome/Edge.	Must	DEC-178
NFR-04	Localization	The system should provide Arabic and English user-facing content.	Language-switch review of delivered Should scope.	Should	CAP-41; DEC-97
NFR-05	Bidirectional Layout	The system should render Arabic in RTL and English in LTR without changing academic symbols.	Visual and task verification in both directions.	Should	CAP-42; DEC-98,212
NFR-06	General Accessibility	The system shall provide clear labels, keyboard-accessible interactions where applicable, readable text and form feedback, sufficient visual clarity, accessible validation and error messages, and consistent navigation.	General accessibility principles are included in this SRS. The detailed Accessibility Checklist, formal accessibility conformance criteria, visual accessibility standards, screen-level accessibility specifications, and detailed usability validation procedures shall be defined and approved later in the UI/UX Design phase. The Must classification applies to the general accessibility and usability essentials stated in this requirement.	Must	OCR-14; DEC-235
NFR-07	Data Integrity	The system shall preserve approved academic relationships, uniqueness rules, calculation consistency, and history across successful operations.	Integrity tests for identifiers, relationships, calculations, and preserved history.	Must	DEC-86,172,187,189
NFR-08	Atomicity	The system shall leave no partial academic change when a sensitive multi-step operation fails.	Failure-injection tests show either complete success or complete rollback.	Must	DEC-189; RSK-06
NFR-09	Concurrent-Update Safety	The system shall detect stale updates and prevent silent overwriting of newer data.	Two-user concurrency test rejects stale save and requests refresh/retry.	Must	DEC-187
NFR-10	Auditability	The system shall make every approved sensitive operation traceable to actor, time, record, reason, values, and outcome.	Audit sampling confirms required fields for each sensitive workflow.	Must	CAP-20; DEC-65
NFR-11	Recoverability	The system may support restoration from an authorized backup with post-restore integrity verification.	Could; restore drill succeeds with consistent fictional dataset and recorded result.	Could	CAP-50; DEC-167-170
NFR-12	Availability During Maintenance	The system shall clearly communicate announced maintenance and prevent new operations while preserving	Maintenance-state test verifies message, operation block, and saved-data preservation.	Must	DEC-182

ID	Category	Requirement Statement	Measurement / Acceptance Threshold	Priority / Status	Source Trace
		saved data.			
NFR-13	Normal Operation Response	The system is proposed to complete normal interactive operations within 3 seconds under defined normal conditions.	Proposed — Pending Architecture Validation. This value is not a mandatory Release 1.0 acceptance blocker and may become final only after technology-stack selection, architecture design, hosting/deployment assumptions, database design, test-data volume definition, performance-test planning, and technical feasibility validation.	Proposed	DEC-179,232; Master Project Control Resolution
NFR-14	Report Generation Response	The system is proposed to generate normal reports within 10 seconds under defined normal conditions.	Proposed — Pending Architecture Validation. This value is not a mandatory Release 1.0 acceptance blocker and may become final only after technology-stack selection, architecture design, hosting/deployment assumptions, database design, test-data volume definition, performance-test planning, and technical feasibility validation.	Proposed	DEC-180,232; Master Project Control Resolution
NFR-15	Concurrent User Capacity	The system is proposed to support 500 concurrent users under an approved representative workload.	Proposed — Pending Architecture Validation. This value is not a mandatory Release 1.0 acceptance blocker and may become final only after technology-stack selection, architecture design, hosting/deployment assumptions, database design, test-data volume definition, performance-test planning, and technical feasibility validation.	Proposed	DEC-181,232; Master Project Control Resolution
NFR-16	Scalability	The approved design shall not hard-code the number of colleges, departments, programs, courses, users, terms, or sections.	Configuration/data-volume review demonstrates variable record counts.	Must	ASM-05; DEC-17
NFR-17	Reliability of Academic Calculations	The system shall produce deterministic registration, attendance, grade, GPA, standing, and eligibility results from the same approved inputs.	Reference-calculation and boundary suites produce repeatable results.	Must	RSK-05,06; DEC-28-35,50,59-66
NFR-18	Error Handling	The system shall reject invalid operations without corrupting data and shall present a clear non-technical explanation while recording technical detail internally.	Negative tests verify rejection, unchanged data, user message, and internal record.	Must	DEC-196
NFR-19	Security and Privacy	The system shall restrict sensitive academic and identity data to approved roles and scopes and minimize such data in reports and exports.	Positive/negative authorization and report-minimization tests.	Must	DEC-118-126,183,186,195
NFR-20	Record Preservation	The system shall preserve permanent academic/transcript records and system-life audit records according to the approved retention rules.	Retention and no-hard-delete tests across lifecycle changes.	Must	DEC-86,146
NFR-21	Maintainability	The delivered system and documentation shall use consistent terminology, controlled identifiers, and traceability sufficient for team review and later change impact analysis.	Artifact consistency and traceability audit passes.	Must	OCR-08,16,20; RSK-12,15
NFR-22	Testability	Each Must requirement shall have linked acceptance criteria and a planned test placeholder before implementation approval.	Traceability audit shows complete FR-to-AC and Must-to-test mapping.	Must	OCR-08; Course Knowledge Baseline testing guidance
NFR-23	Documentation	The project shall maintain professional requirements, design, installation, user, test, and release documentation consistent with the implemented release.	Document checklist and cross-artifact consistency review pass.	Must	OCR-16,23
NFR-24	Connected Operation	The system shall operate as a connected web application and is not required to support offline operation.	Offline-mode inspection confirms no acceptance dependency on offline features.	Must	ASM-09; DEC-19
NFR-25	Originality and Explainability	Project artifacts shall be original, and each team member shall be able to explain the assigned and integrated work.	Originality review plus team walkthrough.	Must	OCR-18-20

Performance control: NFR-13, NFR-14, and NFR-15 are **Proposed - Pending Architecture Validation**. They are not final acceptance thresholds. Approval requires an architecture, deployment-like test environment, fictional dataset size, representative workload mix, percentile/measurement definition, and repeatable performance/load-test evidence.

7. Security Requirements

Table 18 - Security Requirements Catalogue

ID	Requirement	Requirement Statement	Threat/Risk	Protected Asset	Role	Priority	Verification	Source / Relations
SEC-01	Account Authentication	The system shall require successful authentication before granting access to non-public functions.	Unauthorized access	User accounts	All users	Must	Attempt protected operation without session; access denied.	CAP-27,CAP-51; DEC-107
SEC-02	Identity-Specific Login	The system shall use University ID for student login and Employee ID for employee login.	Impersonation/ambiguity	Identity	Students/employees	Must	Wrong identity type is rejected.	CAP-27,CAP-52; DEC-77,108
SEC-03	Password Complexity	The system shall enforce the approved password length and character-category rules on password creation and change.	Weak credential	Credentials	All users	Must	Boundary password tests.	DEC-157
SEC-04	Temporary Credential Change	The system shall require replacement of an initial or reset temporary password before normal access continues.	Persistent shared credential	Credentials	New/reset users	Must	Temporary-password login cannot reach normal functions before change.	DEC-80,161
SEC-05	Secure Password Reset	The system shall permit password reset only after authorized identity verification and shall audit the reset.	Account takeover	Credentials	System Administrator	Must	Unauthorized reset denied; authorized reset audited.	DEC-161
SEC-06	Failed-Login Lockout	The system shall lock an account after three consecutive failures for 15 minutes and support authorized manual unlock.	Brute force	Account	All users/System Administrator	Must	Three-failure and timed-unlock tests.	DEC-158,159
SEC-07	Session Timeout	The system shall terminate a session after 30 minutes of inactivity.	Unattended session misuse	Session	All users	Must	Idle-session test.	DEC-160
SEC-08	Account-State Enforcement	The system shall deny access according to Locked, Disabled, withdrawn, inactive, or archived account policy.	Unauthorized continued access	Account/records	All users	Must	State-transition access matrix.	DEC-72,105,126,163
SEC-09	Role-Based Authorization	The system shall authorize actions by active role and deny every action not granted to that role.	Privilege escalation	Academic functions	All users	Must	Positive/negative role matrix.	CAP-28; DEC-74,162
SEC-10	Least Privilege and Scope	The system shall limit users to the minimum approved self, assigned-student, assigned-section, unit, or aggregate scope.	Excessive data access	Academic data	All users	Must	Cross-scope access attempts denied.	DEC-121-124
SEC-11	Separation of Duties	The system shall prevent technical administrators from performing academic decisions and prevent operational staff from directly changing grades.	Fraud/unauthorized modification	Grades/GPA/decisions	System Administrator/Registration Staff	Must	Attempt prohibited academic actions; deny.	CAP-62; DEC-117,118
SEC-12	Controlled Role Administration	The system shall prevent self-assignment of roles and require authorized actor, reason, and audit for each role change.	Privilege escalation	Roles	System Administrator	Must	Self-assignment denied; authorized change logged.	DEC-162
SEC-13	Input Validation	The system shall validate format, range, required fields, identifiers, file type, file size, and business rules before accepting data.	Malformed/injected/invalid data	All records	All modifying users	Must	Invalid and boundary inputs rejected without change.	DEC-172-175; requirements rules
SEC-14	Sensitive-Change Accountability	The system shall audit grade, attendance, role, registration exception, transcript, account, and administrative changes with old/new values and reason.	Repudiation/tampering	Sensitive records	Authorized modifiers	Must	Audit completeness sampling.	CAP-20; DEC-65,70,192

ID	Requirement	Requirement Statement	Threat/Risk	Protected Asset	Role	Priority	Verification	Source / Relations
SEC-15	Immutable Audit Records	The system shall prevent ordinary users from modifying or deleting audit events.	Log tampering	Audit history	All ordinary users	Must	Modify/delete attempts denied.	DEC-65,166
SEC-16	Login Logging	The system shall record login, logout, success/failure, account, and lock reason and restrict review to the System Administrator.	Undetected access abuse	Authentication log	System Administrator	Must	Log-event and access tests.	DEC-164
SEC-17	No Hard Delete of Approved Records	The system shall use cancel, disable, archive, or corrected-version actions instead of hard deletion for approved academic records.	Evidence destruction	Academic records	Authorized staff	Must	Deletion attempts converted/denied; history preserved.	DEC-86
SEC-18	Concurrent-Modification Protection	The system shall reject stale updates and require refresh rather than overwrite a newer value.	Lost update	Academic records	All modifiers	Must	Concurrent-edit test.	DEC-187
SEC-19	Transactional Integrity	The system shall roll back the full sensitive operation when any required step fails.	Partial unauthorized/inconsistent change	Registration/grade/status records	System	Must	Failure-injection rollback.	DEC-189
SEC-20	Confidential Error Messages	The system shall not expose technical internals, credentials, sensitive identifiers, or unauthorized record details in user-visible errors.	Information disclosure	System/identity data	All users	Must	Error-content review.	DEC-195,196
SEC-21	Protected Attachments and Exports	The system should restrict attachments and exported data to authorized users and approved scope and minimize sensitive content.	Data leakage	Files/reports	Authorized users	Should	Unauthorized download/export denied.	DEC-183,186,195
SEC-22	Backup and Restore Control	The system may restrict backup and restore to System Administrator authority, require confirmation, and audit outcome.	Data loss/unauthorized restore	System data	System Administrator	Could	Unauthorized restore denied; authorized drill logged.	CAP-50; DEC-167-170
SEC-23	Maintenance Operation Control	The system shall block new data-changing operations during announced maintenance while preserving saved data.	Inconsistent maintenance changes	System data	All users	Must	Maintenance access matrix.	DEC-182
SEC-24	Accountable Delegation	The system may limit delegated approval to named operations, scope, and validity period and preserve original/delegated actor accountability.	Unbounded delegated privilege	Approvals	Registrar/Department Head	Could	Expired/out-of-scope delegation denied; valid action audited.	CAP-59; DEC-115,133

8. Data Requirements

The following are conceptual information requirements only; they do not define entities, tables, keys, SQL, or storage technology.

Table 19 - Conceptual Data Requirements Catalogue

ID	Information Category / Purpose	Conceptual Data Elements	Ownership	Creation Source	Readers and Modifiers	Validation	Retention	Classification	Related Requirements
DR-01	User Identity Data	Support account identifier, person association, active roles, account state, credential state, and sign-in/security history.	System Administration	Authorized account creation and identity administration	Self-limited view; System Administrator modifies; academic staff do not view credentials	Uniqueness and approved identifier formats	Account history retained while associated records exist	Restricted/Sensitive	FR-01-FR-14; SEC-01-SEC-12
DR-02	Student Data	Support University ID, official identity, contact data, admission year/date, program, plan version, level, advisor, status, and lifecycle history.	Academic Affairs	Registration Staff after admission	Student self-view and permitted profile update; authorized staff modify official data	University ID format/uniqueness; valid program/advisor/status	Permanent academic identity and history	Restricted	FR-46-FR-51
DR-03	Instructor and Employee Data	Support Employee ID, official identity, department, contact data, instructor status, roles, and assignment history.	University/Department	System Administrator and authorized department staff	Self-view/permitted profile update; authorized staff modify official data	Employee ID format/uniqueness; valid status	Retain while academic history references it	Restricted	FR-52-FR-54
DR-04	Academic Organization Data	Support university display information, colleges, departments, programs, active state, and historical references.	Registrar	Authorized academic governance	Read according to scope; Registrar/authorized heads modify	Valid hierarchy; no orphan active records	Preserve referenced historical values	Internal	FR-15-FR-20
DR-05	Curriculum and Study Plan Data	Support plan versions, required/elective classification, credits, levels, course relationships, and approval history.	Program Governance	Program Coordinator/Department Head	Students/advisors read applicable plans; authorized governance modifies	Version integrity; credit totals; valid course relationships	Permanent once used by student records	Internal/Academic	FR-24-FR-31
DR-06	Course Data	Support code, names, credits, department, level, type, description, minimum-grade rule, active state, prerequisites, and co-requisites.	Department/Program	Authorized academic staff	Authorized users read; academic governance modifies	Unique code; whole positive credits; valid relationships	Retain historical references	Internal/Academic	FR-21-FR-25
DR-07	Academic Term Data	Support term code/type, year, dates, operational state, windows, deadlines, and closure information.	Registrar	Registrar/Registration Staff	All scoped users read relevant dates; authorized staff modify before closure	Unique code; chronological windows; valid state transitions	Permanent after closure	Internal/Academic	FR-32-FR-34,FR-103
DR-08	Section and Schedule Data	Support section identifier, course, term, instructors, capacity, status, meetings, room/lab, changes, and cancellation history.	Department/Registration	Department Head/Registration Staff	Scoped users read; authorized staff modify subject to rules	Unique section ID; active instructor; no prohibited conflicts; capacity rules	Permanent for completed/closed terms	Internal/Academic	FR-35-FR-45
DR-09	Registration Data	Support request, selected sections, validation results, advisor decision, enrollment state, timestamps, actor/reason, exceptions, holds, and waitlist order.	Student/Academic Affairs	Student request; advisor/staff decisions	Student sees own; staff see authorized scope	Rule validation; allowed state transitions; atomic confirmation	Permanent as academic history	Restricted/Academic	FR-55-FR-75
DR-10	Attendance Data	Support session, date/time, status per student, cancellation, edits, reasons, approvals, effective absence, percentage, warnings, bar, and makeup outcome.	Instructor/Department	Assigned Instructor and authorized approver	Student sees own; instructor/department scoped access	Valid enrollment/session/status; approved edit window and exceptions	Permanent for completed term	Restricted/Academic	FR-76-FR-85
DR-11	Assessment and Grade Data	Support assessment definitions, weights, scores, raw result, grading population/statistics, letter/points, workflow state, publication, lock, corrections, and appeals.	Instructor/Department/Registrar	Assigned Instructor; approved reviewers	Student sees own published results; staff scoped access	Weights=100; score ranges/precision; approved workflow	Permanent; no hard delete	Highly Restricted/Academic	FR-86-FR-98
DR-12	GPA and Academic Record	Support attempts, effective GPA attempt, earned/attempted credits, semester/cumulative	Registrar	System calculations from approved records	Student self; authorized academic staff	Approved formulas, precision, and repeat	Permanent	Highly Restricted/Academic	FR-97-FR-103

ID	Information Category / Purpose	Conceptual Data Elements	Ownership	Creation Source	Readers and Modifiers	Validation	Retention	Classification	Related Requirements
	Data	GPA, standing, result states, eligibility, and changes.				rules			
DR-13	Transcript Data	Support transcript type, language, contents, official number, issue time, issuer, status, cancellation, and replacement relationship.	Academic Affairs	System/Registration Staff	Student unofficial self-view; official issuance by authorized staff	Derived from fixed academic record; unique official number	Permanent	Restricted/Official	FR-104-FR-106
DR-14	Degree Progress and Graduation Data	Support applicable plan, completed/remaining requirements, credits, percentage, eligibility status, approvals, and fixed graduation record.	Program/Registrar	System and authorized academic reviewers	Student/advisor read; authorized reviewers modify decisions	Approved eligibility rules and status sequence	Permanent after approval	Restricted/Academic	FR-107-FR-109
DR-15	Notification and Announcement Data	Support sender/system source, audience, subject/content, severity, related action, creation time, read state/time, urgent read count, and retention.	Authorized academic actor/System	Authorized sender or system event	Recipient reads own; authorized sender sees allowed aggregate	Valid target scope; no unauthorized data disclosure	Default two years configurable	Internal/Restricted	FR-115-FR-117
DR-16	Report Data	Support report type, criteria, role/data scope, generation time, issuer, official status, and exported artifact reference where delivered.	Requesting role/System	System from approved source records	Only authorized scoped readers	Source consistency and scope enforcement	Closed-term snapshots permanent; transient reports per policy	Internal/Official as marked	FR-110-FR-114
DR-17	Audit and Login Data	Support actor/account, action/event, time, affected record, old/new values, reason, result, login outcome, and lock reason.	System	System-generated	Authorized reviewers only; no ordinary modification	Completeness and immutability	Audit for system life	Highly Restricted/Security	FR-118-FR-120
DR-18	Profile Data	Support contact, address, optional image, and other permitted non-official profile fields separately from official identity/academic data.	User	User or authorized moderator	User self; authorized staff moderation	Format/file checks; official data protected	While account/record retained	Restricted	FR-122-FR-123
DR-19	Import Data	Support source file metadata, preview rows, validation results, duplicates, accepted/rejected rows, actor, and final import report.	Authorized staff	Authorized import operation	Importer and authorized audit reviewers	File/row/schema/uniqueness/business validation	Import report retained per audit policy	Restricted	FR-124-FR-126
DR-20	Attachment Data	Support file name, type, size, owner, parent record, upload time, and access/retention metadata.	Submitting user	Approved workflow upload	Owner and authorized reviewers	Approved type <=5 MB; parent record exists	Retain with parent record	Restricted	FR-127-FR-128
DR-21	Backup and Restore Data	Support backup creation time/type, scope, retention, operator, restore confirmation, outcome, and integrity-check result.	System Administration	Automatic or authorized manual process	System Administrator only	Retention and integrity rules	Default 30 days configurable for backup copies; logs retained	Highly Restricted/Operational	FR-129-FR-130

9. Interface Requirements

Table 20 - Interface Requirements Catalogue

ID	Interface Requirement	Requirement Statement	Priority	Source / Related
IR-01	Web User Interface	The system shall provide a browser-based interface for all approved interactive functions.	Must	OCR-14; DEC-19,177
IR-02	Authentication Interface	The system shall provide sign-in, first-password-change, password-change, reset-result, lockout, and session-expiry interactions without exposing credentials or technical detail.	Must	FR-01-FR-07; SEC-01-SEC-07
IR-03	Role-Based Navigation	The system shall expose only navigation and actions permitted by the user's active roles and scope.	Must	FR-09-FR-12; SEC-09-SEC-11
IR-04	Data Entry Forms	The system shall provide input interactions that identify required values, validate before acceptance, preserve entered values when safe, and show actionable errors.	Must	FR-134; SEC-13
IR-05	Tabular and Record Views	The system shall provide readable views for authorized lists, details, histories, and status information with permission-scoped search and filtering.	Must	CAP-39; FR-121
IR-06	Validation Feedback	The system shall identify the failed field or business rule and state the corrective action without disclosing unauthorized information.	Must	DEC-196; SEC-20
IR-07	Confirmation for Sensitive Actions	The system shall request explicit confirmation and display the expected consequence before approved destructive-equivalent, restore, closure, publication, cancellation, or status-changing actions.	Must	DEC-86,93,110,169,191
IR-08	Report Interface	The system shall provide criteria, scope indication, generation status, and in-system viewing for approved reports.	Must	CAP-19; FR-110-FR-111
IR-09	Export Interface	The system should provide authorized export controls for approved PDF, Excel, and CSV outputs and clearly identify official status where applicable.	Should	CAP-29; FR-112
IR-10	Responsive Interface	The system shall adapt the delivered interface to supported desktop, tablet, and mobile browser sizes.	Must	DEC-177; NFR-02
IR-11	Browser Compatibility Interface	The system shall avoid requiring a browser-specific interaction outside the approved Chrome, Edge, Firefox, and Safari support set.	Must	DEC-178; NFR-03
IR-12	Bilingual Interface	The system should allow Arabic and English presentation with RTL/LTR direction as applicable.	Should	CAP-41,CAP-42; FR-135-FR-136
IR-13	Notification Interface	The system should provide a recipient view for unread/read, severity, timestamp, related action, and urgent prominence.	Should	CAP-11; FR-115-FR-117
IR-14	Import Interface	The system should provide file selection, preview, validation results, row-error review, authorization to proceed, and final result summary for delivered import scope.	Should/Could	CAP-40,CAP-60; FR-124-FR-125
IR-15	Attachment Interface	The system should communicate permitted attachment types and size, upload result, and access denial without revealing file content.	Should	FR-127-FR-128
IR-16	No External Operational Interface	Release 1.0 shall not depend on payment, SMS/email, biometric, LMS, SSO, government, or other external operational integration.	Must	DEC-26,107,136

10. Use Case Catalogue

Table 21 - Use Case Catalogue

UC ID	Verb-Phrase Name	Primary Actor	Goal	Priority	Related CAPs	Related FRs	Related BRs	Related SEC
UC-01	Sign In	All users	Authenticate and begin an authorized session.	Must	CAP-27,CAP-51,CAP-52	FR-01,FR-03,FR-06,FR-07	BR-62	SEC-01-SEC-08
UC-02	Manage Accounts and Roles	System Administrator	Create, activate, lock, disable, reset, and assign authorized roles.	Must	CAP-27,CAP-28,CAP-62	FR-02-FR-12,FR-14	BR-03,BR-62,BR-63	SEC-03-SEC-12
UC-03	Manage Academic Organization	Registrar	Maintain colleges, departments, programs, and institution display information.	Must	CAP-01,CAP-63	FR-15-FR-20	BR-01-BR-03	SEC-09-SEC-11
UC-04	Manage Course Catalogue	Program Coordinator	Maintain courses, delivery types, credits, prerequisites, and co-requisites.	Must	CAP-02,CAP-14,CAP-32,CAP-33	FR-21-FR-25	BR-17,BR-18	SEC-09,SEC-13
UC-05	Manage Study Plan	Program Coordinator	Create and version a program study plan.	Must	CAP-02,CAP-63	FR-26-FR-28	BR-02,BR-03	SEC-09-SEC-11
UC-06	Manage Academic Term	Registrar/Registration Staff	Create and control a term and its windows.	Must	CAP-03,CAP-31,CAP-37	FR-32-FR-34,FR-103	BR-04,BR-05	SEC-09,SEC-13
UC-07	Manage Section and Schedule	Department Head/Registration Staff	Create sections, meetings, instructors, rooms, capacities, and status.	Must	CAP-04,CAP-05,CAP-15,CAP-56	FR-35-FR-43	BR-06-BR-10	SEC-09-SEC-13
UC-08	Cancel or Change Section	Department Head/Registration Staff	Cancel a section or change instructor/time/place with controls.	Must	CAP-54,CAP-55	FR-44-FR-45	BR-11,BR-12	SEC-13,SEC-14,SEC-19
UC-09	Manage Student Record and Status	Registration Staff/Registrar	Create and maintain student academic identity, profile, leave, suspension, and archive state.	Must	CAP-12,CAP-13,CAP-47,CAP-48,CAP-49,CAP-61,CAP-70	FR-46-FR-51,FR-126	BR-13,BR-15	SEC-09-SEC-17
UC-10	Manage Instructor Record and Status	System Administrator/Department Head	Create and maintain instructor identity, status, and assignment eligibility.	Must	CAP-15,CAP-52	FR-52-FR-54,FR-126	BR-07,BR-14	SEC-09-SEC-13
UC-11	Submit Registration Request	Student	Select sections and submit a valid request.	Must	CAP-06,CAP-07,CAP-22,CAP-32,CAP-53	FR-55-FR-62	BR-16-BR-22	SEC-09,SEC-10,SEC-13
UC-12	Review Registration Request	Academic Advisor	Approve, reject, return, or escalate an assigned student's request.	Must	CAP-06,CAP-65	FR-62-FR-64	BR-16,BR-23	SEC-09,SEC-10,SEC-14
UC-13	Confirm or Intervene in Registration	Registration Staff	Confirm approved registration or perform an authorized intervention/exception.	Must	CAP-06,CAP-07,CAP-61	FR-65-FR-68	BR-16-BR-22	SEC-11,SEC-14,SEC-18,SEC-19
UC-14	Add Drop or Withdraw	Student/Registration Staff	Change enrollment within approved windows and rules.	Must	CAP-23	FR-69-FR-71	BR-18,BR-26	SEC-09,SEC-13,SEC-14
UC-15	Manage Course Repeat	Student/Advisor	Request a repeat and enforce attempt restrictions.	Must	CAP-34	FR-72,FR-98	BR-27,BR-45	SEC-09,SEC-13
UC-16	Join and Manage Waitlist	Student/Registration Staff	Join, order, and review a section waitlist.	Should	CAP-08	FR-73-FR-74	BR-24,BR-25	SEC-09,SEC-10,SEC-13
UC-17	Auto-Promote Waitlisted Student	System	Register the first eligible waitlisted student when a seat opens.	Could	CAP-24	FR-75	BR-24,BR-25	SEC-18,SEC-19
UC-18	Record Attendance	Instructor	Create a session and record attendance for assigned students.	Must	CAP-09,CAP-16,CAP-66	FR-76-FR-83	BR-28-BR-35	SEC-09,SEC-10,SEC-13,SEC-14
UC-19	Manage Laboratory Absence and Makeup	Instructor/Department Head	Review excused laboratory absence and record approved makeup outcome.	Must	CAP-30	FR-80,FR-85,FR-127	BR-31-BR-33	SEC-09,SEC-14,SEC-21
UC-20	Define Assessments and Enter Scores	Instructor	Define weights and maintain draft scores.	Must	CAP-10,CAP-66	FR-86-FR-88	BR-36,BR-37	SEC-09,SEC-10,SEC-13
UC-21	Calculate Submit Approve and Publish Grades	Instructor/Department Head	Calculate grades, submit, approve, publish, and lock a complete section result set.	Must	CAP-10,CAP-64	FR-89-FR-94	BR-38-BR-43	SEC-09-SEC-14,SEC-19
UC-22	Correct Published Grade	Instructor/Department Head/Registrar	Process a formal grade correction and recalculate records.	Must	CAP-25,CAP-20	FR-95,FR-99-FR-103	BR-44,BR-47	SEC-11,SEC-14,SEC-18,SEC-19
UC-23	Appeal Grade	Student	Submit and resolve a grade appeal.	Should	CAP-26	FR-96,FR-127-FR-128	BR-48	SEC-09,SEC-10,SEC-14,SEC-21
UC-24	Close Academic Term	Registrar/Registration Staff	Finalize grades, GPA, standing, records, and snapshots.	Must	CAP-37,CAP-21	FR-99-FR-103	BR-44,BR-49,BR-50	SEC-11,SEC-14,SEC-19
UC-25	View Academic Record and	Student/Advisor	View authorized results, attempts, GPA, and standing.	Must	CAP-17,CAP-21,CAP-	FR-97-FR-102	BR-43-BR-50	SEC-09,SEC-10

UC ID	Verb-Phrase Name	Primary Actor	Goal	Priority	Related CAPs	Related FRs	Related BRs	Related SEC
	GPA				22,CAP-67			
UC-26	View Unofficial Transcript	Student	View own unofficial transcript.	Must	CAP-18,CAP-67	FR-104	BR-53	SEC-09,SEC-10
UC-27	Issue Official Transcript	Registration Staff	Issue, cancel, and reissue an official transcript.	Must	CAP-18,CAP-61	FR-105-FR-106	BR-53,BR-56	SEC-11,SEC-14,SEC-17
UC-28	Review Degree Progress	Student/Advisor/Program Coordinator	Review plan completion and graduation readiness.	Should	CAP-35,CAP-36	FR-29,FR-107-FR-108	BR-51,BR-52	SEC-09,SEC-10
UC-29	Approve Graduation	Registrar/Program Coordinator	Review eligibility and create the fixed graduation record.	Should	CAP-35	FR-108-FR-109	BR-51,BR-52	SEC-11,SEC-14,SEC-17
UC-30	View Academic Report	Authorized User	Generate and view a role-scoped academic report.	Must	CAP-19,CAP-39	FR-110-FR-111,FR-121	BR-54	SEC-09,SEC-10,SEC-20
UC-31	Export Report	Authorized User	Export an approved report or audit result.	Should	CAP-29	FR-112	BR-54	SEC-09,SEC-10,SEC-21
UC-32	Review Audit History	Registrar/System Administrator/Department Head	Search and review authorized audit history.	Must	CAP-20	FR-118-FR-120	BR-55,BR-56	SEC-14-SEC-16
UC-33	Use Role Dashboard	Authorized User	View role-specific tasks and indicators.	Should	CAP-38,CAP-68	FR-113-FR-114	BR-03,BR-54	SEC-09,SEC-10
UC-34	Receive Notification	Authorized User	View and acknowledge an in-app notification.	Should	CAP-11	FR-115,FR-117	BR-57	SEC-09,SEC-10
UC-35	Publish Academic Announcement	Authorized Academic Staff	Publish a targeted announcement.	Should	CAP-58	FR-116	BR-57	SEC-09,SEC-10,SEC-20
UC-36	Search Authorized Records	Authorized User	Search and filter records within scope.	Must	CAP-39	FR-121	BR-03	SEC-09,SEC-10,SEC-13
UC-37	Manage Own Profile	Authorized User	Update permitted profile data.	Must	CAP-43,CAP-44	FR-04,FR-122-FR-123	BR-64	SEC-03,SEC-09,SEC-13
UC-38	Transfer Academic Program	Student/Advisor/Program Coordinator/Registrar	Process an approved program transfer and re-evaluate the plan.	Should	CAP-45,CAP-46	FR-30-FR-31,FR-127-FR-128	BR-03	SEC-09,SEC-10,SEC-14,SEC-21
UC-39	Maintain Advising Notes	Academic Advisor	Create private notes and optional student-facing summaries.	Should	CAP-57	FR-139	BR-03	SEC-09,SEC-10,SEC-20
UC-40	Import Student or Course Data	Authorized Staff	Preview, validate, and apply an approved bulk import.	Should/Could	CAP-40,CAP-60	FR-124-FR-126	BR-58	SEC-09,SEC-13,SEC-14
UC-41	Back Up and Restore	System Administrator	Create or restore an authorized backup.	Could	CAP-50	FR-129-FR-130	BR-60	SEC-22
UC-42	Delegate Approval	Registrar/Department Head	Delegate a defined approval for a limited scope and period.	Could	CAP-59	FR-140	BR-03,BR-63	SEC-24
UC-43	Access Archived Records	Graduate/Archived User	Access permitted read-only history or receive blocked access according to policy.	Must/Should	CAP-49,CAP-69,CAP-70	FR-13-FR-14,FR-51	BR-15,BR-56	SEC-08-SEC-10

All Must capabilities are represented by one or more use cases except supporting quality/data constraints that are inherently non-interactive; those are realized by NFR, SEC, DR, and system-process requirements.

11. Detailed Use Case Specifications

UC-01 - Sign In

Goal: Establish an authorized session for an approved account.

Primary actor: Any human user

Supporting actors: System Administrator for reset/unlock; System Clock for timeout

Stakeholder interests: Users need reliable access; administrators need controlled identity; the university needs traceable access.

Preconditions: Account exists and is not Disabled; connected access is available.

Trigger: User submits account identifier and password.

Main Success Scenario

1. User enters the assigned University ID or Employee ID and password.
2. The system validates identifier type, account state, and credential.
3. The system records the login result.
4. If the credential is temporary, the user replaces it under the password rules.
5. The system activates the permitted role context and opens an authorized session.

Alternative Flows

- A1: User with multiple roles selects or switches among assigned roles without a second account.
- A2: Authorized reset provides a temporary password and returns to step 4.

Exception / Validation Flows

- E1: Invalid credential increments the failure count and returns a safe error.
- E2: Third consecutive failure locks the account for 15 minutes.
- E3: Locked or Disabled account is denied.
- E4: Session inactivity of 30 minutes ends the session.

Postconditions: Successful session and login event, or no session and recorded failure.

Success guarantee: Only the correct approved account receives its authorized scope.

Minimal guarantee: Credentials and technical details are not exposed; account/data remain unchanged except security counters/logs.

Priority: Must

Frequency: High

Related FRs: FR-01,FR-03,FR-06,FR-07

Related BRs: BR-62,BR-63

Related SEC: SEC-01-SEC-08

Related DRs: DR-01,DR-17

Acceptance criteria: AC-001,AC-003,AC-006,AC-007

Open issues: None.

UC-02 - Manage Accounts and Roles

Goal: Administer accounts, states, credentials, and authorized roles without academic authority leakage.

Primary actor: System Administrator

Supporting actors: Authorized academic owner for role approval where required

Stakeholder interests: Users need correct access; academic owners need separation of duties; audit reviewers need accountability.

Preconditions: Administrator is authenticated and authorized; person identity exists or is being created under approved rules.

Trigger: Administrator opens an account administration action.

Main Success Scenario

1. Administrator identifies the person and requested account action.
2. The system validates University ID or Employee ID uniqueness and format.
3. Administrator creates/updates the account state or requests a role change with reason.
4. The system validates the administrator's authority and separation-of-duties rules.
5. The system applies the change, issues a temporary credential if applicable, and records the audit event.

Alternative Flows

- A1: One account receives multiple approved roles.
- A2: Locked account is manually unlocked after authorized review.
- A3: Archived account is set to read-only or blocked according to administrative reason.

Exception / Validation Flows

- E1: Duplicate identifier/account is rejected.
- E2: Self-assignment or unapproved role is denied.
- E3: Technical administrator attempt to gain academic decision authority is denied.

Postconditions: Account and role state reflect the authorized change; history is preserved.

Success guarantee: Access matches approved roles and scope.

Minimal guarantee: No unauthorized role/state change occurs; request and denial are safe and traceable.

Priority: Must

Frequency: Medium

Related FRs: FR-02-FR-12,FR-14

Related BRs: BR-03,BR-62,BR-63

Related SEC: SEC-03-SEC-12,SEC-14

Related DRs: DR-01,DR-17

Acceptance criteria: AC-002-AC-012,AC-014

Open issues: Team operating procedure for role approval must be documented later; no new product policy is required.

UC-03 - Manage Academic Organization

Goal: Maintain the single-university academic hierarchy.

Primary actor: Registrar

Supporting actors: Department Head; Program Coordinator; System Administrator for display configuration only

Stakeholder interests: Academic governance needs valid hierarchy; historical records must remain interpretable.

Preconditions: Actor is authenticated with academic-governance authority.

Trigger: Actor creates or changes institution, college, department, or program information.

Main Success Scenario

1. Actor selects the organization level and action.
2. The system displays the permitted parent context and current state.
3. Actor enters or revises the authorized information.
4. The system validates hierarchy, uniqueness, active references, and authority.
5. The system saves the change and preserves historical references where used.

Alternative Flows

- A1: Record is deactivated instead of deleted.
- A2: Configurable demonstration name is updated without changing academic policy.

Exception / Validation Flows

- E1: Invalid parent relationship or orphaning change is rejected.
- E2: Unauthorized program/department scope is denied.
- E3: Hard deletion of referenced history is denied.

Postconditions: A valid active/inactive hierarchy exists with preserved references.

Success guarantee: Organization data support later curriculum, users, sections, and reporting.

Minimal guarantee: Existing records remain unchanged on failure.

Priority: Must

Frequency: Low-Medium

Related FRs: FR-15-FR-20

Related BRs: BR-01-BR-03,BR-56

Related SEC: SEC-09-SEC-11,SEC-14,SEC-17

Related DRs: DR-04

Acceptance criteria: AC-015-AC-020

Open issues: None.

UC-04 - Manage Course Catalogue

Goal: Maintain approved course definitions and academic relationships.

Primary actor: Program Coordinator

Supporting actors: Department Head

Stakeholder interests: Programs need correct rules; advisors/students need reliable eligibility; historical results need stable course references.

Preconditions: Academic organization exists; actor has program/department authority.

Trigger: Actor creates or changes a course or relationship.

Main Success Scenario

1. Actor selects the course or creates a new course definition.
2. Actor enters code, name, credits, department, level, type, description, and minimum-grade rules.
3. Actor defines prerequisite and co-requisite relationships.
4. The system validates unique code, whole positive credits, valid relationship references, and prohibited cycles/invalid self-reference.
5. Department Head approves the governed change where required, and the system preserves the resulting version/history.

Alternative Flows

- A1: Course is deactivated while historical references remain.
- A2: Co-requisite withdrawal effect is recorded for later enforcement.

Exception / Validation Flows

- E1: Duplicate code, fractional credits, invalid type, or invalid relationship is rejected.
- E2: Unauthorized plan/course scope is denied.

Postconditions: Catalogue and relationships are valid and traceable.

Success guarantee: Registration can evaluate the approved rules.

Minimal guarantee: No partial course/rule change is saved.

Priority: Must

Frequency: Medium

Related FRs: FR-21-FR-25

Related BRs: BR-17,BR-18,BR-56

Related SEC: SEC-09-SEC-13,SEC-17

Related DRs: DR-06

Acceptance criteria: AC-021-AC-025

Open issues: None.

UC-05 - Manage Study Plan

Goal: Create and preserve approved program study-plan versions.

Primary actor: Program Coordinator

Supporting actors: Department Head; Registrar for activation authority

Stakeholder interests: Students need applicable plans; governance needs history; degree progress needs consistent rules.

Preconditions: Program and course catalogue exist; actor is authorized.

Trigger: Actor creates or revises a study plan.

Main Success Scenario

1. Actor selects the program and creates a new plan version.
2. Actor adds required/elective courses, levels, and required credits.
3. Actor reviews prerequisite/co-requisite and minimum-grade dependencies.
4. The system validates references, classifications, totals, and version consistency.
5. Authorized approver activates the plan version; the system preserves previous versions and student associations.

Alternative Flows

- A1: Draft version is revised before activation.
- A2: Existing students remain on an earlier approved version.

Exception / Validation Flows

- E1: Invalid course, duplicate placement, inconsistent total, or unauthorized activation is rejected.

Postconditions: One approved plan version is available without altering historical versions.

Success guarantee: Students can be associated with a stable applicable plan.

Minimal guarantee: Previous approved plan remains unchanged.

Priority: Must

Frequency: Low-Medium

Related FRs: FR-26-FR-28

Related BRs: BR-02,BR-03,BR-56

Related SEC: SEC-09-SEC-11,SEC-13,SEC-17

Related DRs: DR-05

Acceptance criteria: AC-026-AC-028

Open issues: None.

UC-06 - Manage Academic Term

Goal: Create and control a term, its dates, windows, and closure state.

Primary actor: Registrar / Registration Staff

Supporting actors: Department Head at closure

Stakeholder interests: All users need authoritative windows; records need a controlled lifecycle.

Preconditions: Authorized actor is authenticated; calendar settings apply.

Trigger: Actor creates or changes a term or initiates closure.

Main Success Scenario

1. Actor creates/selects a First, Second, or Summer term.
2. Actor sets code, dates, registration/add-drop/withdrawal/appeal/grade windows, and initial state.
3. The system validates code uniqueness, chronological consistency, and allowed state transition.
4. Actor activates the term for permitted operations.
5. At closure, authorized actors confirm prerequisites and the system finalizes results, snapshots, and locks the term.

Alternative Flows

- A1: Future window dates are adjusted before their dependent operation is closed.
- A2: Summer-specific load rules apply automatically.

Exception / Validation Flows

- E1: Duplicate code, invalid date order, prohibited state transition, or incomplete closure prerequisite is rejected.

Postconditions: Term state and windows govern all dependent operations; closure fixes approved records.

Success guarantee: No operation occurs outside an allowed term state/window without approved exception.

Minimal guarantee: Prior term state remains unchanged on failure.

Priority: Must

Frequency: Several times per year

Related FRs: FR-32-FR-34,FR-103

Related BRs: BR-04,BR-05,BR-20,BR-21

Related SEC: SEC-09,SEC-13,SEC-14,SEC-19

Related DRs: DR-07,DR-12,DR-16

Acceptance criteria: AC-032-AC-034,AC-103

Open issues: None for the course submission cutoff. ASM-12 is resolved as Thursday, 30 July 2026, 11:00 PM, platform-displayed local time; this project deadline remains separate from academic-term configuration.

UC-07 - Manage Section and Schedule

Goal: Create a valid course section with instructors, meetings, place, capacity, and status.

Primary actor: Department Head / Registration Staff

Supporting actors: Instructor

Stakeholder interests: Students need conflict-free offerings; departments need teaching control; facilities need capacity compliance.

Preconditions: Course and term exist; actors, rooms/labs, and instructors are available.

Trigger: Authorized actor creates or edits a section.

Main Success Scenario

1. Actor selects course and term and creates the section.
2. The system assigns or validates the approved unique section identifier.
3. Actor assigns active Primary/Co-Instructors, capacity, status, and one or more meetings.
4. Actor assigns room/lab to each meeting.
5. The system validates instructor/room conflicts, place compatibility, and capacity and saves the section.

Alternative Flows

- A1: Section is created as inactive/draft before release.
- A2: Multiple meetings and co-instructors are provided.

Exception / Validation Flows

- E1: Inactive instructor, duplicate identifier, conflict, or capacity violation without exception is rejected.
- E2: Unauthorized department access is denied.

Postconditions: A valid section is available for registration according to state.

Success guarantee: All section constraints pass.

Minimal guarantee: No partial schedule is committed.

Priority: Must

Frequency: Each term

Related FRs: FR-35-FR-43

Related BRs: BR-06-BR-10

Related SEC: SEC-09-SEC-13,SEC-19

Related DRs: DR-08

Acceptance criteria: AC-035-AC-043

Open issues: None.

UC-08 - Cancel or Change Section

Goal: Control cancellation, instructor reassignment, and time/place changes.

Primary actor: Department Head / Registration Staff

Supporting actors: Instructor; Student; System notification process

Stakeholder interests: Affected users need notice; records need history; schedules must remain valid.

Preconditions: Section exists; actor has authority.

Trigger: Actor requests cancellation or approved change.

Main Success Scenario

1. Actor selects the section and change type.
2. Actor records reason and proposed effective data.
3. The system validates authority, section/term state, and affected records.
4. For time/place/instructor change, the system rechecks active status, capacity, and conflicts.
5. The system applies the approved change, preserves old/new values, handles registrations as required, and creates audit/notification records according to delivered priority.

Alternative Flows

- A1: Pre-start cancellation stops registration and marks existing enrollments Cancelled.
- A2: Post-start cancellation transfers students or cancels without W/GPA effect.
- A3: Reassignment preserves prior instructor history.

Exception / Validation Flows

- E1: Conflict/capacity failure blocks change.
- E2: Attempt to delete history is denied.
- E3: Any multi-step failure rolls back the full operation.

Postconditions: Section and affected registrations are consistent and traceable.

Success guarantee: No invalid schedule or unexplained record change remains.

Minimal guarantee: Original section remains valid on failure.

Priority: Must

Frequency: Low-Medium

Related FRs: FR-44-FR-45

Related BRs: BR-11,BR-12,BR-56,BR-61

Related SEC: SEC-13,SEC-14,SEC-17-SEC-19

Related DRs: DR-08,DR-09,DR-15,DR-17

Acceptance criteria: AC-044,AC-045

Open issues: Notification delivery is conditional on CAP-11, but the core section action is Must.

UC-09 - Manage Student Record and Status

Goal: Create and maintain the official post-admission student record and lifecycle state.

Primary actor: Registration Staff / Registrar

Supporting actors: Advisor; Student

Stakeholder interests: Student needs accurate self-record; university needs permanent official history.

Preconditions: Admission has occurred outside SUMS; actor is authorized; program/plan/advisor exist.

Trigger: Actor creates the record or changes official status.

Main Success Scenario

1. Actor enters official identity and post-admission academic information.
2. The system validates University ID format/uniqueness and academic references.
3. Actor assigns program, plan version, level, advisor, and initial Active status.
4. For leave, suspension, withdrawal, graduation, or reactivation, actor records reason, dates, and approval.
5. The system applies registration restrictions, preserves history, and records the sensitive change.

Alternative Flows

- A1: User updates only permitted contact/profile data through UC-37.
- A2: Archived access becomes read-only or blocked according to reason.

Exception / Validation Flows

- E1: Duplicate/invalid ID or invalid status transition is rejected.
- E2: Unauthorized official-data change is denied.

Postconditions: Official student profile and lifecycle history are consistent.

Success guarantee: Downstream registration/records use the approved state.

Minimal guarantee: No partial or unapproved status change occurs.

Priority: Must

Frequency: Medium

Related FRs: FR-46-FR-51,FR-126

Related BRs: BR-13,BR-15,BR-56,BR-64

Related SEC: SEC-08-SEC-14,SEC-17

Related DRs: DR-02,DR-17,DR-18

Acceptance criteria: AC-046-AC-051,AC-126

Open issues: None.

UC-10 - Manage Instructor Record and Status

Goal: Maintain employee identity and instructor assignment eligibility.

Primary actor: System Administrator / Department Head

Supporting actors: Instructor

Stakeholder interests: Department needs eligible staff; instructor needs correct identity/access; records need assignment history.

Preconditions: Authorized actor; department exists.

Trigger: Actor creates or changes instructor record/status.

Main Success Scenario

1. Actor creates/selects the instructor record.
2. The system validates Employee ID format and uniqueness.
3. Actor maintains official details, department, roles, and Active/On Leave/Inactive status within authority.
4. The system prevents new assignment when status is not Active.
5. The system preserves status and assignment history and audits sensitive changes.

Alternative Flows

- A1: Existing historical teaching remains visible after Inactive status.

Exception / Validation Flows

- E1: Duplicate ID or unauthorized role/status change is rejected.
- E2: Attempted section assignment to non-Active instructor is rejected.

Postconditions: Instructor identity/status and assignment eligibility are consistent.

Success guarantee: Only Active instructors receive new assignments.

Minimal guarantee: Prior state remains unchanged on failure.

Priority: Must

Frequency: Medium

Related FRs: FR-52-FR-54,FR-126

Related BRs: BR-07,BR-14,BR-56

Related SEC: SEC-09-SEC-14,SEC-17

Related DRs: DR-03,DR-17

Acceptance criteria: AC-052-AC-054,AC-126

Open issues: None.

UC-11 - Submit Registration Request

Goal: Submit a rule-valid set of sections for mandatory advisor review.

Primary actor: Student

Supporting actors: System; Academic Advisor

Stakeholder interests: Student needs clear eligibility; university needs compliant enrollment.

Preconditions: Student is authenticated, Active, within open registration window, and has no blocking hold; sections are offered.

Trigger: Student submits selected sections.

Main Success Scenario

1. Student reviews offered/eligible sections and selects desired sections.
2. The system validates term/student/section state, holds, prerequisite, co-requisite, conflicts, capacity, load, repeat, and one-section rules.
3. The system displays each validation result and any permitted waitlist option.
4. Student confirms the valid request.
5. The system records Pending Advisor Approval and presents it to the assigned advisor.

Alternative Flows

- A1: Full section enters delivered Should waitlist path.
- A2: Final-term or other permitted exception is requested separately.

Exception / Validation Flows

- E1: Prerequisite failure is rejected without override.
- E2: Conflict, load, hold, inactive state, or closed window blocks submission.
- E3: Unauthorized access to another student's request is denied.

Postconditions: Valid request awaits advisor action; no seat is finally confirmed yet unless later flow succeeds.

Success guarantee: Request contains only rule-valid selections or explicit permitted exception paths.

Minimal guarantee: No enrollment or seat corruption occurs.

Priority: Must

Frequency: High during registration

Related FRs: FR-55-FR-62

Related BRs: BR-16-BR-22,BR-27

Related SEC: SEC-09,SEC-10,SEC-13,SEC-18,SEC-19

Related DRs: DR-09

Acceptance criteria: AC-055-AC-062

Open issues: Waitlist is conditional Should; direct registration remains complete without it.

UC-12 - Review Registration Request

Goal: Provide mandatory advisor decision for an assigned student.

Primary actor: Academic Advisor

Supporting actors: Department Head for delay/exception

Stakeholder interests: Student needs guidance; program needs controlled progress; advisor scope must be limited.

Preconditions: Request is Pending Advisor Approval and belongs to assigned student or approved temporary scope.

Trigger: Advisor opens the pending request.

Main Success Scenario

1. Advisor reviews student plan, record, standing, load, and validation results.
2. Advisor may return the request with guidance or reject it with reason.
3. Advisor approves selections that satisfy policy and records the decision.
4. The system rechecks advisor scope and request state and marks it Approved for confirmation.

5. If pending reaches two working days, the system exposes the approved escalation path.

Alternative Flows

- A1: Advisor recommends a permitted exception to Department Head/Registrar.
- A2: Request is returned for student changes.

Exception / Validation Flows

- E1: Unassigned-student access is denied.
- E2: Prerequisite override request is denied.
- E3: Stale request state requires refresh.

Postconditions: Request has documented advisor outcome or escalation status.

Success guarantee: Only an authorized advisor decision advances confirmation.

Minimal guarantee: Enrollment remains unchanged on denial/failure.

Priority: Must

Frequency: High during registration

Related FRs: FR-62-FR-64,FR-67

Related BRs: BR-16,BR-17,BR-23

Related SEC: SEC-09,SEC-10,SEC-14,SEC-18

Related DRs: DR-09,DR-17

Acceptance criteria: AC-062-AC-064,AC-067

Open issues: None.

UC-13 - Confirm or Intervene in Registration

Goal: Create a confirmed enrollment atomically or perform a controlled staff action.

Primary actor: Registration Staff / System

Supporting actors: Advisor; Department Head; Registrar

Stakeholder interests: Student needs correct seat; staff need operational control; audit needs accountability.

Preconditions: Request is advisor-approved or staff action is explicitly authorized.

Trigger: Confirmation or intervention is initiated.

Main Success Scenario

1. Actor/system selects the approved request or staff action.
2. The system revalidates seat, conflict, load, status, hold, term, and section conditions.
3. For exception, the system verifies required recommendation/approval and prohibition of prerequisite override.
4. The system atomically changes enrollment to Registered or the approved target state and updates capacity.
5. The system records actor, reason, approvals, result, and any affected waitlist state.

Alternative Flows

- A1: If full and Should waitlist is delivered, move to Waitlisted.
- A2: Staff corrects an operational enrollment error within authority.

Exception / Validation Flows

- E1: Last-seat race loses revalidation; operation is rejected without partial change.
- E2: Any step failure rolls back the full action.
- E3: Unauthorized staff grade/academic decision is denied.

Postconditions: Enrollment/capacity are consistent and auditable.

Success guarantee: Exactly one valid enrollment outcome is committed.

Minimal guarantee: No seat or record inconsistency occurs.

Priority: Must

Frequency: High during registration

Related FRs: FR-65-FR-68,FR-131-FR-132

Related SEC: SEC-11,SEC-13,SEC-14,SEC-18,SEC-19

Acceptance criteria: AC-065-AC-068,AC-131,AC-132

Related BRs: BR-16-BR-22,BR-61

Related DRs: DR-09,DR-17

Open issues: None.

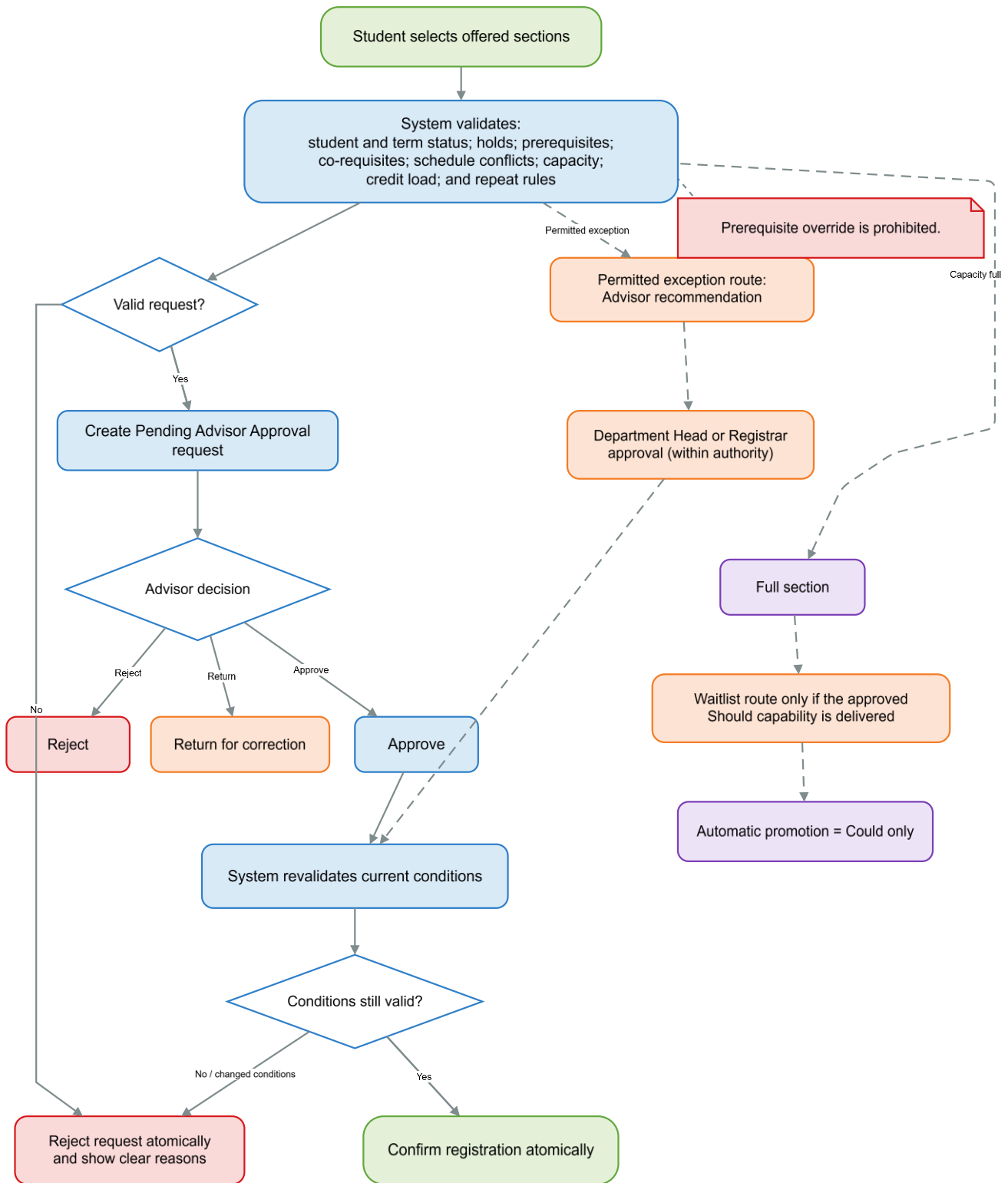


Figure 4 - High-Level Course Registration Workflow

Figure 4 summarizes the approved student-driven registration process: comprehensive validation, advisor review, revalidation, and atomic confirmation. It also preserves the permitted exception route, the conditional waitlist route, the optional automatic-promotion route, and the prohibition on prerequisite override.

UC-14 - Add Drop or Withdraw

Goal: Change enrollment according to configured windows and result rules.

Primary actor: Student / Registration Staff

Supporting actors: Advisor; approver for late exception

Stakeholder interests: Student needs controlled schedule change; official record needs correct W/history.

Preconditions: Enrollment/section exists; actor is authorized; applicable term window or approved exception exists.

Trigger: Actor selects add, drop, or withdrawal.

Main Success Scenario

1. Actor selects the enrollment action.
2. The system checks term window, student/section status, co-requisite effects, load, conflict, and approval requirements.
3. For add, the system follows registration confirmation rules.
4. For drop within add/drop, the system records Dropped without W.
5. For later authorized withdrawal, the system records Withdrawn with W and preserves academic history.

Alternative Flows

- A1: Late action routes through approved exception.
- A2: Co-requisite withdrawal requires advisor review and relationship-specific outcome.

Exception / Validation Flows

- E1: Closed deadline without exception is rejected.
- E2: Unauthorized or inconsistent co-requisite action is rejected.
- E3: Failure rolls back all affected enrollments.

Postconditions: Enrollment state and academic impact match the approved period/action.

Success guarantee: No wrong W/GPA effect is introduced.

Minimal guarantee: Original enrollment remains on failure.

Priority: Must

Frequency: Medium

Related FRs: FR-69-FR-71

Related BRs: BR-18,BR-26,BR-61

Related SEC: SEC-09,SEC-13,SEC-14,SEC-19

Related DRs: DR-09,DR-12

Acceptance criteria: AC-069-AC-071

Open issues: None.

UC-15 - Manage Course Repeat

Goal: Register a permitted repeat and apply the approved academic effect.

Primary actor: Student / Academic Advisor

Supporting actors: Department Head after limit

Stakeholder interests: Student needs improvement opportunity; academic record must preserve all attempts and correct GPA.

Preconditions: Student has prior attempt; registration window is open.

Trigger: Student requests a course previously attempted.

Main Success Scenario

1. The system identifies prior attempts and current effective result.
2. The system checks additional-attempt count, parallel-attempt rule, and one-section-per-term rule.
3. If beyond the permitted two additional attempts, Department Head approval is required.
4. The normal registration flow completes for an allowed repeat.

- After completion, the system preserves all attempts, uses the last completed attempt for GPA, and awards credit once.

Alternative Flows

- A1: Passed course may be repeated for improvement.
- A2: Department Head approves an additional attempt beyond limit.

Exception / Validation Flows

- E1: Parallel active attempt or second section in same term is rejected.
- E2: Missing required approval is rejected.

Postconditions: Repeat enrollment and later GPA/credit effects follow policy.

Success guarantee: Attempt history and effective GPA attempt are correct.

Minimal guarantee: Prior record remains unchanged on failure.

Priority: Must

Frequency: Medium

Related FRs: FR-72,FR-98

Related BRs: BR-27,BR-45

Related SEC: SEC-09,SEC-13,SEC-14

Related DRs: DR-09,DR-12

Acceptance criteria: AC-072,AC-098

Open issues: None.

UC-18 - Record Attendance

Goal: Record and evaluate attendance for an assigned section.

Primary actor: Instructor

Supporting actors: Department Head; Student; System

Stakeholder interests: Instructor needs efficient entry; student needs transparent status; department needs policy enforcement.

Preconditions: Instructor is assigned; section/session and enrolled students exist.

Trigger: Instructor opens an attendance session.

Main Success Scenario

- Instructor creates/confirms the scheduled session or marks it Cancelled with reason.
- For each enrolled student, instructor records Present, Absent, Late, or Excused.
- The system validates assignment, enrollment, status, and edit window.
- The system recalculates effective absences, Late conversion, attendance percentage, warning, and laboratory rules.
- The system saves the session and exposes each student's own result.

Alternative Flows

- A1: Excused evidence/review is attached through delivered supporting scope.
- A2: Within seven days instructor corrects entry.
- A3: Later correction routes to Department Head approval.

Exception / Validation Flows

- E1: Unassigned section access is denied.
- E2: Invalid status, student, or late unapproved edit is rejected.
- E3: Cancelled session is retained but excluded.

Postconditions: Attendance and derived risk are consistent and traceable.

Success guarantee: Policy calculation matches approved session data.

Minimal guarantee: No unauthorized change or percentage corruption.

Priority: Must

Frequency: Very high

Related FRs: FR-76-FR-84

Related BRs: BR-28-BR-35

Related SEC: SEC-09,SEC-10,SEC-13,SEC-14

Related DRs: DR-10,DR-17

Acceptance criteria: AC-076-AC-084

Open issues: None.

UC-19 - Manage Laboratory Absence and Makeup

Goal: Resolve excused laboratory absence and record makeup outcome.

Primary actor: Instructor / Department Head

Supporting actors: Student

Stakeholder interests: Student needs fair review; department needs strict laboratory rule; record needs evidence.

Preconditions: Laboratory attendance record exists; student submits evidence where applicable.

Trigger: Excused laboratory absence or makeup request is reviewed.

Main Success Scenario

1. Instructor reviews the absence record and submitted evidence.
2. The system applies file/access validation when attachment scope is delivered.
3. Instructor recommends whether makeup is required.
4. Department Head approves or rejects the makeup.
5. Instructor records the approved makeup outcome; the system recalculates laboratory status.

Alternative Flows

- A1: Absence remains unexcused and counts toward the one-absence limit.
- A2: Mixed course theory percentage remains independent.

Exception / Validation Flows

- E1: Unauthorized evidence access is denied.
- E2: Makeup result without Department Head approval is rejected.

Postconditions: Lab absence/makeup status is approved and traceable.

Success guarantee: Laboratory eligibility reflects approved outcome.

Minimal guarantee: Original absence record remains if request fails.

Priority: Must

Frequency: Low-Medium

Related FRs: FR-80,FR-85,FR-127-FR-128

Related BRs: BR-31-BR-33,BR-59

Related SEC: SEC-09,SEC-14,SEC-21

Related DRs: DR-10,DR-20

Acceptance criteria: AC-080,AC-085,AC-127,AC-128

Open issues: Attachments are Should and must not block the Must core workflow if not delivered; evidence may be referenced administratively in the minimal release.

UC-20 - Define Assessments and Enter Scores

Goal: Prepare a valid section assessment scheme and draft scores.

Primary actor: Instructor

Supporting actors: Department Head for alternative approval

Stakeholder interests: Instructor needs flexible scheme; students need correct calculation; department needs 100-percent integrity.

Preconditions: Instructor is assigned to active section; grade-entry period permits draft work.

Trigger: Instructor defines assessments or enters scores.

Main Success Scenario

1. Instructor selects the section and defines assessment items.
2. The system presents default 20/30/50 weights or accepts an approved alternative.
3. The system validates that weights total 100 percent.
4. Instructor enters draft scores up to two decimals for enrolled students.
5. The system validates ranges/precision and calculates exact raw scores without publishing them.

Alternative Flows

- A1: Draft scores are revised before submission.
- A2: Approved alternative weighting replaces default.

Exception / Validation Flows

- E1: Weight total not 100 or invalid score is rejected.
- E2: Unassigned-section access is denied.

Postconditions: Valid draft scheme and scores exist.

Success guarantee: Raw results are reproducible from exact inputs.

Minimal guarantee: Published records remain unchanged.

Priority: Must

Frequency: High per term

Related FRs: FR-86-FR-88

Related BRs: BR-36,BR-37

Related SEC: SEC-09,SEC-10,SEC-13

Related DRs: DR-11

Acceptance criteria: AC-086-AC-088

Open issues: None.

UC-21 - Calculate Submit Approve and Publish Grades

Goal: Convert complete numeric results to approved published locked section grades.

Primary actor: Instructor / Department Head

Supporting actors: System

Stakeholder interests: Students need correct simultaneous publication; department needs approval; records need lock/history.

Preconditions: Assessment scheme and complete draft scores exist; actor has assigned/department scope.

Trigger: Instructor submits the section result set.

Main Success Scenario

1. The system determines the completed numeric population, mean, and standard deviation.
2. The system applies relative grading or absolute fallback and enforces raw-score fail floor.
3. Instructor reviews the resulting raw, letter, and point values and submits the complete set.
4. Department Head reviews and approves or returns the set.
5. The system publishes all section results together, records publication, and locks them.

Alternative Flows

- A1: Fewer than 10 numeric grades or zero standard deviation invokes absolute fallback.
- A2: Returned set goes back to draft for correction before approval.

Exception / Validation Flows

- E1: Incomplete grades, invalid population, or unauthorized approval is rejected.
- E2: Any failure before publication leaves no partial published results.
- E3: Post-lock direct edit is denied and must use UC-22.

Postconditions: All section results are approved, simultaneously visible, locked, and auditable.

Success guarantee: Each result matches formula/threshold/floor and workflow.

Minimal guarantee: No partial publication or unlocked approved grade remains.

Priority: Must

Frequency: High at grading periods

Related FRs: FR-89-FR-94

Related BRs: BR-38-BR-43

Related SEC: SEC-09-SEC-14, SEC-19

Related DRs: DR-11, DR-17

Acceptance criteria: AC-089-AC-094

Open issues: None.

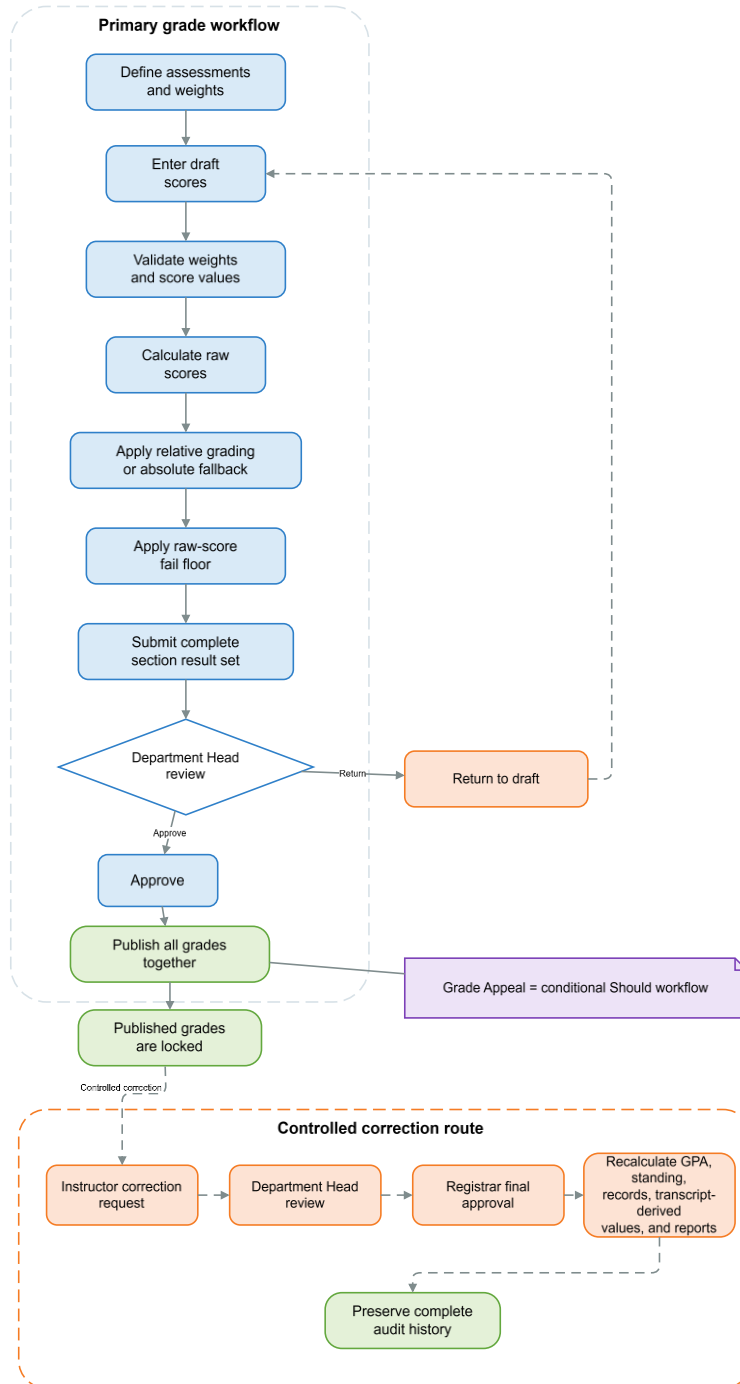


Figure 5 - High-Level Grade Workflow

Figure 5 summarizes assessment setup, draft scoring, validation, grade calculation, department review, publication, locking, and the controlled correction route. Grade Appeal remains a conditional Should workflow.

UC-22 - Correct Published Grade

Goal: Apply an authorized correction and recalculate dependent records.

Primary actor: Instructor / Department Head / Registrar

Supporting actors: Registration Staff for viewing updated official record

Stakeholder interests: Student needs corrected result; university needs high-control audit and consistency.

Preconditions: Grade is published/locked; instructor identifies a justified correction.

Trigger: Instructor submits correction request with reason.

Main Success Scenario

1. Instructor selects the published grade and submits proposed correction with reason.
2. Department Head reviews and approves or rejects the request.
3. Registrar gives final approval for an accepted correction.
4. The system records old/new grade and approvals, applies the correction, and recalculates GPA, standing, eligibility, transcript-derived data, and affected reports.
5. The system preserves complete correction history and notifies the student when notification scope is delivered.

Alternative Flows

- A1: Department/Registrar rejects with reason; grade remains unchanged.

Exception / Validation Flows

- E1: Direct post-lock edit is denied.
- E2: Unauthorized actor or missing reason/approval is rejected.
- E3: Recalculation failure rolls back the entire correction.

Postconditions: Corrected grade and every dependent record agree; audit is complete.

Success guarantee: No stale GPA/standing remains.

Minimal guarantee: Original grade remains if workflow fails.

Priority: Must

Frequency: Low-Medium

Related FRs: FR-95,FR-99,FR-103,FR-118,FR-132

Related BRs: BR-44,BR-47,BR-49,BR-55,BR-61

Related SEC: SEC-11,SEC-14,SEC-17-SEC-19

Related DRs: DR-11,DR-12,DR-17

Acceptance criteria: AC-095,AC-099-AC-103,AC-118,AC-132

Open issues: None.

UC-24 - Close Academic Term

Goal: Finalize the term and fix official academic outcomes.

Primary actor: Registrar / Registration Staff

Supporting actors: Department Head; Program Coordinator; System

Stakeholder interests: All stakeholders need consistent final results; history must be immutable except formal correction.

Preconditions: Required section grades are approved/published; closure prerequisites are satisfied.

Trigger: Authorized actor initiates term closure.

Main Success Scenario

1. The system checks completeness of required grades and term prerequisites.
2. Authorized actor reviews unresolved exceptions and confirms closure.
3. The system calculates semester/cumulative GPA using approved repeat/result rules and truncation.
4. The system determines standing, updates degree progress where delivered, and creates fixed grade/registration/attendance/standing report snapshots.
5. The system marks the term closed and locks term records.

Alternative Flows

- A1: Closure is postponed because required data is incomplete.
- A2: Later grade correction uses UC-22 and creates controlled revised derived values/history.

Exception / Validation Flows

- E1: Unauthorized closure or incomplete mandatory result blocks closure.
- E2: Any calculation/snapshot failure rolls back closure.

Postconditions: Term is closed with consistent fixed results/snapshots.

Success guarantee: All dependent official values match approved records.

Minimal guarantee: Term remains open and unchanged if closure fails.

Priority: Must

Frequency: Each term

Related FRs: FR-99-FR-103

Related BRs: BR-44-BR-50,BR-54,BR-56,BR-61

Related SEC: SEC-11,SEC-14,SEC-17,SEC-19

Related DRs: DR-07,DR-12,DR-16,DR-17

Acceptance criteria: AC-099-AC-103

Open issues: Degree progress/graduation UI is Should; core GPA/standing/record closure is Must.

UC-25 - View Academic Record and GPA

Goal: View an authorized student's academic history and derived standing.

Primary actor: Student / Advisor / Authorized Academic Staff

Supporting actors: System

Stakeholder interests: Student needs transparency; advisors need correct decisions; privacy requires strict scope.

Preconditions: Authenticated user; records exist; requester has self/assigned/authorized scope.

Trigger: Requester opens academic record.

Main Success Scenario

1. Requester identifies self or an authorized student.
2. The system verifies data scope.
3. The system retrieves all attempts, **W/I/P/F** states, effective repeat, credits, semester/cumulative GPA, standing, and eligibility data.
4. The system presents fixed historical values and any current derived status.
5. Requester may filter by term within authorization.

Alternative Flows

- A1: Graduate receives limited read access.
- A2: Advisor views only assigned students.

Exception / Validation Flows

- E1: Cross-student or unassigned access is denied without revealing record existence.
- E2: Archived policy may block the user.

Postconditions: No data changes; authorized view is traceable as required.

Success guarantee: Displayed values match official calculations and history.

Minimal guarantee: No unauthorized data is disclosed.

Priority: Must

Frequency: High

Related FRs: FR-97-FR-102

Related BRs: BR-43-BR-50,BR-56

Related SEC: SEC-08-SEC-10,SEC-20

Related DRs: DR-12

Acceptance criteria: AC-097-AC-102

Open issues: None.

UC-26 - View Unofficial Transcript

Goal: Allow a student to view the student's own unofficial transcript.

Primary actor: Student

Supporting actors: System

Stakeholder interests: Student needs consolidated record; university needs clear unofficial status.

Preconditions: Student authenticated and access policy allows record view.

Trigger: Student requests unofficial transcript.

Main Success Scenario

1. Student opens transcript function.
2. The system verifies self-scope and archived access policy.
3. The system derives transcript contents from official academic record.
4. The system labels the output Unofficial and displays it to the student.

Alternative Flows

- A1: Graduate uses limited record access.

Exception / Validation Flows

- E1: Blocked archived account or other-student request is denied.

Postconditions: No official issuance record is created unless later requested through staff.

Success guarantee: Displayed transcript matches official academic history and is clearly Unofficial.

Minimal guarantee: No unauthorized record is disclosed.

Priority: Must

Frequency: Medium-High

Related FRs: FR-104

Related BRs: BR-53, BR-56

Related SEC: SEC-08-SEC-10, SEC-20

Related DRs: DR-12, DR-13

Acceptance criteria: AC-104

Open issues: Arabic/English variant is Should under bilingual delivery.

UC-27 - Issue Official Transcript

Goal: Issue, cancel, and reissue a controlled official transcript.

Primary actor: Registration Staff

Supporting actors: Registrar; Student recipient

Stakeholder interests: Student needs official evidence; university needs issuer/status/history control.

Preconditions: Staff authenticated and authorized; academic record available.

Trigger: Staff requests official transcript issuance or correction.

Main Success Scenario

1. Staff selects the student and transcript language.
2. The system verifies scope and derives contents from fixed academic record.
3. The system assigns unique number, issue time, issuer, and Valid status and labels the output Official.
4. Staff releases the official transcript through the approved process.
5. If incorrect, authorized staff cancels it with reason and issues a replacement while preserving both records.

Alternative Flows

- A1: Staff issues Arabic or English while academic symbols remain unchanged.

Exception / Validation Flows

- E1: Unauthorized issuance/cancellation is denied.

- E2: Attempt to delete an issued transcript is denied.
- E3: Data inconsistency blocks issuance.

Postconditions: Valid or Cancelled official issuance history is permanent and traceable.

Success guarantee: Official output corresponds exactly to academic record.

Minimal guarantee: No incorrect official artifact is silently replaced/deleted.

Priority: Must

Frequency: Medium

Related FRs: FR-105-FR-106

Related BRs: BR-53,BR-56,BR-65

Related SEC: SEC-11,SEC-14,SEC-17,SEC-20

Related DRs: DR-12,DR-13,DR-17

Acceptance criteria: AC-105,AC-106

Open issues: Electronic signature/QR remains future enhancement.

UC-30 - View Academic Report

Goal: Generate and view an authorized role-scoped report.

Primary actor: Authorized user

Supporting actors: System

Stakeholder interests: Operational/executive users need decision information; privacy requires minimization.

Preconditions: Requester authenticated; report and source data exist.

Trigger: Requester selects a report and criteria.

Main Success Scenario

1. Requester selects an approved report type and criteria.
2. The system determines the requester's role/data scope.
3. The system validates criteria and retrieves only permitted source data.
4. The system generates the in-system report and displays scope, generation time, and status.
5. Requester reviews results and may use delivered export path.

Alternative Flows

- A1: Student sees self report; instructor own sections; department own unit; management aggregate.
- A2: Closed-term snapshot is used for fixed historical reporting.

Exception / Validation Flows

- E1: Out-of-scope criteria are rejected or constrained.
- E2: Sensitive identities are minimized in aggregate report.
- E3: Generation failure returns safe error without partial official output.

Postconditions: No source data change; generated result is scoped and consistent.

Success guarantee: Report values reconcile with approved source records.

Minimal guarantee: No unauthorized data disclosure.

Priority: Must

Frequency: High

Related FRs: FR-110-FR-111,FR-121

Related BRs: BR-54

Related SEC: SEC-09,SEC-10,SEC-20,SEC-21

Related DRs: DR-16

Acceptance criteria: AC-110,AC-111,AC-121

Open issues: 3-second/10-second targets remain Proposed pending architecture validation.

UC-32 - Review Audit History

Goal: Review immutable audit/login evidence within authority.

Primary actor: Registrar / System Administrator / Department Head

Supporting actors: System

Stakeholder interests: University needs accountability; users need protected logs; technical/academic scopes differ.

Preconditions: Reviewer authenticated with relevant log-review authority.

Trigger: Reviewer searches audit or login history.

Main Success Scenario

1. Reviewer selects audit type and permitted filters.
2. The system verifies reviewer role and data scope.
3. The system returns immutable matching events with required fields.
4. Reviewer inspects details and may use delivered authorized export.
5. The system records access to sensitive logs where required by later design.

Alternative Flows

- A1: System Administrator reviews login log only under approved authority.
- A2: Registrar/Department Head reviews academic audit within scope.

Exception / Validation Flows

- E1: Ordinary user or out-of-scope reviewer is denied.
- E2: Modify/delete attempt is denied.
- E3: Export not delivered or unauthorized remains unavailable.

Postconditions: Logs remain unchanged.

Success guarantee: Reviewer obtains complete authorized evidence only.

Minimal guarantee: No log modification or unauthorized disclosure.

Priority: Must

Frequency: Medium

Related FRs: FR-118-FR-120

Related BRs: BR-55,BR-56

Related SEC: SEC-14-SEC-16,SEC-20,SEC-21

Related DRs: DR-17

Acceptance criteria: AC-118-AC-120

Open issues: Audit export is Should through CAP-29; in-system review is Must.

UC-36 - Search Authorized Records

Goal: Find relevant records within role/data scope.

Primary actor: Authorized user

Supporting actors: System

Stakeholder interests: Users need efficiency; privacy requires scope isolation.

Preconditions: Authenticated user; searchable records exist.

Trigger: User enters criteria/filters.

Main Success Scenario

1. User selects an authorized record type.
2. User enters search criteria or filters.
3. The system validates criteria and applies role/data scope.
4. The system returns only matching authorized records with safe empty/no-access behavior.

Alternative Flows

- A1: User refines/sorts/paginates as later UI design permits without changing requirement meaning.

Exception / Validation Flows

- E1: Invalid criteria gets clear corrective feedback.
- E2: Out-of-scope identifiers do not reveal protected record details.

Postconditions: No record changes.

Success guarantee: Results are relevant and authorized.

Minimal guarantee: No unauthorized disclosure.

Priority: Must

Frequency: High

Related FRs: FR-121

Related BRs: BR-03

Related SEC: SEC-09,SEC-10,SEC-13,SEC-20

Related DRs: All scoped data categories

Acceptance criteria: AC-121

Open issues: Exact search fields and layout are deferred to UI/UX design.

UC-37 - Manage Own Profile

Goal: Update permitted personal profile data without changing official academic identity.

Primary actor: Authorized user

Supporting actors: Registration Staff for official-data correction; moderator for optional image

Stakeholder interests: User needs current contact data; university protects official data.

Preconditions: User authenticated.

Trigger: User edits profile or password.

Main Success Scenario

1. User opens own profile.
2. The system distinguishes editable contact/address/image/password fields from staff-controlled official data.
3. User submits permitted changes.
4. The system validates format, password, and optional file rules.
5. The system saves permitted changes and records sensitive security changes.

Alternative Flows

- A1: User changes password after current-password validation.
- A2: Optional image is removed by authorized moderator.

Exception / Validation Flows

- E1: Attempt to change official ID/program/grade data is denied.
- E2: Invalid password/file/format is rejected.

Postconditions: Permitted profile data is updated; official data unchanged.

Success guarantee: Only owned permitted fields change.

Minimal guarantee: No unauthorized official-data change.

Priority: Must

Frequency: Medium

Related FRs: FR-04,FR-122-FR-123

Related BRs: BR-59,BR-62,BR-64

Related SEC: SEC-03,SEC-09,SEC-13,SEC-20,SEC-21

Related DRs: DR-01,DR-18,DR-20

Acceptance criteria: AC-004,AC-122,AC-123

Open issues: Profile image is Could and not a Release 1.0 blocker.

12. Acceptance Criteria

12.1 Rule

AC-001 through AC-140 map one-to-one to FR-01 through FR-140. Each criterion contains:

6. positive authorized scenario;
7. invalid-input scenario;
8. authorization scenario;
9. boundary condition;
10. linked business-rule violation;
11. expected response with no unintended change and audit where sensitive.

12.2 Complete Acceptance-Criteria Catalogue

Table 22 - Acceptance Criteria Catalogue

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-001	FR-01	Must	Given an authorized user context and valid prerequisites/inputs, when Authenticate User is performed, then the system shall authenticate each approved user using the account identifier assigned to that user and a password. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-002	FR-02	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Enforce Unique Accounts is performed, then the system shall maintain one unique account for each person and prevent duplicate active accounts for the same university or employee identity. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-003	FR-03	Must	Given an authorized System Administrator, All users context and valid prerequisites/inputs, when Issue Temporary Initial Password is performed, then the system shall issue a temporary password for a newly activated account and require its replacement at first successful sign-in. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-004	FR-04	Must	Given an authorized user context and valid prerequisites/inputs, when Change Password is performed, then the system shall allow an authenticated user to change the user's own password after validating the current credential and the approved password rules. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			create audit evidence where the operation is sensitive.
AC-005	FR-05	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Reset Password is performed, then the system shall allow an authorized administrator to reset a password after identity verification and shall require the user to replace the temporary reset password at the next sign-in. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-006	FR-06	Must	Given an authorized System Administrator, All users context and valid prerequisites/inputs, when Apply Failed-Login Lockout is performed, then the system shall lock an account for the approved period after the approved number of consecutive failed sign-in attempts and shall permit authorized manual unlock. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-007	FR-07	Must	Given an authorized user context and valid prerequisites/inputs, when End Inactive Sessions is performed, then the system shall end an authenticated session after the approved inactivity period and require re-authentication. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-008	FR-08	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Manage Account State is performed, then the system shall allow authorized administrators to set an account to Active, Locked, or Disabled and shall enforce the access effect of the selected state. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-009	FR-09	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Assign Multiple Roles is performed, then the system shall allow an authorized account to hold one or more approved roles without creating duplicate accounts. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-010	FR-10	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Control Role Changes is performed, then the system shall allow only authorized administrators to add or remove roles, require a reason, and record the change in the audit history. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-011	FR-11	Must	Given an authorized user context and valid prerequisites/inputs, when Enforce Role and Data Scope is performed, then the system shall authorize each requested operation according to the user's active roles and the approved data

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			scope for those roles. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-012	FR-12	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Separate Technical and Academic Authority is performed, then the system shall prevent technical administration roles from approving or modifying grades, GPA, registration exceptions, standing, graduation, or other academic decisions. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-013	FR-13	Should	Given an authorized Graduate context and valid prerequisites/inputs, when Provide Graduate Record Access is performed, then the system should provide a graduate with restricted access to the graduate's own records, transcript, notifications, and permitted profile data. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-014	FR-14	Must	Given an authorized System Administrator, Registration Staff context and valid prerequisites/inputs, when Apply Archived-Account Access Policy is performed, then the system shall apply read-only or blocked access to a withdrawn, inactive, or otherwise archived account according to the recorded administrative reason. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-015	FR-15	Must	Given an authorized Registrar, System Administrator context and valid prerequisites/inputs, when Manage Institution Display Information is performed, then the system shall allow authorized staff to maintain the configurable institution display name and general institutional information for the single-university installation. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-016	FR-16	Must	Given an authorized Registrar context and valid prerequisites/inputs, when Manage Colleges is performed, then the system shall allow authorized academic staff to create, update, activate, and deactivate college records within the single university. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-017	FR-17	Must	Given an authorized Registrar, Department Head context and valid prerequisites/inputs, when Manage Departments is performed, then the system shall allow authorized academic staff to create, update, activate, and deactivate departments and associate each department with one college. Invalid input, an unauthorized actor, an applicable

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-018	FR-18	Must	Given an authorized Registrar, Program Coordinator context and valid prerequisites/inputs, when Manage Academic Programs is performed, then the system shall allow authorized academic staff to create, update, activate, and deactivate academic programs and associate each program with one department. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-019	FR-19	Must	Given an authorized Registrar context and valid prerequisites/inputs, when Preserve Organization History is performed, then the system shall preserve historical references to academic organization records that are used by approved academic records even after those organization records are deactivated. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-020	FR-20	Must	Given an authorized Registrar, Department Head, Program Coordinator context and valid prerequisites/inputs, when Enforce Organization Authority is performed, then the system shall restrict academic organization and program-governance changes to the approved Registrar, Department Head, and Program Coordinator authority boundaries. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-021	FR-21	Must	Given an authorized Department Head, Program Coordinator context and valid prerequisites/inputs, when Manage Course Catalogue is performed, then the system shall allow authorized academic staff to create, update, activate, and deactivate course definitions containing the approved conceptual course information. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-022	FR-22	Must	Given an authorized Department Head, Program Coordinator context and valid prerequisites/inputs, when Classify Course Delivery Type is performed, then the system shall classify each course as Theoretical, Practical/Laboratory, or Mixed. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-023	FR-23	Must	Given an authorized Department Head, Program Coordinator context and valid prerequisites/inputs, when Validate Course Credits is performed, then the system shall accept only positive whole-number credit hours for a course. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-024	FR-24	Must	Given an authorized Program Coordinator, Department Head context and valid prerequisites/inputs, when Manage Prerequisites is performed, then the system shall allow authorized program staff to define and revise prerequisite relationships and the required minimum grade for each relationship. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-025	FR-25	Must	Given an authorized Program Coordinator, Department Head context and valid prerequisites/inputs, when Manage Co-requisites is performed, then the system shall allow authorized program staff to define and revise co-requisite relationships and the approved effect of withdrawing from a co-requisite. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-026	FR-26	Must	Given an authorized Program Coordinator, Department Head context and valid prerequisites/inputs, when Manage Study Plans is performed, then the system shall allow authorized program staff to define a study plan containing required and elective courses, levels, required credits, and applicable academic rules. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-027	FR-27	Must	Given an authorized Program Coordinator, Department Head context and valid prerequisites/inputs, when Version Study Plans is performed, then the system shall preserve approved historical study-plan versions and associate each student with the applicable plan version. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-028	FR-28	Must	Given an authorized Program Coordinator context and valid prerequisites/inputs, when Classify Plan Courses is performed, then the system shall allow authorized program staff to classify plan courses as required or elective and assign their program level and credit contribution. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-029	FR-29	Should	Given an authorized Student, Advisor context and valid prerequisites/inputs, when Identify Eligible Courses is performed, then the system should identify courses that a student is eligible to request based on the approved plan, completion history, prerequisite, co-requisite, status, and term rules. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-030	FR-30	Should	Given an authorized Student, Advisor, Program Coordinator, Registrar context and valid prerequisites/inputs, when Manage Program Transfer Request is performed, then the system should support the authorized processing of a student's transfer from one academic program to another. Invalid input, an unauthorized actor, an applicable

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-031	FR-31	Should	Given an authorized Program Coordinator, Registration Staff context and valid prerequisites/inputs, when Re-evaluate Plan after Transfer is performed, then the system should preserve the previous program history and determine which completed courses apply to the new approved study plan. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-032	FR-32	Must	Given an authorized Registrar, Registration Staff context and valid prerequisites/inputs, when Manage Academic Terms is performed, then the system shall allow authorized staff to create and maintain academic terms for First Semester, Second Semester, and Summer Semester using the approved code convention. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-033	FR-33	Must	Given an authorized Registrar, Registration Staff context and valid prerequisites/inputs, when Manage Term Dates and Windows is performed, then the system shall allow authorized staff to configure each term's start and end dates and the registration, add/drop, withdrawal, appeal, and grade deadlines. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-034	FR-34	Must	Given an authorized Registrar, Registration Staff context and valid prerequisites/inputs, when Manage Term State is performed, then the system shall allow authorized staff to control the approved lifecycle state of an academic term and shall permit term-dependent operations only in a compatible state. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-035	FR-35	Must	Given an authorized Department Head, Registration Staff context and valid prerequisites/inputs, when Create Course Offerings and Sections is performed, then the system shall allow authorized staff to create a course offering and one or more sections for an approved course and academic term. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-036	FR-36	Must	Given an authorized Registration Staff context and valid prerequisites/inputs, when Generate and Validate Section Identity is performed, then the system shall assign or validate a unique section identifier following the approved course-term-section convention. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-037	FR-37	Must	Given an authorized Department Head, Registration Staff context and valid prerequisites/inputs, when Manage Section Capacity and Status is performed, then the system shall allow authorized staff to maintain a section's capacity and operational status subject to approved room and registration constraints. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-038	FR-38	Must	Given an authorized Department Head context and valid prerequisites/inputs, when Manage Section Instructors is performed, then the system shall allow authorized department staff to assign one active Primary Instructor and zero or more active Co-Instructors to a section. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-039	FR-39	Must	Given an authorized Department Head, Registration Staff context and valid prerequisites/inputs, when Manage Section Meetings is performed, then the system shall allow authorized staff to maintain one or more weekly meeting schedules for each section. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-040	FR-40	Must	Given an authorized Registration Staff, Department Head context and valid prerequisites/inputs, when Manage Rooms and Laboratories is performed, then the system shall allow authorized staff to maintain buildings, rooms, and laboratories with their type, capacity, and active status. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-041	FR-41	Must	Given an authorized Registration Staff, Department Head context and valid prerequisites/inputs, when Assign Meeting Places is performed, then the system shall allow authorized staff to assign an active room or laboratory to each section meeting and validate compatibility with meeting type. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-042	FR-42	Must	Given an authorized Registration Staff, Department Head, Advisor, Student context and valid prerequisites/inputs, when Validate Schedule Conflicts is performed, then the system shall detect overlapping student, instructor, room, and laboratory meetings during section scheduling and registration. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-043	FR-43	Must	Given an authorized Registration Staff, Department Head context and valid prerequisites/inputs, when Validate Room Capacity is performed, then the system shall prevent a section capacity from exceeding the assigned room or laboratory capacity unless an approved exception is recorded. Invalid input, an unauthorized actor, an applicable

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-044	FR-44	Must	Given an authorized Department Head, Registration Staff context and valid prerequisites/inputs, when Cancel a Section is performed, then the system shall allow authorized staff to cancel a section with a reason and date, stop further registration, preserve history, and initiate the approved handling of affected registrations. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-045	FR-45	Must	Given an authorized Department Head, Registration Staff context and valid prerequisites/inputs, when Change Section Instructor or Meeting is performed, then the system shall allow an approved instructor reassignment or meeting time/place change, preserve old and new values, revalidate conflicts, and notify affected users according to priority. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-046	FR-46	Must	Given an authorized Registration Staff context and valid prerequisites/inputs, when Create Student Academic Record is performed, then the system shall allow authorized registration staff to create a student academic record after admission using the approved university identifier and required official academic information. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-047	FR-47	Must	Given an authorized Registration Staff, Advisor, Student context and valid prerequisites/inputs, when Maintain Student Academic Profile is performed, then the system shall maintain each student's program, plan version, level, admission date, advisor, status, earned credits, semester GPA, and cumulative GPA. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-048	FR-48	Must	Given an authorized Registrar, Registration Staff context and valid prerequisites/inputs, when Manage Student Status is performed, then the system shall allow authorized staff to change a student among Active, Suspended, On Leave, Graduated, and Withdrawn while preserving the effective date, reason, authority, and history. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-049	FR-49	Must	Given an authorized Registrar, Registration Staff context and valid prerequisites/inputs, when Manage Student Leave is performed, then the system shall record an approved leave period and reason, block registration during leave, and

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			support approved reactivation. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-050	FR-50	Must	Given an authorized Registrar, Registration Staff context and valid prerequisites/inputs, when Manage Student Suspension is performed, then the system shall record an approved suspension reason and duration, block registration during suspension, and support approved reactivation. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-051	FR-51	Must	Given an authorized System Administrator, Registration Staff context and valid prerequisites/inputs, when Archive Academic Accounts and Records is performed, then the system shall preserve graduate, withdrawn, inactive-student, and inactive-instructor records while preventing unauthorized new academic operations. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-052	FR-52	Must	Given an authorized System Administrator, Department Head context and valid prerequisites/inputs, when Create and Maintain Instructor Record is performed, then the system shall allow authorized staff to create and maintain an instructor record using the approved employee identifier and official assignment information. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-053	FR-53	Must	Given an authorized System Administrator, Department Head context and valid prerequisites/inputs, when Manage Instructor Status is performed, then the system shall allow authorized staff to set an instructor to Active, On Leave, or Inactive and preserve status history. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-054	FR-54	Must	Given an authorized Department Head context and valid prerequisites/inputs, when Prevent Assignment of Non-Active Instructors is performed, then the system shall prevent a new section assignment to an instructor whose status is not Active. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-055	FR-55	Must	Given an authorized Student context and valid prerequisites/inputs, when Start Registration Request is performed, then the system shall allow an Active student to select offered sections and submit a registration request during an open registration window. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-056	FR-56	Must	Given an authorized Student,Advisor,Registration Staff context and valid prerequisites/inputs, when Validate Registration Preconditions is performed, then the system shall validate term state, student status, registration holds, section status, and other approved preconditions before accepting a registration request. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-057	FR-57	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Validate Prerequisites is performed, then the system shall reject registration when the student has not completed each prerequisite with at least the approved minimum grade and shall not permit a prerequisite override. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-058	FR-58	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Validate Co-requisites is performed, then the system shall validate each co-requisite relationship during registration and during withdrawal from a related course. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-059	FR-59	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Validate Registration Conflicts is performed, then the system shall reject a registration request that creates an unapproved schedule conflict for the student. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-060	FR-60	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Validate Section Capacity is performed, then the system shall reject direct registration in a full section and apply the approved waitlist behavior when the waitlist capability is available. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-061	FR-61	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Validate Credit Load is performed, then the system shall calculate the requested term load and enforce the approved regular, high-GPA, warning, probation, summer, and final-term limits. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-062	FR-62	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Require Advisor Approval is performed, then the system shall place a valid student registration request in Pending Advisor Approval and shall not confirm it until the assigned advisor approves it. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-063	FR-63	Must	Given an authorized Advisor context and valid prerequisites/inputs, when Review Registration Request is performed, then the system shall allow the assigned advisor to approve, reject, or return a pending registration request with a recorded reason and advice. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-064	FR-64	Must	Given an authorized Advisor,Department Head context and valid prerequisites/inputs, when Escalate Delayed Advising is performed, then the system shall identify a registration request that remains pending for two working days and make it available for approved Department Head escalation. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-065	FR-65	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Confirm Registration is performed, then the system shall convert an approved and still-valid request into Registered enrollment atomically and revalidate seat, conflict, load, status, and hold conditions at confirmation time. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-066	FR-66	Must	Given an authorized Registration Staff context and valid prerequisites/inputs, when Support Authorized Registration Intervention is performed, then the system shall allow authorized registration staff to perform or correct registration operations within their authority and shall record the actor, reason, and result. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-067	FR-67	Must	Given an authorized Advisor,Department Head,Registrar context and valid prerequisites/inputs, when Process Registration Exceptions is performed, then the system shall route a permitted registration exception through advisor recommendation and Department Head or Registrar approval according to scope, while prohibiting prerequisite override. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-068	FR-68	Must	Given an authorized Registrar,Registration Staff context and valid prerequisites/inputs, when Manage Registration Holds is performed, then the system shall allow authorized staff to apply, review, and release Academic, Disciplinary,

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			Missing Documents, and Administrative holds and shall block registration while an applicable hold is active. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-069	FR-69	Must	Given an authorized Student,Registration Staff context and valid prerequisites/inputs, when Add or Drop during Window is performed, then the system shall allow an eligible student or authorized staff member to add or drop a course during the configured add/drop window without recording W. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-070	FR-70	Must	Given an authorized Student,Advisor,Registration Staff context and valid prerequisites/inputs, when Withdraw after Add/Drop is performed, then the system shall record W when an authorized withdrawal is completed after add/drop and before the configured withdrawal deadline, without GPA or earned-credit effect. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-071	FR-71	Must	Given an authorized Advisor,Department Head,Registrar,Registration Staff context and valid prerequisites/inputs, when Process Late Registration Action is performed, then the system shall require the approved exception workflow for an add, drop, or withdrawal requested after the applicable deadline. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-072	FR-72	Must	Given an authorized Student,Advisor context and valid prerequisites/inputs, when Enforce Repeat and Parallel-Attempt Rules is performed, then the system shall enforce the approved repeat-attempt limit, prevent parallel active attempts, and prevent enrollment in more than one section of the same course in one term. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-073	FR-73	Should	Given an authorized Student context and valid prerequisites/inputs, when Join Section Waitlist is performed, then the system should allow an otherwise eligible student to join a full section's waitlist in first-come-first-served order. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-074	FR-74	Should	Given an authorized Student,Advisor,Registration Staff context and valid prerequisites/inputs, when Maintain Waitlist Eligibility and Order is performed, then the system should preserve the approved waitlist ordering, permit temporary load excess while waiting, and reassess eligibility when a seat becomes available. Invalid input, an unauthorized actor,

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-075	FR-75	Could	Given an authorized Student,Registration Staff context and valid prerequisites/inputs, when Automatically Promote Waitlisted Student is performed, then the system may atomically register the first eligible waitlisted student when a seat becomes available and stop automatic promotion after add/drop. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-076	FR-76	Must	Given an authorized Instructor context and valid prerequisites/inputs, when Create Attendance Session is performed, then the system shall allow an assigned instructor to create or confirm each scheduled attendance session for the instructor's section. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-077	FR-77	Must	Given an authorized Instructor context and valid prerequisites/inputs, when Record Attendance Status is performed, then the system shall allow an assigned instructor to record Present, Absent, Late, or Excused for each enrolled student in a non-cancelled session. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-078	FR-78	Must	Given an authorized Instructor,Student,Department Head context and valid prerequisites/inputs, when Calculate Attendance Percentage is performed, then the system shall calculate attendance percentage using counted sessions, excluding Cancelled sessions and approved Excused absences from effective absence as defined by policy. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-079	FR-79	Must	Given an authorized Instructor,Student context and valid prerequisites/inputs, when Convert Late Records is performed, then the system shall treat every three Late records as one absence for attendance-rule evaluation. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-080	FR-80	Must	Given an authorized Instructor,Department Head,Student context and valid prerequisites/inputs, when Apply Laboratory Attendance Rules is performed, then the system shall separately evaluate the approved unexcused-absence and makeup requirements for Practical/Laboratory and Mixed course laboratory meetings. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			sensitive.
AC-081	FR-81	Must	Given an authorized Instructor,Student,Department Head context and valid prerequisites/inputs, when Issue Attendance Warning is performed, then the system shall identify an approaching or breached attendance threshold and present the approved warning before an Attendance Barred decision. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-082	FR-82	Must	Given an authorized Department Head,Registrar context and valid prerequisites/inputs, when Record Attendance Bar is performed, then the system shall allow an authorized decision to set Attendance Barred after a threshold breach and shall preserve the reason and approval history. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-083	FR-83	Must	Given an authorized Instructor,Department Head context and valid prerequisites/inputs, when Control Attendance Edits is performed, then the system shall allow an assigned instructor to edit attendance within seven days and shall require reason, Department Head approval, and audit for a later change. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-084	FR-84	Must	Given an authorized Instructor,Department Head context and valid prerequisites/inputs, when Cancel Attendance Session is performed, then the system shall retain a cancelled session as Cancelled, exclude it from attendance calculation, record a reason, and notify affected students when notifications are delivered. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-085	FR-85	Must	Given an authorized Instructor,Department Head,Student context and valid prerequisites/inputs, when Process Makeup Laboratory is performed, then the system shall support instructor review, Department Head approval, and instructor recording of the approved makeup-laboratory outcome. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-086	FR-86	Must	Given an authorized Instructor,Department Head context and valid prerequisites/inputs, when Define Assessment Scheme is performed, then the system shall allow an assigned instructor to define section assessments and weights subject to the approved default or an approved alternative totaling 100 percent. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-087	FR-87	Must	Given an authorized Instructor context and valid prerequisites/inputs, when Enter Assessment Scores is performed,

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			then the system shall allow an assigned instructor to enter and revise draft assessment scores with up to two decimal places before grade submission. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-088	FR-88	Must	Given an authorized Instructor, Student context and valid prerequisites/inputs, when Calculate Raw Course Score is performed, then the system shall calculate each completed student's raw course score from the approved assessment weights using the exact stored assessment values. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-089	FR-89	Must	Given an authorized Instructor, Department Head context and valid prerequisites/inputs, when Select Grading Model is performed, then the system shall apply relative grading when the approved population and variation conditions are satisfied and otherwise apply the approved absolute fallback. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-090	FR-90	Must	Given an authorized Instructor, Department Head context and valid prerequisites/inputs, when Calculate Relative Letter Grade is performed, then the system shall calculate Z and T scores from the approved section population and map the T score to the approved letter-grade thresholds. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-091	FR-91	Must	Given an authorized Instructor, Department Head context and valid prerequisites/inputs, when Apply Absolute Letter Grade is performed, then the system shall map the raw score to the approved absolute letter-grade thresholds when the fallback conditions apply. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-092	FR-92	Must	Given an authorized Instructor, Department Head context and valid prerequisites/inputs, when Enforce Fail Floor is performed, then the system shall assign F to a raw score below 45 regardless of the relative result while retaining that numeric score in the approved section statistics. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-093	FR-93	Must	Given an authorized Instructor context and valid prerequisites/inputs, when Submit Section Grades is performed, then the system shall allow the assigned instructor to submit a complete section grade set for department review and shall prevent publication while required results are incomplete. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			shall return clear feedback and create audit evidence where the operation is sensitive.
AC-094	FR-94	Must	Given an authorized Department Head context and valid prerequisites/inputs, when Approve and Publish Section Grades is performed, then the system shall allow the Department Head to approve the complete section result set, publish all results together, and lock the published grades. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-095	FR-95	Must	Given an authorized Instructor, Department Head, Registrar context and valid prerequisites/inputs, when Process Grade Correction is performed, then the system shall process a grade correction through instructor request, Department Head review, Registrar approval, audit history, and automatic recalculation of affected academic results. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-096	FR-96	Should	Given an authorized Student, Instructor, Department Head, Registrar context and valid prerequisites/inputs, when Process Grade Appeal is performed, then the system should allow a student to submit one appeal within seven days of publication and route it through Instructor review, Department Head recommendation, and Registrar final decision. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-097	FR-97	Must	Given an authorized Instructor, Registrar, Registration Staff context and valid prerequisites/inputs, when Manage W, I, and Pass/Fail Results is performed, then the system shall record and apply the approved GPA, credit, completion, and deadline effects of W, I, P, and F results. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-098	FR-98	Must	Given an authorized Student, Advisor, Registration Staff context and valid prerequisites/inputs, when Apply Course Repeat to GPA is performed, then the system shall use the last completed attempt for GPA, retain every attempt, and award earned credit only once according to the approved repeat policy. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-099	FR-99	Must	Given an authorized Student, Registration Staff context and valid prerequisites/inputs, when Calculate Semester and Cumulative GPA is performed, then the system shall calculate credit-weighted semester and cumulative GPA on the 4.00 scale and truncate the result to two decimal places without rounding. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-100	FR-100	Must	Given an authorized Registrar,Registration Staff,Advisor,Student context and valid prerequisites/inputs, when Determine Academic Standing is performed, then the system shall determine Good Standing, Academic Warning, or Academic Probation from the official truncated cumulative GPA after semester closure. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-101	FR-101	Must	Given an authorized Registrar,Registration Staff context and valid prerequisites/inputs, when Recommend Possible Dismissal is performed, then the system shall identify two consecutive Probation terms for human review and shall not dismiss a student automatically. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-102	FR-102	Must	Given an authorized Registrar,Registration Staff,Student context and valid prerequisites/inputs, when Maintain Complete Academic Record is performed, then the system shall preserve all attempts, result states, credits, GPA values, standing, eligibility, and approved changes as a fixed academic history. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-103	FR-103	Must	Given an authorized Registrar,Registration Staff,Department Head context and valid prerequisites/inputs, when Close Academic Term is performed, then the system shall finalize approved grades, GPA, standing, attendance, progress, and required report snapshots and lock closed-term records. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-104	FR-104	Must	Given an authorized Student context and valid prerequisites/inputs, when Provide Unofficial Transcript is performed, then the system shall allow a student to view the student's own unofficial transcript containing the approved academic record information. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-105	FR-105	Must	Given an authorized Registration Staff context and valid prerequisites/inputs, when Issue Official Transcript is performed, then the system shall allow authorized registration staff to issue an official transcript with a unique number, issue time, issuer, language, and Valid status. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-106	FR-106	Must	Given an authorized Registration Staff,Registrar context and valid prerequisites/inputs, when Cancel and Reissue Official Transcript is performed, then the system shall allow authorized staff to cancel an incorrect official transcript without deletion and issue a corrected replacement while preserving both records. Invalid input, an unauthorized

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-107	FR-107	Should	Given an authorized Student,Advisor,Program Coordinator context and valid prerequisites/inputs, when Provide Degree Progress is performed, then the system should show required and earned credits, completed and remaining courses, completion percentage, and graduation readiness under the applicable study plan. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-108	FR-108	Should	Given an authorized Registration Staff,Program Coordinator,Student context and valid prerequisites/inputs, when Evaluate Graduation Eligibility is performed, then the system should evaluate the approved program-completion, GPA, result-state, and minimum-grade conditions and assign the appropriate graduation-eligibility status. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-109	FR-109	Should	Given an authorized Registration Staff,Program Coordinator,Registrar context and valid prerequisites/inputs, when Process Graduation Approval is performed, then the system should support staff review, Program Coordinator confirmation, Registrar approval, and creation of a fixed graduation record. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-110	FR-110	Must	Given an authorized user context and valid prerequisites/inputs, when Provide Academic Reports is performed, then the system shall provide the approved performance, registration, enrollment, attendance-risk, grade-distribution, GPA/standing, graduation, teaching-load, and program/department reports within the user's scope. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-111	FR-111	Must	Given an authorized user context and valid prerequisites/inputs, when Apply Report Scope is performed, then the system shall automatically restrict each report to the approved self, section, department, program, or institution aggregate scope of the requesting user. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-112	FR-112	Should	Given an authorized user context and valid prerequisites/inputs, when View and Export Reports is performed, then the system should allow authorized users to view reports in the system and export permitted outputs as PDF, Excel, or CSV with the approved metadata and data minimization. Invalid input, an unauthorized actor, an applicable boundary

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-113	FR-113	Should	Given an authorized direct-user context and valid prerequisites/inputs, when Provide Role Dashboards is performed, then the system should provide each major role with a dashboard summarizing permitted tasks, alerts, and academic indicators. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-114	FR-114	Could	Given an authorized University Management context and valid prerequisites/inputs, when Provide Executive Read-Only Analytics is performed, then the system may provide University Management with read-only aggregated dashboards and reports without record-modification actions. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-115	FR-115	Should	Given an authorized operational-user context and valid prerequisites/inputs, when Deliver In-App Notifications is performed, then the system should deliver in-app notifications with recipient scope, severity, time, read/unread state, and a permitted related-action reference. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-116	FR-116	Should	Given an authorized Registrar,Registration Staff,Department Head,Program Coordinator,Instructor context and valid prerequisites/inputs, when Publish Targeted Announcements is performed, then the system should allow authorized academic staff to publish announcements to an approved university, college, department, program, section, or role audience. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-117	FR-117	Should	Given an authorized Authorized sender,Recipient context and valid prerequisites/inputs, when Manage Urgent Notification Accountability is performed, then the system should display urgent notifications prominently, record the read time, and allow the authorized sender to view the recipient read count. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-118	FR-118	Must	Given an authorized System,Authorized reviewer context and valid prerequisites/inputs, when Record Audit Events is performed, then the system shall create an immutable audit event for each approved sensitive operation containing actor, action, time, affected record, old and new values, reason, and result. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-119	FR-119	Must	Given an authorized Registrar, System Administrator, Department Head context and valid prerequisites/inputs, when Review Audit History is performed, then the system shall allow authorized reviewers to search and filter audit history without modifying or deleting it. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-120	FR-120	Must	Given an authorized System Administrator context and valid prerequisites/inputs, when Record Login Activity is performed, then the system shall record login, logout, result, account, and lock reason and restrict login-log review to the System Administrator. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-121	FR-121	Must	Given an authorized direct-user context and valid prerequisites/inputs, when Search and Filter Authorized Data is performed, then the system shall provide search and filtering over relevant records while enforcing the requester's authorization and data scope. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-122	FR-122	Must	Given an authorized All users, Registration Staff context and valid prerequisites/inputs, when Manage User Profile is performed, then the system shall allow a user to update permitted contact, address, password, and profile information while reserving official identity and academic data for authorized staff. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-123	FR-123	Could	Given an authorized All users, Registration Staff context and valid prerequisites/inputs, when Manage Profile Image is performed, then the system may allow an optional profile image and permit authorized moderation or removal. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-124	FR-124	Should	Given an authorized Registration Staff, System Administrator, Program Coordinator context and valid prerequisites/inputs, when Bulk Import Students and Courses is performed, then the system should allow authorized staff to import student and course data from CSV or spreadsheet files. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-125	FR-125	Could	Given an authorized Registration Staff, System Administrator, Program Coordinator context and valid prerequisites/inputs, when Validate and Partially Apply Import is performed, then the system may preview an import, detect duplicates and row errors, apply only valid rows when authorized, and produce a final result report. Invalid

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-126	FR-126	Must	Given an authorized staff context and valid prerequisites/inputs, when Validate Controlled Identifiers is performed, then the system shall enforce uniqueness of University ID, Employee ID, Course Code, Semester Code, and section identifier within the approved scope. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-127	FR-127	Should	Given an authorized Student, Instructor, Advisor, Authorized reviewers context and valid prerequisites/inputs, when Manage Supporting Attachments is performed, then the system should allow approved PDF, JPG, JPEG, or PNG attachments of at most 5 MB for appeals, excused absences, transfers, and exceptions. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-128	FR-128	Should	Given an authorized user context and valid prerequisites/inputs, when Protect and Retain Attachments is performed, then the system should restrict an attachment to its owner and authorized reviewers and retain it with the associated record. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-129	FR-129	Could	Given an authorized System Administrator context and valid prerequisites/inputs, when Create Backup is performed, then the system may allow the System Administrator to create daily automatic and authorized manual backups with configurable retention defaulting to 30 days. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-130	FR-130	Could	Given an authorized System Administrator context and valid prerequisites/inputs, when Restore Backup is performed, then the system may allow only the System Administrator to restore an approved backup after confirmation and shall record the result and verify post-restore integrity. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-131	FR-131	Must	Given an authorized All modifying users context and valid prerequisites/inputs, when Reject Stale Concurrent Update is performed, then the system shall reject a save based on stale data and require the user to refresh and retry rather than silently overwriting a newer change. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
AC-132	FR-132	Must	Given an authorized System context and valid prerequisites/inputs, when Apply Atomic Sensitive Operations is performed, then the system shall roll back the complete sensitive operation when any required step fails, leaving no partial academic change. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-133	FR-133	Must	Given an authorized System Administrator, All users context and valid prerequisites/inputs, when Control Maintenance State is performed, then the system shall announce a maintenance window, prevent new operations during the window, and preserve already saved data. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-134	FR-134	Must	Given an authorized user context and valid prerequisites/inputs, when Provide Clear Error Feedback is performed, then the system shall present clear, localized, non-technical error messages to users and keep technical details in authorized internal records. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-135	FR-135	Should	Given an authorized user context and valid prerequisites/inputs, when Support Arabic and English is performed, then the system should support Arabic and English user-facing content. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-136	FR-136	Should	Given an authorized user context and valid prerequisites/inputs, when Support RTL and LTR Direction is performed, then the system should use right-to-left direction for Arabic and left-to-right direction for English. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-137	FR-137	Must	Given an authorized user context and valid prerequisites/inputs, when Support Responsive Web Access is performed, then the system shall provide connected web access on supported desktop, tablet, and mobile browser sizes without requiring a native application. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-138	FR-138	Must	Given an authorized user context and valid prerequisites/inputs, when Support Approved Browsers is performed, then the system shall support current mainstream versions of Chrome, Edge, Firefox, and Safari, with release verification prioritizing Chrome and Edge. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear

AC ID	Requirement	Priority	Given / When / Then and Negative Conditions
			feedback and create audit evidence where the operation is sensitive.
AC-139	FR-139	Should	Given an authorized Advisor, Student context and valid prerequisites/inputs, when Maintain Academic Advising Notes is performed, then the system should allow an assigned advisor to maintain private advising notes and, when explicitly chosen, publish a separate student-facing summary. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.
AC-140	FR-140	Could	Given an authorized Registrar, Department Head, Registration Staff context and valid prerequisites/inputs, when Delegate Approval Authority is performed, then the system may allow an authorized approver to delegate specified approval operations for a limited period and scope while preserving accountability. Invalid input, an unauthorized actor, an applicable boundary violation, or a linked business-rule violation shall be rejected with no unintended data change; the system shall return clear feedback and create audit evidence where the operation is sensitive.

13. Requirements Prioritization

Table 23 - Requirements Prioritization - Catalogue

Catalogue	Total	Must	Should	Could	Proposed/Pending
Functional Requirements	140	112	21	7	0
Non-Functional Requirements	25	19	2	1	3
Business Rules	66	Derived from linked Must rules	Conditional rules retained	Optional rules retained	0
Security Requirements	24	21	1	2	0
Data Requirements	21	Priority inherited from linked requirements	-	-	0
Interface Requirements	16	11	4	Included in one Should/Could import requirement	0
Use Cases	43	28	10	3	2 mixed-priority catalog entries
Acceptance Criteria	140	112	21	7	0

Priority inherits from the highest governing CAP and approved policy. A requirement that supports both a Must and a conditional capability is mandatory only for the Must behavior; conditional extensions remain non-blocking.

14. Requirements Traceability

14.A OCR --> CAP -> Requirement

Table 24 - Traceability Matrix - 14.A OCR -> CAP -> Requirement

OCR	Official Obligation	CAP / Scope Effect	SRS Coverage
OCR-01	Project selection	CAP catalogue as selected SUMS scope	Document identity; Section 16
OCR-02	SUMS selected	CAP-01-CAP-70	All product requirements
OCR-03	Seven official areas	CAP-01-CAP-70	FR-15-FR-121; UC-03-UC-36
OCR-04	Project cannot change	Scope control	Section 16; Appendix G
OCR-05	Team of 3-4	Project constraint	NFR-25; ASM-11
OCR-06	Equal contribution	Project constraint	NFR-25; Section 16
OCR-07	Analyze requirements	All capabilities	Entire SRS
OCR-08	Apply Software Engineering	All capabilities	NFR-21,NFR-22; Sections 14,17
OCR-09	Follow SDLC	Phase gate	Section 16; Appendix G
OCR-10	Professional UML	Deferred design input	Section 16; Appendix G
OCR-11	Use OOP	Implementation constraint	Section 16
OCR-12	Relational database	Implementation constraint	Section 16; DR-01-DR-21 conceptual input
OCR-13	Complete application	Must CAPs	All Must FRs; UC-01-UC-43 as prioritized
OCR-14	User-friendly interface	CAP-38,39,41-43	NFR-01-NFR-06; IR-01-IR-15
OCR-15	Effective teamwork	Project constraint	NFR-21,NFR-25; RSK-02,04,15
OCR-16	Professional documentation	Project constraint	NFR-21-NFR-23
OCR-17	Present/demonstrate	Must scope + demo data	ASM-01,ASM-10; Section 16
OCR-18	Original work	Project constraint	NFR-25; ASM-02

OCR	Official Obligation	CAP / Scope Effect	SRS Coverage
OCR-19	No copying	Project constraint	NFR-25
OCR-20	Members explain work	Project constraint	NFR-21,NFR-25
OCR-21	GitHub encouraged	Project management	Section 16; DEC-20
OCR-22	Submit source code	Final deliverable	Section 16
OCR-23	Submit PDF report	Final deliverable	NFR-23; Section 16
OCR-24	Submit SQL file	Final deliverable	Section 16; later database phase
OCR-25	Submit slides	Final deliverable	Section 16
OCR-26	Video if requested	Resolved as conditional/optional	ASM-01; Sections 15 and 16
OCR-27	Repository link if available	Project management	Section 16
OCR-28	One ZIP upload	Final delivery	Section 16
OCR-29	Deadline 30-07-2026	Project constraint	ASM-12; Section 15/16

14.B CAP -> FR / UC Coverage

Table 25 - Traceability Matrix - 14.B CAP -> FR / UC Coverage

CAP	Capability	Priority	Release Treatment	FR Coverage	UC Coverage	Approved Decision
CAP-01	Academic Organization Management	Must	Realized - Release 1.0 committed	FR-15, FR-16, FR-17, FR-18, FR-19	UC-03	DEC-45
CAP-02	Curriculum and Study Plan Management	Must	Realized - Release 1.0 committed	FR-24, FR-25, FR-26, FR-27, FR-28	UC-04, UC-05	DEC-46
CAP-03	Academic Term Management	Must	Realized - Release 1.0 committed	FR-32, FR-33, FR-34, FR-126	UC-06	DEC-47, 138-139
CAP-04	Course Offering and Section Management	Must	Realized - Release 1.0 committed	FR-35, FR-36, FR-37, FR-38, FR-39, FR-43, FR-45, FR-126	UC-07	DEC-48
CAP-05	Schedule Conflict Validation	Must	Realized - Release 1.0 committed	FR-42, FR-59	UC-07	DEC-48, 50
CAP-06	Student Registration and Advising	Must	Realized - Release 1.0 committed	FR-55, FR-56, FR-62, FR-63, FR-64, FR-65, FR-66, FR-132	UC-11, UC-12, UC-13	DEC-49
CAP-07	Registration Rules and Exceptions	Must	Realized - Release 1.0 committed	FR-56, FR-57, FR-58, FR-59, FR-60, FR-61, FR-65, FR-67, FR-71, FR-127	UC-11, UC-13	DEC-41, 50, 109, 128
CAP-08	Section Waitlist Management	Should	Deferred conditional - after Must stability	FR-60, FR-73, FR-74	UC-16	DEC-51, 69, 141, 150-151
CAP-09	Attendance Management	Must	Realized - Release 1.0 committed	FR-76, FR-77, FR-78, FR-83, FR-84	UC-18	DEC-52
CAP-10	Assessment and Grade Workflow	Must	Realized - Release 1.0 committed	FR-86, FR-87, FR-88, FR-89, FR-90, FR-91, FR-92, FR-93, FR-94, FR-97, FR-132	UC-20, UC-21	DEC-53
CAP-11	In-App Notification Management	Should	Deferred conditional - after Must stability	FR-115, FR-117	UC-34	DEC-24,54,85
CAP-12	Student Lifecycle Management	Must	Realized - Release 1.0 committed	FR-46, FR-48	UC-09	DEC-55
CAP-13	Student Academic Profile	Must	Realized - Release 1.0 committed	FR-46, FR-47	UC-09	DEC-56

CAP	Capability	Priority	Release Treatment	FR Coverage	UC Coverage	Approved Decision
CAP-14	Course Catalog Management	Must	Realized - Release 1.0 committed	FR-21, FR-23, FR-24, FR-126	UC-04	DEC-57
CAP-15	Instructor Status and Assignment Control	Must	Realized - Release 1.0 committed	FR-38, FR-52, FR-53, FR-54	UC-07, UC-10	DEC-58
CAP-16	Attendance Policy Management	Must	Realized - Release 1.0 committed	FR-78, FR-79, FR-80, FR-81, FR-82, FR-84	UC-18	DEC-59-61,76,143,152-153
CAP-17	Academic Record Management	Must	Realized - Release 1.0 committed	FR-47, FR-97, FR-98, FR-99, FR-102	UC-25	DEC-62
CAP-18	Transcript Management	Must	Realized - Release 1.0 committed	FR-104, FR-105, FR-106	UC-26, UC-27	DEC-63, 212-214
CAP-19	Academic Reporting	Must	Realized - Release 1.0 committed	FR-110, FR-111	UC-30	DEC-64
CAP-20	Audit and Accountability	Must	Realized - Release 1.0 committed	FR-10, FR-45, FR-66, FR-83, FR-95, FR-106, FR-118, FR-119, FR-120, FR-128, FR-131, FR-132	UC-22, UC-32	DEC-65
CAP-21	Academic Standing Management	Must	Realized - Release 1.0 committed	FR-99, FR-100, FR-101	UC-24, UC-25	DEC-66,217-218
CAP-22	Restricted Registration Management	Must	Realized - Release 1.0 committed	FR-61	UC-11, UC-25	DEC-67,81,219
CAP-23	Add, Drop and Withdrawal Management	Must	Realized - Release 1.0 committed	FR-33, FR-69, FR-70, FR-71	UC-14	DEC-68,139
CAP-24	Automatic Waitlist Enrollment	Could	Optional - not Release 1.0 blocker	FR-75	UC-17	DEC-69,132,141,188,190
CAP-25	Grade Correction Management	Must	Realized - Release 1.0 committed	FR-95	UC-22	DEC-70,211
CAP-26	Grade Appeal Management	Should	Deferred conditional - after Must stability	FR-96, FR-127	UC-23	DEC-71,82-83,220-221
CAP-27	User Account and Access Management	Must	Realized - Release 1.0 committed	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-10, FR-11, FR-120, FR-126	UC-01, UC-02	DEC-72,77,80,157-164
CAP-28	Multi-Role Authorization	Must	Realized - Release 1.0 committed	FR-09, FR-10	UC-02	DEC-74
CAP-29	Report Viewing and Export	Should	Deferred conditional - after Must stability	FR-112	UC-31	DEC-75,166,186
CAP-30	Laboratory Attendance Exceptions	Must	Realized - Release 1.0 committed	FR-80, FR-85, FR-127	UC-19	DEC-76,84,143
CAP-31	Summer Semester Management	Must	Realized - Release 1.0 committed	FR-32, FR-61	UC-06	DEC-87,138,219
CAP-32	Co-requisite Validation	Must	Realized - Release 1.0 committed	FR-25, FR-58	UC-04, UC-11	DEC-88,200
CAP-33	Course Delivery Type Management	Must	Realized - Release 1.0 committed	FR-22	UC-04	DEC-89
CAP-34	Course Repeat and Improvement	Must	Realized - Release 1.0 committed	FR-72, FR-98	UC-15	DEC-30,90,197-199
CAP-35	Degree Progress and Audit	Should	Deferred conditional - after Must stability	FR-107, FR-108, FR-109	UC-28, UC-29	DEC-91
CAP-36	Eligible Course Identification	Should	Deferred conditional - after Must stability	FR-29	UC-28	DEC-92
CAP-37	Semester Closure	Must	Realized - Release 1.0 committed	FR-103	UC-06, UC-24	DEC-93,215
CAP-38	Role-Based Dashboards	Should	Deferred conditional - after Must stability	FR-113	UC-33	DEC-94
CAP-39	Authorized Search and Filtering	Must	Realized - Release 1.0 committed	FR-121, FR-134	UC-30, UC-36	DEC-95
CAP-40	Student and Course Bulk Import	Should	Deferred conditional - after Must stability	FR-124	UC-40	DEC-96,116,171
CAP-41	Multilingual System Support	Should	Deferred conditional - after Must stability	FR-134, FR-135	Non-interactive/supporting	DEC-97
CAP-42	Bidirectional Interface Support	Should	Deferred conditional - after Must stability	FR-136	Non-interactive/supporting	DEC-98
CAP-43	User Profile Management	Must	Realized - Release 1.0 committed	FR-04, FR-122	UC-37	DEC-99
CAP-44	Profile Image Management	Could	Optional - not Release 1.0 blocker	FR-123	UC-37	DEC-100
CAP-45	Academic Program Transfer	Should	Deferred conditional - after Must stability	FR-30, FR-127	UC-38	DEC-101

CAP	Capability	Priority	Release Treatment	FR Coverage	UC Coverage	Approved Decision
CAP-46	Transfer History and Plan Re-evaluation	Should	Deferred conditional - after Must stability	FR-31	UC-38	DEC-102
CAP-47	Student Leave Management	Must	Realized - Release 1.0 committed	FR-49	UC-09	DEC-103
CAP-48	Student Suspension Management	Must	Realized - Release 1.0 committed	FR-50	UC-09	DEC-104
CAP-49	Account and Record Archiving	Must	Realized - Release 1.0 committed	FR-08, FR-14, FR-19, FR-51, FR-102	UC-09, UC-43	DEC-105,125-126
CAP-50	Backup and Restore	Could	Optional - not Release 1.0 blocker	FR-129, FR-130	UC-41	DEC-106,167-170
CAP-51	Controlled System Access	Must	Realized - Release 1.0 committed	FR-01, FR-11, FR-133	UC-01	DEC-107
CAP-52	Employee Identity and Login	Must	Realized - Release 1.0 committed	FR-01, FR-02, FR-52, FR-126	UC-01, UC-10	DEC-108,225
CAP-53	Registration Hold Management	Must	Realized - Release 1.0 committed	FR-56, FR-68	UC-11	DEC-109
CAP-54	Course Section Cancellation	Must	Realized - Release 1.0 committed	FR-44	UC-08	DEC-110,201
CAP-55	Instructor Reassignment	Must	Realized - Release 1.0 committed	FR-45	UC-08	DEC-111
CAP-56	Room and Laboratory Scheduling	Must	Realized - Release 1.0 committed	FR-40, FR-41, FR-42, FR-43	UC-07	DEC-112,140
CAP-57	Academic Advising Notes	Should	Deferred conditional - after Must stability	FR-139	UC-39	DEC-113
CAP-58	Academic Announcements	Should	Deferred conditional - after Must stability	FR-116	UC-35	DEC-114
CAP-59	Temporary Approval Delegation	Could	Optional - not Release 1.0 blocker	FR-140	UC-42	DEC-115,133
CAP-60	Validated Bulk Import	Could	Optional - not Release 1.0 blocker	FR-125	UC-40	DEC-116,171
CAP-61	Operational Student Record Administration	Must	Realized - Release 1.0 committed	FR-46, FR-66, FR-105	UC-09, UC-13, UC-27	DEC-117
CAP-62	Technical and Academic Separation of Duties	Must	Realized - Release 1.0 committed	FR-11, FR-12, FR-111	UC-02	DEC-118
CAP-63	Academic Program Governance	Must	Realized - Release 1.0 committed	FR-18, FR-20, FR-26, FR-27	UC-03, UC-05	DEC-119,129
CAP-64	Department Academic Oversight	Must	Realized - Release 1.0 committed	FR-20, FR-94	UC-21	DEC-120
CAP-65	Scoped Academic Advising Access	Must	Realized - Release 1.0 committed	FR-11, FR-62, FR-63, FR-111	UC-12	DEC-121
CAP-66	Scoped Instructor Access	Must	Realized - Release 1.0 committed	FR-11, FR-76, FR-87, FR-111	UC-18, UC-20	DEC-122
CAP-67	Student Data Isolation	Must	Realized - Release 1.0 committed	FR-11, FR-104, FR-111	UC-25, UC-26	DEC-123
CAP-68	Executive Read-Only Analytics	Could	Optional - not Release 1.0 blocker	FR-114	UC-33	DEC-124
CAP-69	Graduate Academic Record Access	Should	Deferred conditional - after Must stability	FR-13	UC-43	DEC-125
CAP-70	Conditional Archived Account Access	Must	Realized - Release 1.0 committed	FR-08, FR-14	UC-09, UC-43	DEC-126

14.C Stakeholder -> Requirement

Table 26 - Traceability Matrix - 14.C Stakeholder -> Requirement

Stakeholder	Role	Requirement Coverage	Use Cases
STK-01	Registrar	FR-15-FR-20,FR-32-FR-34,FR-48-FR-50,FR-67,FR-82,FR-95,FR-100-FR-109,FR-118-FR-119	UC-03,UC-06,UC-09,UC-22,UC-24,UC-27,UC-29,UC-32
STK-02	Registration Staff	FR-32-FR-45,FR-46-FR-51,FR-65-FR-71,FR-103,FR-105-FR-106,FR-110-FR-112,FR-124-FR-128	UC-06-UC-09,UC-13,UC-14,UC-24,UC-27,UC-30,UC-31,UC-40
STK-03	System Administration	FR-01-FR-14,FR-52-FR-54,FR-120,FR-129-FR-133	UC-01,UC-02,UC-10,UC-32,UC-41,UC-43
STK-04	Department Head	FR-17-FR-20,FR-21-FR-28,FR-35-FR-45,FR-64,FR-67,FR-82-FR-85,FR-86,FR-94-FR-96,FR-110-FR-119	UC-03-UC-08,UC-12,UC-18-UC-23,UC-30-UC-32
STK-05	Program Coordinator	FR-18-FR-31,FR-107-FR-109,FR-124-FR-126	UC-03-UC-05,UC-28,UC-29,UC-38,UC-40

Stakeholder	Role	Requirement Coverage	Use Cases
STK-06	Academic Advisor	FR-29-FR-31,FR-47,FR-55-FR-75,FR-98-FR-102,FR-107-FR-109,FR-139	UC-11-UC-16,UC-25,UC-28,UC-38,UC-39
STK-07	Instructor	FR-38,FR-45,FR-52-FR-54,FR-76-FR-96,FR-110-FR-121	UC-07,UC-08,UC-10,UC-18-UC-23,UC-30-UC-37
STK-08	Student	FR-13-FR-14,FR-29-FR-31,FR-46-FR-51,FR-55-FR-85,FR-87-FR-117,FR-121-FR-128,FR-135-FR-139	UC-01,UC-09,UC-11,UC-14-UC-19,UC-23,UC-25-UC-38,UC-43
STK-09	University Management	FR-15-FR-20,FR-110-FR-114	UC-03,UC-30,UC-33
STK-10	SUMS Project Team	NFR-21-NFR-25; all requirements as delivery responsibility	All use cases as analysis/design/test responsibility
STK-11	Course Instructor/Evaluator	OCR/constraint requirements; ASM-01,ASM-12; approval gate	No operational system use case unless separately assigned

14.D Use Case -> FR / BR / SEC / DR / CAP

Table 27 - Traceability Matrix - 14.D Use Case -> FR / BR / SEC / DR / CAP

UC	Name	Priority	FR	BR	SEC	DR	CAP
UC-01	Sign In	Must	FR-01,FR-03,FR-06,FR-07	BR-62	SEC-01-SEC-08	DR-01,DR-17	CAP-27,CAP-51,CAP-52
UC-02	Manage Accounts and Roles	Must	FR-02-FR-12,FR-14	BR-03,BR-62,BR-63	SEC-03-SEC-12	DR-01,DR-17	CAP-27,CAP-28,CAP-62
UC-03	Manage Academic Organization	Must	FR-15-FR-20	BR-01-BR-03	SEC-09-SEC-11	DR-04-DR-06	CAP-01,CAP-63
UC-04	Manage Course Catalogue	Must	FR-21-FR-25	BR-17,BR-18	SEC-09,SEC-13	DR-04-DR-06	CAP-02,CAP-14,CAP-32,CAP-33
UC-05	Manage Study Plan	Must	FR-26-FR-28	BR-02,BR-03	SEC-09-SEC-11	DR-04-DR-06	CAP-02,CAP-63
UC-06	Manage Academic Term	Must	FR-32-FR-34,FR-103	BR-04,BR-05	SEC-09,SEC-13	DR-07,DR-08,DR-17	CAP-03,CAP-31,CAP-37
UC-07	Manage Section and Schedule	Must	FR-35-FR-43	BR-06-BR-10	SEC-09-SEC-13	DR-07,DR-08,DR-17	CAP-04,CAP-05,CAP-15,CAP-56
UC-08	Cancel or Change Section	Must	FR-44-FR-45	BR-11,BR-12	SEC-13,SEC-14,SEC-19	DR-07,DR-08,DR-17	CAP-54,CAP-55
UC-09	Manage Student Record and Status	Must	FR-46-FR-51,FR-126	BR-13,BR-15	SEC-09-SEC-17	DR-02,DR-03,DR-17	CAP-12,CAP-13,CAP-47,CAP-48,CAP-49,CAP-61,CAP-70
UC-10	Manage Instructor Record and Status	Must	FR-52-FR-54,FR-126	BR-07,BR-14	SEC-09-SEC-13	DR-02,DR-03,DR-17	CAP-15,CAP-52
UC-11	Submit Registration Request	Must	FR-55-FR-62	BR-16-BR-22	SEC-09,SEC-10,SEC-13	DR-09,DR-12,DR-17	CAP-06,CAP-07,CAP-22,CAP-32,CAP-53
UC-12	Review Registration Request	Must	FR-62-FR-64	BR-16,BR-23	SEC-09,SEC-10,SEC-14	DR-09,DR-12,DR-17	CAP-06,CAP-65
UC-13	Confirm or Intervene in Registration	Must	FR-65-FR-68	BR-16-BR-22	SEC-11,SEC-14,SEC-18,SEC-19	DR-09,DR-12,DR-17	CAP-06,CAP-07,CAP-61
UC-14	Add Drop or Withdraw	Must	FR-69-FR-71	BR-18,BR-26	SEC-09,SEC-13,SEC-14	DR-09,DR-12,DR-17	CAP-23
UC-15	Manage Course Repeat	Must	FR-72,FR-98	BR-27,BR-45	SEC-09,SEC-13	DR-09,DR-12,DR-17	CAP-34
UC-16	Join and Manage Waitlist	Should	FR-73-FR-74	BR-24,BR-25	SEC-09,SEC-10,SEC-13	DR-09,DR-12,DR-17	CAP-08
UC-17	Auto-Promote Waitlisted Student	Could	FR-75	BR-24,BR-25	SEC-18,SEC-19	DR-09,DR-12,DR-17	CAP-24
UC-18	Record Attendance	Must	FR-76-FR-83	BR-28-BR-35	SEC-09,SEC-10,SEC-13,SEC-14	DR-10,DR-20	CAP-09,CAP-16,CAP-66
UC-19	Manage Laboratory Absence and Makeup	Must	FR-80,FR-85,FR-127	BR-31-BR-33	SEC-09,SEC-14,SEC-21	DR-10,DR-20	CAP-30
UC-20	Define Assessments and Enter Scores	Must	FR-86-FR-88	BR-36,BR-37	SEC-09,SEC-10,SEC-13	DR-11,DR-12,DR-17	CAP-10,CAP-66
UC-21	Calculate Submit Approve and Publish	Must	FR-89-FR-94	BR-38-BR-43	SEC-09-SEC-14,SEC-19	DR-11,DR-12,DR-17	CAP-10,CAP-64

UC	Name	Priority	FR	BR	SEC	DR	CAP
	Grades						
UC-22	Correct Published Grade	Must	FR-95,FR-99-FR-103	BR-44,BR-47	SEC-11,SEC-14,SEC-18,SEC-19	DR-11,DR-12,DR-17	CAP-25,CAP-20
UC-23	Appeal Grade	Should	FR-96,FR-127-FR-128	BR-48	SEC-09,SEC-10,SEC-14,SEC-21	DR-11,DR-12,DR-17	CAP-26
UC-24	Close Academic Term	Must	FR-99-FR-103	BR-44,BR-49,BR-50	SEC-11,SEC-14,SEC-19	DR-11,DR-12,DR-17	CAP-37,CAP-21
UC-25	View Academic Record and GPA	Must	FR-97-FR-102	BR-43-BR-50	SEC-09,SEC-10	DR-12-DR-14	CAP-17,CAP-21,CAP-22,CAP-67
UC-26	View Unofficial Transcript	Must	FR-104	BR-53	SEC-09,SEC-10	DR-12-DR-14	CAP-18,CAP-67
UC-27	Issue Official Transcript	Must	FR-105-FR-106	BR-53,BR-56	SEC-11,SEC-14,SEC-17	DR-12-DR-14	CAP-18,CAP-61
UC-28	Review Degree Progress	Should	FR-29,FR-107-FR-108	BR-51,BR-52	SEC-09,SEC-10	DR-12-DR-14	CAP-35,CAP-36
UC-29	Approve Graduation	Should	FR-108-FR-109	BR-51,BR-52	SEC-11,SEC-14,SEC-17	DR-12-DR-14	CAP-35
UC-30	View Academic Report	Must	FR-110-FR-111,FR-121	BR-54	SEC-09,SEC-10,SEC-20	DR-15-DR-17	CAP-19,CAP-39
UC-31	Export Report	Should	FR-112	BR-54	SEC-09,SEC-10,SEC-21	DR-15-DR-17	CAP-29
UC-32	Review Audit History	Must	FR-118-FR-120	BR-55,BR-56	SEC-14-SEC-16	DR-15-DR-17	CAP-20
UC-33	Use Role Dashboard	Should	FR-113-FR-114	BR-03,BR-54	SEC-09,SEC-10	DR-15-DR-17	CAP-38,CAP-68
UC-34	Receive Notification	Should	FR-115,FR-117	BR-57	SEC-09,SEC-10	DR-15-DR-17	CAP-11
UC-35	Publish Academic Announcement	Should	FR-116	BR-57	SEC-09,SEC-10,SEC-20	DR-15-DR-17	CAP-58
UC-36	Search Authorized Records	Must	FR-121	BR-03	SEC-09,SEC-10,SEC-13	DR-15-DR-17	CAP-39
UC-37	Manage Own Profile	Must	FR-04,FR-122-FR-123	BR-64	SEC-03,SEC-09,SEC-13	DR-18-DR-21	CAP-43,CAP-44
UC-38	Transfer Academic Program	Should	FR-30-FR-31,FR-127-FR-128	BR-03	SEC-09,SEC-10,SEC-14,SEC-21	DR-18-DR-21	CAP-45,CAP-46
UC-39	Maintain Advising Notes	Should	FR-139	BR-03	SEC-09,SEC-10,SEC-20	DR-18-DR-21	CAP-57
UC-40	Import Student or Course Data	Should/Could	FR-124-FR-126	BR-58	SEC-09,SEC-13,SEC-14	DR-18-DR-21	CAP-40,CAP-60
UC-41	Back Up and Restore	Could	FR-129-FR-130	BR-60	SEC-22	DR-18-DR-21	CAP-50
UC-42	Delegate Approval	Could	FR-140	BR-03,BR-63	SEC-24	DR-18-DR-21	CAP-59
UC-43	Access Archived Records	Must/Should	FR-13-FR-14,FR-51	BR-15,BR-56	SEC-08-SEC-10	DR-18-DR-21	CAP-49,CAP-69,CAP-70

14.E Requirement -> Acceptance Criteria

The mapping is complete and one-to-one: $FR - 01 \rightarrow AC - 001$, $FR - 02 \rightarrow AC - 002, \dots$, $FR - 140 \rightarrow AC - 140$ Section 12 contains every row.

14.F Must Requirement -> Planned Test Coverage Placeholder

Each Must FR has a reserved placeholder $TC - PH - FR - ID$ > Example: $FR - 01 \rightarrow TC - PH - FR - 01$ These placeholders are not executable tests; they reserve traceability until the authorized Test Design phase. Should/Could tests may be added without becoming Release 1.0 blockers.

14.G Decision / Assumption -> Requirement

Decision Coverage

Table 28 - Decision Coverage - Decision Range

Decision Range	Controlled Area	SRS Coverage
DEC-15-DEC-27	Governance, scope, boundary, platform, Product Owner	Sections 1,2,15,16; FR-15,FR-137; NFR-24
DEC-28-DEC-44	GPA/grading, graduation, stakeholder authority and project governance	FR-86-FR-109; BR-36-BR-53; UC-20-UC-29
DEC-45-DEC-58	Organization, plans, terms, sections, registration, attendance, grades, students, courses, instructors	FR-15-FR-54; UC-03-UC-10
DEC-59-DEC-71	Attendance thresholds, records, reports, audit, standing, registration, corrections, appeals	FR-55-FR-121; BR-28-BR-50
DEC-72-DEC-86	Accounts, identity, roles, exports, lab, passwords, loads, appeal, notifications, no hard delete	FR-01-FR-14,FR-61,FR-80-FR-85,FR-96,FR-112,FR-115-FR-120; SEC-01-SEC-17
DEC-87-DEC-100	Summer, co-requisites, course types, repeats, degree audit, eligibility, closure, dashboards, search, import, language, profile	FR-22-FR-31,FR-61,FR-72,FR-98,FR-103,FR-107-FR-109,FR-113,FR-121-FR-125,FR-135-FR-136
DEC-101-DEC-126	Transfer, leave/suspension, archive, backup, closed access, holds, cancellation/reassignment, rooms, notes, announcements, delegation, scoped roles	FR-13-FR-14,FR-30-FR-31,FR-40-FR-51,FR-68,FR-139-FR-140; SEC-08-SEC-12,SEC-22,SEC-24
DEC-127-DEC-156	Authority refinements, academic codes/windows/waitlist/attendance/grading precision/GPA truncation	BR-03-BR-05,BR-23-BR-25,BR-28-BR-44; FR-32-FR-109
DEC-157-DEC-176	Password/session/security, logs, retention, backup/import/identifiers/attachments/timezone	FR-01-FR-10,FR-118-FR-130,FR-134; SEC-03-SEC-07,SEC-13-SEC-22
DEC-177-DEC-196	Responsive/browser/performance proposals, maintenance, attachments, reports, concurrency, atomicity, publication, attendance edits, urgent notifications, privacy/errors	FR-83-FR-94,FR-112-FR-138; NFR-02-NFR-19
DEC-197-DEC-216	Repeat/co-requisite/cancellation/section changes/graduation/relative grading/stats/corrections/transcripts/snapshots/report scope	FR-44-FR-45,FR-58,FR-72,FR-89-FR-117; BR-27,BR-38-BR-54
DEC-217-DEC-233	Dismissal/standing/load/appeals/statistics/identity/demo name/performance classification/approval corrections	FR-15,FR-61,FR-96,FR-100-FR-101,FR-108-FR-109; NFR-13-NFR-15
DEC-234-DEC-235	Approved supporting-attachment priority and approved accessibility-basics priority	FR-127-FR-128; NFR-06; SEC-21; DR-20; IR-15; Section 15.4

Assumption Coverage

Table 29 - Assumption Coverage - Assumption

Assumption	Meaning	SRS Coverage	Status
ASM-01	Demonstration video is optional unless explicitly requested by the instructor or course platform	Sections 15 and 16; no product FR	Resolved
ASM-02	Fictional data only	DR-01-DR-21; NFR-19,NFR-25	Approved
ASM-03	No real-university interviews	Source authority and change control; all BR/DEC traces	Approved
ASM-04	No Campus level	FR-40; DR-08	Approved
ASM-05	Configurable record counts	NFR-16; FR-16-FR-18	Approved
ASM-06	Whole-number course credits	FR-23; BR-17/plan calculations as applicable	Approved

Assumption	Meaning	SRS Coverage	Status
ASM-07	Multiple weekly meetings	FR-39; BR-08	Approved
ASM-08	Primary plus Co-Instructors	FR-38; BR-08	Approved
ASM-09	Connected operation only	FR-137; NFR-24; IR-01	Approved
ASM-10	Complete fictional demo dataset	DR-01-DR-21; validation/test planning	Approved

15. Assumptions, Dependencies, and Open Issues

15.1 Preserved Assumptions

Table 30 - Preserved Assumptions Log

ID	Assumption / Clarified Information	Status	Impact / Handling
ASM-01	A Project Demonstration Video is optional and is required only if the course instructor or course platform explicitly requests it.	Resolved	Do not treat the video as a mandatory Release 1.0 deliverable unless a new official instruction is received.
ASM-02	Only fictional student and employee data will be used.	Approved	Applies to all development, testing, demonstration, reports, exports, and backups.
ASM-03	No formal real-university interviews; policies approved internally through virtual Registrar perspective.	Approved	Do not claim external institutional validation.
ASM-04	Campus is not a separate Release 1.0 level.	Approved	Buildings, rooms, and laboratories are sufficient.
ASM-05	Counts of colleges, departments, and programs are configurable.	Approved	No hard-coded counts.
ASM-06	Course credits are whole numbers.	Approved	FR-23 and calculations enforce it.
ASM-07	A section may have multiple weekly meetings.	Approved	FR-39.
ASM-08	A section may have Primary plus Co-Instructors.	Approved	FR-38.
ASM-09	Connected operation only; no offline mode.	Approved	FR-137; NFR-24.
ASM-10	A complete fictional demonstration dataset will be prepared.	Approved	Required for validation and demonstration.

15.2 Dependencies

- Technology-stack selection, architecture design, hosting/deployment assumptions, database design, test-data volume definition, performance-test planning, and technical feasibility validation are prerequisites for approving NFR-13-NFR-15 as final acceptance thresholds.
- UI/UX Design is the prerequisite for the detailed Accessibility Checklist, visual standards, and screen-level criteria under NFR-06.
- Team-role, responsibility, module, testing, and documentation allocations must be completed before repository setup and implementation begin; they do not block SRS approval.
- Approved organization and curriculum records precede offerings; offerings precede registration; registration precedes attendance and grading; approved grades precede GPA, standing, transcripts, and graduation records.
- Later design artifacts must preserve all controlled identifiers and traceability.

15.3 Open Issues Requiring Master Project Control

1. Complete team-role and responsibility allocation before repository setup and implementation; this is not an SRS-approval blocker.
2. Validate or revise NFR-13-NFR-15 after the authorized technical design and performance-planning prerequisites are complete.
3. Create and approve the detailed Accessibility Checklist during the UI/UX Design phase.
4. Confirm whether any new rubric, diagram-count, naming, or submission instruction has been issued.
5. Monitor Must-scope feasibility through a quantified implementation plan and submit any required scope change to Master Project Control instead of changing requirements silently.

15.4 Approved SRS-Phase Decisions

DEC-234 - Approved - Supporting Attachment Priority

File-upload support for appeals, excused absences, program transfers, and exception requests is approved as Should.

Release 1.0 Must workflows shall remain operable without uploaded-file storage by preserving, where applicable:

- an evidence reference
- a reviewer note
- the decision
- and the complete audit history

If attachment upload is implemented, it shall comply with:

BR-59, FR-127, FR-128, SEC-21, DR-20, and IR-15

Decision Status:

Approved by Master Project Control.

DEC-235 — Approved with Modification — Accessibility Priority

Basic accessibility and usability essentials are approved as Must for Release 1.0.

These essentials include:

- clear and meaningful labels
- readable text
- clear form feedback
- accessible validation and error messages
- consistent navigation
- sufficient visual clarity
- and keyboard-accessible interaction for core workflows where applicable.

The following remain Should and shall be defined and approved later during the UI/UX Design phase:

- the detailed Accessibility Checklist
- formal accessibility conformance criteria
- visual accessibility standards
- screen-level accessibility specifications
- and detailed usability validation procedures.

This decision does not claim full compliance with a specific accessibility standard unless such compliance is separately approved and verified.

Decision Status: Approved with Modification by Master Project Control.

No new assumption is required. ASM-13 remains unused.

16. Constraints and Exclusions

- SDLC, Software Engineering, OOP in implementation, a relational database, a user-friendly interface, original work, equal contribution, and explainable work remain official constraints.
- SUMS is a modern connected web application for one generic university.
- No mandatory external integrations or public portal are included.
- Admissions, finance, HR, full LMS, native app, offline mode, automatic timetable generation, and multi-university tenancy remain excluded.
- Technology stack, architecture, classes, database schema, UI screens, deployment, API design, SQL, and source code remain deferred to later authorized phases.
- All Must capabilities remain committed to Release 1.0. No Must capability may be deleted, downgraded, merged, or silently simplified.
- NFR-13-NFR-15 remain Proposed — Pending Architecture Validation and are not mandatory Release 1.0 acceptance blockers.
- Implementation remains unauthorized by this approved SRS baseline. The next authorized work is the controlled design phase; implementation may begin only after the required UML, architecture, database, and UI/UX baselines are approved.

17. Validation and Quality Review

Table 31 - SRS Quality Audit

Audit Check	Result	Evidence / Required Action
1. Identifier continuity	Pass	FR, BR, NFR, SEC, DR, IR, UC, AC, ACT, OCR, STK, CAP, ASM, RSK, and DEC-15-DEC-235 are continuous and preserved.
2. Duplicate requirements	Pass	No exact or high-similarity duplicate FR statement was identified; related policies remain centralized in BRs and profiles.
3. Contradictory requirements	Pass	No direct contradiction was identified. Must, Should, Could, and Proposed treatments remain distinct.
4. Missing acceptance criteria	Pass	All FR-01-FR-140 have exactly one linked AC-001-AC-140.
5. Broken traces	Pass	FR-to-AC, UC-to-FR/BR/SEC/DR/CAP, CAP-to-FR/UC, OCR, stakeholder, decision, and assumption traces were validated.
6. CAP-01-CAP-70 coverage	Pass	All 70 capabilities appear in Section 14.B and Appendix E with Release treatment and requirement coverage.
7. Must capability coverage	Pass	All 49 Must capabilities are realized by Release 1.0 requirements; no Must capability was deleted, downgraded, merged, or silently simplified.
8. Actor-authority consistency	Pass	ACT-01-ACT-10 and the Section 3.1 matrix preserve separation of duties, scope limits, and read-only boundaries.
9. Priority consistency	Pass	49 Must, 15 Should, and 6 Could CAP priorities are preserved; no Should/Could item is an acceptance blocker.
10. Requirement atomicity	Pass with Review	FRs are individually identified and testable; approval reviewers may request editorial splits only through controlled review without changing meaning.
11. Testability	Pass	Every FR has an AC; every Must FR has the reserved TC-PH-FR-ID verification path pending the authorized Test Design phase.
12. Feasibility	At Risk - Not Yet Demonstrated	The committed Must scope requires the quantified work-package, ownership, dependency, integration, and test plan in Section 17.2.
13. Security coverage	Pass	Authentication, authorization, least privilege, separation of duties, validation, audit, integrity, privacy, retention, and accountability are covered by SEC-01-SEC-24 and linked requirements.
14. Business-rule consistency	Pass	BR-01-BR-66 align with the approved academic policies and functional profiles; mixed-priority rules remain explicitly mixed.
15. Terminology consistency	Pass	Registrar, section, term, plan, attempt, published grade, GPA, standing, and priority terms are used consistently.

Audit Check	Result	Evidence / Required Action
16. Cross-reference accuracy	Pass	Section, figure, table, use-case, requirement, and appendix references were reviewed; detailed CAP and trace matrices remain authoritative.
17. Grammar and professional English	Pass	Editorial normalization was applied without changing controlled meanings.
18. Diagram-to-text consistency	Pass	Figures 1-5 restate approved scope and workflows, contain the required informative note, and introduce no design baseline.
19. Table completeness	Pass	Actor, requirement, traceability, acceptance, feasibility, work-package, team, assumption, and approved-decision tables are present as selectable editable text.
20. No accidental implementation decisions	Pass	The SRS does not select a language, framework, database product/schema, API design, class model, architecture, deployment technology, or screen layout.

17.1 Quality Conclusion

The approved baseline is structurally complete and traceable. ASM-01 and ASM-12 are resolved, while ASM-11 is partially resolved without blocking SRS approval. The principal remaining management concern is the feasibility of delivering and fully testing the committed Must scope before the official deadline. The three performance values remain proposals and are not Release 1.0 acceptance blockers until architecture validation is approved.

17.2 Must-Scope Feasibility Review

Table 32 - Must-Scope Feasibility Review

Review Factor	Current Evidence	Assessment
Total Must Functional Requirements	112 Must FRs	Large committed implementation and test surface.
Total Must Use Cases	28 Must use cases, fully specified	Significant end-to-end workflow and integration workload.
Business-Rule Complexity	66 BRs total; 54 are purely Must and two additional mixed rules include Must treatment	High complexity, especially registration, attendance, grading, GPA, standing, transcript, audit, and record-preservation rules.
Security Coverage	24 SEC requirements total; 21 Must	Broad mandatory authentication, authorization, scope, privacy, audit, integrity, and accountability coverage.
System Actors	10 actors: nine human roles and one supporting system/scheduled-process actor	High authorization and scenario-coverage burden.
Integration Dependencies	No mandatory external integration; 11 internal dependency profiles and a critical academic chain from organization/curriculum through offerings, registration, attendance/grading, records, and reports	External risk is limited, but internal integration risk is high.
Available Team Size	Four confirmed members; roles and ownership not yet assigned	Official size is compliant, but unallocated work increases immediate planning risk.
Time Remaining	18 calendar days between the 12-07-2026 review date and the official 30-07-2026 deadline; internal safety cutoff is three hours earlier on the deadline date	Very compressed schedule for professional implementation, integration, security testing, documentation, and presentation preparation.

Feasibility result: At Risk — feasibility is not yet demonstrated. The Must scope is not changed, reduced, merged, downgraded, or simplified by this conclusion. There are no mandatory external integrations, which improves feasibility, but the combination of 112 Must FRs and 28 pure Must use cases, 66 business rules, 21 Must security requirements, 10 actors, internal dependency depth, four team members, and the remaining time creates a high schedule and integration risk.

Impact: Without immediate decomposition and ownership, the project risks incomplete integration, weak security coverage, insufficient regression testing, inconsistent documentation, or late delivery. These risks affect delivery confidence but do not authorize requirement changes.

Controlled response required:

1. Freeze the approved Must requirement set and defer all Should and Could implementation until every Must requirement is complete, integrated, and tested.
2. Complete team role, responsibility, module, testing, and documentation allocation before repository setup and implementation.
3. Produce a quantified work-breakdown and critical-path plan mapping all 112 Must FRs and 28 Must use cases to owners, estimates, dependencies, integration checkpoints, and planned tests.
4. Use daily integration and requirement-to-test tracking against the internal safety deadline.
5. If the quantified plan forecasts that Must delivery is not feasible, record the specific affected identifiers, explain the effect on acceptance, propose explicit controlled options, and submit a formal Change Decision request to Master Project Control.
6. No proposed change becomes effective until Master Project Control approves it.

17.2.1 Must-Scope Feasibility Control

- **Current Must FR count:** 112.
- **Current Must UC count:** 28 pure Must use cases. UC-43 is a separate mixed Must/Should catalogue entry that realizes archived-record controls.
- **Current Must BR count:** 54 purely Must rules; BR-03 and BR-64 also contain Must treatment in mixed-priority rules.
- **Current Must SEC count:** 21.
- **Team members:** Four confirmed members; ownership remains unassigned.
- **Official deadline:** Thursday, 30 July 2026, 11:00 PM - platform-displayed local time.
- **Major dependency groups:** P1 identity/access; P2 organization; P3 curriculum; P4 terms/sections; P5 people records; P6 registration/advising; P7 attendance; P8 grading/GPA; P9 records/transcripts; P10 reporting/audit; P11 supporting operations.
- **Principal schedule risks:** compressed calendar, unassigned ownership, deep internal dependencies, integration of rule-heavy workflows, security verification, regression testing, documentation, presentation preparation, and the risk of late scope discovery.
- **Required mitigation actions:** approve ownership before repository setup; freeze Must scope; defer Should/Could implementation; map all Must FRs and UCs to estimates, dependencies, integration checkpoints, and tests; integrate daily; track against the internal safety deadline; and use formal change control if the quantified plan identifies infeasibility.

Table 33 - Must-Scope Work-Package Planning

Work Package	Related Profile / Requirements	Dependency	Assigned Member	Status
Identity and Access	P1	None / organization identity	To be assigned	Pending
Academic Organization	P2	P1 authorization	To be assigned	Pending
Curriculum and Plans	P3	P2	To be assigned	Pending
Terms and Sections	P4	P2 and P3	To be assigned	Pending
Students and Instructors	P5	P1 and P2	To be assigned	Pending
Registration and Advising	P6	P3, P4, P5	To be assigned	Pending
Attendance	P7	P4, P5, P6	To be assigned	Pending
Grading, GPA, and Standing	P8	P3, P4, P6	To be assigned	Pending
Records and Transcripts	P9	P3 and P8	To be assigned	Pending
Reports, Search, and Audit	P10	P1-P9	To be assigned	Pending
Supporting Operations	P11	Relevant source modules	To be assigned	Pending

17.3 SRS Approval and Next-Phase Gate

This Software Requirements Specification is approved as the requirements baseline for the Smart University Management System - SUMS.

The requirements, business rules, security requirements, data requirements, interface requirements, use cases, acceptance criteria, priorities, and traceability relationships defined in this document shall remain controlled through the approved change-control process.

The next authorized project phase is: Behavioral UML Design

The Behavioral UML Design phase shall derive its models directly from the approved requirements and use cases in this SRS.

The next phase include:

- Main Use Case Diagram.
- Subsystem Use Case Diagrams where necessary.
- Activity Diagrams for major or complex workflows.
- Sequence Diagrams for approved use-case scenarios.
- State Diagrams only where an entity has a meaningful lifecycle.
- Behavioral traceability to FR, BR, SEC, DR, and UC identifiers.

The five figures included in this SRS are informative requirement overviews only. They are not the approved Behavioral UML Design baseline.

The following activities remain deferred until their authorized project phases:

- System Architecture.
- Database Design.
- OOP and Class Design.
- UI/UX Design.
- Technology-stack selection.
- Repository and environment setup.
- Source-code implementation.
- Test implementation.

No implementation shall begin solely on the basis of this SRS without completion and approval of the required design phases.

Approval Status: Approved Baseline v1.0

Approval Authority: Master Project Control

Next Authorized Phase: Behavioral UML Design

Appendix A - Glossary and Acronyms

This appendix provides the controlled glossary used throughout the SUMS requirements baseline. Where a term is governed by a specific requirement or business rule, that controlled statement remains authoritative.

Term / Acronym	Approved Definition
SUMS	Smart University Management System.
GPA	Grade Point Average calculated according to the approved academic rules.
RTL	Right-to-left presentation direction used for Arabic content.
LTR	Left-to-right presentation direction used for English content.
Academic Term	A controlled academic period with configured dates, windows, state, and deadlines.
Course Offering	An approved course made available during a specific academic term.
Section	A term-specific delivery instance of an approved course.
Registration Request	A student-selected set of sections submitted for validation and advisor approval.
Enrollment	The confirmed relationship between a student and a course section.
Advisor Approval	Mandatory academic review before normal registration confirmation.
Hold	An approved restriction that may prevent registration while active.
Waitlist	An ordered conditional queue for an otherwise eligible student when a section is full.
Attendance Barred	An approved academic restriction resulting from an attendance-threshold breach.
Raw Score	The weighted numeric course result before letter-grade mapping.
Academic Standing	The student's approved status of Good Standing, Academic Warning, or Academic Probation.
W	Withdrawn result with no GPA or earned-credit effect.
I	Temporary Incomplete result resolved according to the approved deadline and policy.
P/F	Pass/Fail result classification.
Closed Term	A finalized academic term whose approved academic records are locked.
Closed-Term Snapshot	A fixed report state preserved when an academic term is closed.
Official Transcript	A controlled staff-issued academic record with a unique number and issuance metadata.
Unofficial Transcript	A student-accessible view of the student's own academic record.
Audit Event	An immutable record of a sensitive operation, actor, time, reason, values, and result.
Must	A committed Release 1.0 obligation and acceptance scope.
Should	A conditional capability delivered only after Must-scope stability.
Could	An optional capability that does not block Release 1.0 acceptance.
Proposed	A value awaiting later validation and not yet an approved acceptance threshold.

Appendix B - Requirement Writing Rules

The following rules govern the wording, identification, traceability, and controlled maintenance of SUMS requirements.

1. Each controlled requirement shall have one unique and stable identifier.
2. Mandatory product obligations shall use “The system shall”.
3. Conditional Should obligations shall use “The system should”.
4. Optional Could obligations shall use “The system may”.
5. Each requirement shall express one primary testable obligation.
6. Requirements shall describe WHAT the system must do, not HOW it will be designed or implemented.
7. Every requirement shall be necessary, clear, unambiguous, consistent, feasible, testable, prioritized, and traceable.
8. Every functional requirement shall be linked to an acceptance criterion.
9. Every requirement shall identify its approved source or controlling relationship.
10. Vague words such as fast, easy, suitable, sufficient, or appropriate shall not be used without an approved measurable meaning.
11. Must, Should, Could, and Proposed classifications shall retain their approved Release treatment.
12. Should and Could requirements shall not become Release 1.0 acceptance blockers.
13. Controlled identifiers shall not be deleted, reused, renumbered, or reinterpreted without approved change control.
14. Any approved change shall be reflected in all affected requirements, use cases, business rules, security requirements, data requirements, acceptance criteria, and traceability records.

Appendix C - Requirement Catalogue Index

The following index identifies the controlled requirement catalogues and their authoritative locations. It does not duplicate or replace the controlling statements in the referenced sections.

Requirement Type	Controlled ID Range	Count	Authoritative Location
Actors	ACT-01-ACT-10	10	Section 3
Functional Requirements	FR-01-FR-140	140	Section 4
Business Rules	BR-01-BR-66	66	Section 5
Non-Functional Requirements	NFR-01-NFR-25	25	Section 6
Security Requirements	SEC-01-SEC-24	24	Section 7
Data Requirements	DR-01-DR-21	21	Section 8
Interface Requirements	IR-01-IR-16	16	Section 9
Use Cases	UC-01-UC-43	43	Sections 10-11
Acceptance Criteria	AC-001-AC-140	140	Section 12
Capabilities	CAP-01-CAP-70	70	Section 14.B and Appendix E

The authoritative requirement statements remain in the referenced sections. This appendix is a controlled catalogue index and does not create additional requirements.

Appendix D - Use Case Catalogue Summary

Section 10 contains the complete controlled use-case catalogue. Section 11 contains the approved detailed specifications required for Release 1.0 Must behavior.

Use Case Classification	Count	Authoritative Location
Total Use Cases	43	Section 10
Must	28	Sections 10 and 11
Should	10	Section 10
Could	3	Section 10
Must / Should	1	UC-43
Should / Could	1	UC-40

Detailed specifications cover all pure Must use cases and UC-43 because UC-43 includes approved Must behavior. Should and Could use cases remain catalogued and traceable without becoming Release 1.0 acceptance blockers.
UC-43 - Access Archived Records

Goal: Allow a graduate or archived user to access permitted read-only academic history or receive blocked access according to the recorded archive policy.

Primary actor: Graduate / Archived User

Supporting actors: System Administrator; Registration Staff; System

Stakeholder interests: The user needs controlled access to permitted personal academic records. The university must preserve official history and prevent unauthorized new academic operations. Security requires strict identity, scope, and confidentiality controls.

Preconditions: An archived academic account or record exists. The account status, archive reason, and applicable access policy have been recorded. The user provides valid authentication credentials where access is permitted.

Trigger: The graduate or archived user signs in or requests access to an archived academic record.

Main Success Scenario

1. The user provides the assigned account identifier and valid credentials.
2. The system retrieves the account status, archived-record state, and recorded administrative reason.
3. The system evaluates whether the applicable policy permits read-only access or requires blocked access.
4. When read-only access is permitted, the system restricts the user to the approved personal records, transcript, notifications, and permitted profile information.
5. The system prevents modification of official records and prevents new academic operations that are not permitted for the archived status.
6. The system records the login or access activity and displays only the authorized archived information.

Alternative Flows

A1: Where graduate record access is delivered, the graduate receives the restricted access defined by FR-13.

A2: An authorized administrator may process a separate account-state or record-status change through the applicable account or academic-record management use case. Exception / Validation Flows.

E1: Invalid credentials cause access to be denied using a safe error message.

E2: When the recorded archive policy requires blocked access, the system denies the request without revealing unauthorized record information.

E3: Any attempt to modify an archived official record or initiate an unauthorized academic operation is rejected and recorded as required.

Postconditions: No official academic data is modified. The recorded archived-account access policy remains enforced.

Success guarantee: The user receives only the permitted read-only archived information within the approved data scope.

Minimal guarantee: No unauthorized academic information is disclosed or changed.

Priority: Must / Should

Frequency: As needed

Related FRs: FR-13-FR-14, FR-51

Related BRs: BR-15, BR-56

Related SEC: SEC-08-SEC-10

Related DRs: DR-18-DR-21

Acceptance criteria: AC-013-AC-014, AC-051

Open issues: FR-13 graduate record access remains Should. FR-14 and FR-51 remain Must.

Appendix E - CAP-01-CAP-70 Coverage Matrix

This appendix summarizes the approved capability coverage. The complete controlling CAP-to-FR/UC matrix remains in Section 14.B.

CAP Priority	Count	Release Treatment	Coverage
Must	49	Committed Release 1.0 scope	49/49 covered
Should	15	Conditional after Must stability	15/15 documented
Could	6	Optional	6/6 documented
Total	70	-	70/70 covered

No Must capability is uncovered.

Any later CAP mapping change requires approved change control and an update to every affected traceability relationship.

Appendix F - Open Issues and Pending Validation Targets

The following controlled items remain pending or deferred after SRS approval. They do not alter the approved requirement meanings or priorities.

Pending Item	Current Status	Effect	Required Resolution
Final team-role allocation	Pending under ASM-11	Does not block SRS approval; blocks repository setup and implementation.	Master Project Control assigns roles, responsibilities, ownership, testing, and documentation duties.
Detailed Accessibility Checklist	Deferred to UI/UX Design	General accessibility requirements remain valid.	Create and approve the detailed checklist during UI/UX Design.
NFR-13-NFR-15	Proposed - Pending Architecture Validation	Not Release 1.0 acceptance blockers.	Validate technology, architecture, deployment, dataset, workload, and performance-test plan.
Quantified Must-scope feasibility plan	Pending	Must-scope feasibility remains At Risk.	Approve work packages, ownership, estimates, dependencies, integration, and test plan.
Later official instruction or rubric	Monitored	May require a controlled document update.	Record official evidence and process the change through approved change control.

DEC-234 and DEC-235 are approved and are not open issues.

Appendix G - Baseline Preservation and Change-Control Rules

The approved baseline shall be preserved through the following change-control rules.

1. Approved controlled identifiers and meanings shall not be deleted, reused, renumbered, downgraded, promoted, merged, or reinterpreted without approved change control.
2. The next available decision identifier is DEC-236.
3. The next available assumption identifier is ASM-13.
4. Every requested change shall record the following information:
 - Change request identifier.
 - Requester and date.
 - Reason and supporting evidence.
 - Affected controlled identifiers.
 - Scope and priority impact.
 - Actor and authority impact.
 - Business-rule impact.
 - Security and privacy impact.
 - Data and interface impact.
 - Use-case and acceptance impact.
 - Testing and documentation impact.
 - Master Project Control decision.
 - Approved version and effective date.
5. No proposed change becomes effective before Master Project Control approval.
6. Approved changes shall update every affected traceability relationship.
7. Should and Could requirements shall not be promoted to Must, and Must requirements shall not be downgraded, without an approved Change Decision.
8. No implementation shall begin before completion of the required approved planning and design gates.
9. The authoritative editable baseline shall remain the approved DOCX; the PDF shall remain the corresponding read-only reference copy.